

From Recordings to Ecology: Turning Bird Sounds into Ecological Insights

by

Yi-Chin (Sunny) Tseng

M.Sc., University of British Columbia, 2019

B.A., National Taiwan University, 2017

THESIS SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY
IN
NATURAL RESOURCES AND ENVIRONMENTAL STUDIES

UNIVERSITY OF NORTHERN BRITISH COLUMBIA

April 2026

© Yi-Chin (Sunny) Tseng, 2026

Abstract

Passive acoustic monitoring is increasingly used to study bird populations across large spatiotemporal scales; however, converting large volumes of audio data into reliable ecological information remains a major challenge. Using three years of passive acoustic monitoring data collected in the John Prince Research Forest, British Columbia, Canada, I evaluate the performance of machine learning tools for avian acoustic identification, specifically BirdNET, and demonstrate their application to ecological questions across multiple ecological scales. First, I develop a workflow for setting species-specific confidence thresholds for BirdNET detections, comparing this approach with the use of a single universal threshold across species. I show that species-specific thresholds increase precision (the proportion of true positives among predicted detections) while also retaining a greater number of usable detections, thereby improving data quality without unnecessary data loss. Building on this workflow, I then demonstrate the feasibility of identifying individual Barred Owls (*Strix varia*) from passive acoustic monitoring data. Recordings collected under typical field conditions contain sufficient acoustic detail to extract individually distinctive vocal characteristics, highlighting the potential for passive acoustics to support individual-level analyses in nocturnal species. Third, I apply passive acoustic monitoring to quantify seasonal vocal activity patterns and habitat associations using a case-study species, the Olive-sided Flycatcher (*Contopus cooperi*). The resulting activity patterns and habitat relationships provide insight into species-specific phenology. Finally, I integrate passive acoustic monitoring data with LiDAR-derived habitat metrics to assess breeding bird richness. This study illustrates both the opportunities and the limitations of combining emerging monitoring technologies to investigate bird–habitat relationships at broader spatial scales.

Together, this thesis demonstrates that passive acoustic monitoring coupled with automated identification tools, when carefully evaluated and applied, can move beyond simple presence–absence inference to address a wide range of ecological and management-relevant questions. The approaches developed here support more transparent, reproducible, and scalable use of passive acoustic monitoring for avian research and conservation.

From Recordings to Ecology

Turning Bird Sounds into Ecological Insights

Recent advances in machine learning-based tools, such as BirdNET, have enabled unprecedented scale in passive acoustic monitoring for avian studies; however, they also generate large volumes of detections that are challenging to validate and analyze. **This thesis addresses this gap by developing validation workflows and integrating statistical methods to better support reliable ecological inference from BirdNET outputs.**



John Prince Research Forest

May-Jul

Feb-Apr

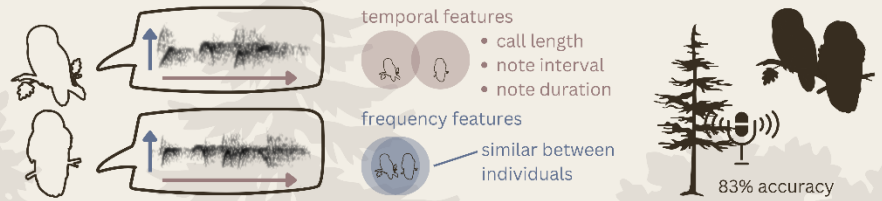
2020

2021

2022

Within species

Passive acoustic recordings contain sufficient detail to **capture individually distinctive vocal characteristics** for Barred Owls, demonstrating the potential for individual-level analyses.



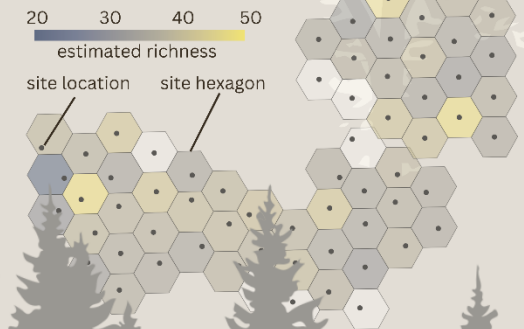
Single species

Integrating **generalized additive and occupancy modelling** with passive acoustic monitoring **reveals species-specific phenology and habitat associations.**



Multiple species

Integrating **rarefaction curves** with passive acoustic monitoring **enables the assessment of avian species richness variation across landscapes.**



Threshold

Species-specific BirdNET confidence thresholds improve **precision** and **retain more detections** than universal thresholds.



Putting it all together: birdnetTools

This thesis integrates **methodological innovation** with **passive acoustic monitoring** to improve inference on avian communities and vocal behaviour, with all developed approaches summarized in the open-source **birdnetTools R package**, providing complete **post-processing tools for BirdNET detection outputs**. - Sunny Tseng, PhD thesis.



Table of Contents

Abstract.....	i
Table of Contents	iv
List of Tables	viii
List of Figures.....	ix
Acknowledgements	xi
Dedication	xiv
Preface.....	xv
Chapter 1 General Introduction.....	1
1-1 Passive acoustic monitoring.....	1
1-2 Motivation and research context.....	3
1-3 Study system monitoring design.....	4
1-4 Thesis scope and structure.....	7
Chapter 2 Setting BirdNET Confidence Thresholds: Species-specific vs. Universal Approaches	9
2-1 Abstract	9
2-2 Introduction	10
2-3 Methods	12
2-3-1 Study area.....	12
2-3-2 Audio data processing	14
2-3-3 Target species and song/call complexity.....	15
2-3-4 Validation dataset.....	18
2-3-5 Calibration curves – linking confidence value to true positive rate	19
2-3-6 Confidence threshold setting	19
2-4 Results.....	21

2-4-1 Linking confidence value to true positive rate	21
2-4-2 Impact of confidence threshold setting	24
2-4-3 Song/call complexity vs BirdNET performance	29
2-5 Discussion	31
2-5-1 Overview of BirdNET performance and thresholds	31
2-5-2 Applications of two types of thresholds	32
2-5-3 Workflow to define species-specific threshold	33
Chapter 3 Using Autonomous Recording Units for Vocal Individuality: Insights from Barred Owl Identification	36
3-1 Abstract	36
3-2 Introduction	37
3-3 Methods	41
3-3-1 Target species and study area	41
3-3-2 Audio data collection	44
3-3-3 Sound feature processing	45
3-3-4 Statistical modelling	49
3-3-5 Spatial distance and vocal similarity	50
3-4 Results	51
3-4-1 Temporal features outperform frequency features	51
3-4-2 LDA model	54
3-4-3 Identification performance	56
3-4-4 Low correlation between spatial distance and vocal similarity	58
3-5 Discussion	59
3-6 Conclusion	62
Chapter 4 Quantifying Seasonal and Habitat-related Bird Vocal Activity Using Passive Acoustic Monitoring with GAMs and Occupancy Models	63
4-1 Abstract	63

4-2 Introduction	64
4-3 Methods	67
4-3-1 Target species	67
4-3-2 Study area	68
4-3-3 Audio processing	68
4-3-4 Modelling temporal pattern – generalized additive models	70
4-3-5 Modelling spatial pattern – occupancy models	72
4-4 Results	74
4-4-1 Temporal vocal activity pattern	74
4-4-2 Spatial vocal activity pattern	77
4-5 Discussion	79
4-5-1 Temporal vocal activity pattern	79
4-5-2 Spatial vocal activity pattern	80
4-6 Conclusion	82
 Chapter 5 Integrating Passive Acoustic Monitoring and LiDAR for Breeding Bird Habitat Modelling in A Working Forest	 83
5-1 Abstract	83
5-2 Introduction	84
5-3 Methods	86
5-3-1 Study area and acoustic recording sites	86
5-3-2 Acoustic data processing and target species	88
5-3-3 Asymptotic richness detected by ARUs	92
5-3-4 LiDAR covariates and model selection	94
5-4 Results	96
5-4-1 Asymptotic richness estimates	96
5-4-2 LiDAR covariate pre-selection	97
5-4-3 Asymptotic richness model with LiDAR covariates	98

5-5 Discussion	101
5-6 Conclusion	103
Chapter 6 General Conclusion	104
6-1 Summary of contributions	104
6-2 Recommendations for the passive acoustic monitoring program	105
6-3 Future directions.....	107
6-3-1 BirdNET post-processing and threshold setting.....	107
6-3-2 Understanding the causes of species-specific BirdNET performance	107
6-3-3 Expanding BirdNET beyond species-level classification	108
6-3-4 Integrating passive acoustic monitoring with ecological modelling	109
6-3-5 Integrating passive acoustic monitoring with camera traps and other monitoring methods	110
References	112
Appendix.....	123
Appendix for Chapter 2.....	123
Appendix for Chapter 3.....	127
Appendix for Chapter 4.....	129
Appendix for Chapter 5.....	130

List of Tables

Table 2-1 Parameters used in this study for the BirdNET analyzer.....	14
Table 2-2 Target species in this study.....	16
Table 2-3 Statistics for each target species.	25
Table 3-1 Statistics, including mean, standard deviation, and coefficient of variation	52
Table 3-2 Correlations between discriminant function and predictor variables	55
Table 4-1 Candidate generalized additive models of daily Olive-sided Flycatcher detections ...	71
Table 4-2 AICc-based comparison of candidate GAM models.	75
Table 5-1 BirdNET arguments with default values and the values used in this study.....	89
Table 5-2 The 40 breeding passerine species detected during the dawn chorus.....	90
Table 5-3 LiDAR-derived habitat covariates used in the analysis.....	95

List of Figures

Figure 1-1 Spatial distribution of the 66 Autonomous Recording Units monitoring sites	6
Figure 1-2 Conceptual structure of the thesis.	7
Figure 2-1 Locations of the 66 recording sites	13
Figure 2-2 Spectrogram illustrating the detection of song/call signals from a sound recording .	17
Figure 2-3 Calibration curves to estimate appropriate number of recordings	22
Figure 2-4 Calibration curves for the 19 target species	23
Figure 2-5 The number of detections identified by BirdNET.....	28
Figure 2-6 Vocal complexity of the 19 target species ranked by the BirdNET's performance...	30
Figure 2-7 Workflow for defining species-specific thresholds.....	35
Figure 3-1 Locations of the 66 recording sites	43
Figure 3-2 Barred Owl calls collected during the 2021 breeding season	46
Figure 3-3 Spectrogram of a Barred Owl call, consisting of eight notes.....	48
Figure 3-4 The change of overall accuracy and Kappa statistic	54
Figure 3-5 The ordination plot of the final LDA model	56
Figure 3-6 The confusion matrix summarizes the final linear discriminant model's outcomes ..	57
Figure 3-7 Low correlation between the distance and the vocal similarity	58
Figure 4-1 Site map of the John Prince Research Forest in British Columbia, Canada	69
Figure 4-2 Detection history for Olive-sided Flycatcher	72
Figure 4-3 Seasonal activity of Olive-sided Flycatcher within a site-year	74
Figure 4-4 Model predictions.....	76
Figure 4-5 Posterior effect sizes of covariates on occupancy and detection probabilities	77
Figure 4-6 Spatial predictions from the occupancy model for Olive-sided Flycatcher	78

Figure 5-1 Spatiotemporal distribution of survey effort	87
Figure 5-2 Number of days each species was detected at each site using ARUs	93
Figure 5-3 Rarefaction curves for sites with different survey efforts	96
Figure 5-4 Pre-selection of LiDAR covariates	97
Figure 5-5 Sum of model weights for each LiDAR covariate across different species groups ...	99
Figure 5-6 K-means clustering of target species based on site-use patterns.....	100
Figure 5-7 Simple linear regression of the most influential LiDAR covariate.....	100
Figure 6-1 Overview of the workflow implemented in the R package birdnetTools	105

Acknowledgements

It takes a village to raise a child, and the same is true for finishing a PhD. Although this thesis is written as an individual piece of work, completing it would have been impossible without the generous support of family, friends, colleagues, and funding agencies. The people who supported me throughout this PhD journey are far more than I can list individually here, and I am deeply grateful to all who contributed to this special life experience in different ways.

I would like to express my deepest gratitude to my supervisor, Ken Otter, who has shown unwavering trust in me from the very beginning to the end of my PhD. Your strong belief in me, and your steady support during challenging times, have been a source of reassurance and strength. Thank you for being open and curious when I proposed new ideas and methods, and for trusting my decisions, which encouraged me to grow into an independent researcher. I am fortunate to have had you as my supervisor.

My supervisory committee consists of Heather Bryan, Dexter Hodder, and Joe Shea. Thank you, Heather, for your insightful guidance on my statistical analyses, and for the many stimulating collaborations in the Applied Analysis Hub. Beyond the academic support, you also helped me through many difficult moments. Thank you, Dexter, for being a role model to me as a field ecologist. I am deeply grateful for the time you spent showing me how to do fieldwork in the research forest and for encouraging me to think from the perspectives of forests and wildlife. Thank you, Joe, for your encouragement and for providing me with valuable opportunities to interact with other PhD students, as well as for equipping me with a strong foundation in remote sensing and data analysis.

No research can be completed without financial support and collaborative partnerships. This project was funded by the Natural Sciences and Engineering Research Council of Canada,

the Habitat Conservation Trust Foundation, Mitacs Accelerate, and the John Prince Research Forest. I respectfully thank Tl'azt'en Nation, Binche Whut'en, and Nak'azdli Whut'en for supporting this work on their territories and for their collaboration in data collection. I extend my sincere thanks to Shannon Crowley, Lauren Wheelhouse, Morgan Conrad, Steven Murdock, and Amanda Carriere for their invaluable assistance with data collection. Numerous scholarships and fellowships also played an important role in supporting my PhD, including those provided by the University of Northern British Columbia, BC Nature, Nature Vancouver, the R Consortium, rOpenSci, and the Society of Canadian Ornithologists.

Thank you to all the WildCoers at UBC! I am especially grateful to Cole Burton for adopting me as a visiting student and giving me a second academic home. Thank you to Erin, Katie, and Maddy for being the first friends I made in WildCo and for all the fun times we shared in the office, in the field, and at the climbing gym. I have witnessed the growth of the lab and am sincerely thankful to all the kind and warm members: Zoe, Taz, Ali, Hannah, Tristen, Lindsay, Jag, Murat, Larissa, Elena, Vanessa, and Mary. Also, Nick, Jenny, Katie, and Kibo (!) from Gaynor's lab. Thank you for accepting me to be part of this wonderful group. I cannot overstate how meaningful these interactions and friendships have been to me.

Thank you to the BirdNET team in Chemnitz, Germany. I am especially grateful to Stefan Kahl. Thank you for creating BirdNET, and for being such an open-minded computer scientist, including sharing your insights and strong support through the development of birdnetTools. I also thank Josef, Max, and Raja for being such fun companions during my stay in Germany. I enjoyed our time in the office, as well as the climbing gym, festivals, and walks with Nala (the dog). Working with the BirdNET team in person in Germany was my dream come true.

To my dearest friends, thank you. Faezeh, I am grateful to have shared this PhD journey. Colin, thank you for always welcoming me so warmly in Prince George. Hannah, I was so lucky to have you as a conference buddy, and as a friend to share new findings and excitements in research. Kaitlyn, thank you for the pancake recipe that became my Saturday morning routine. Jasmine, thank you for always being there in both good and difficult times. Jerome, thank you for the strong remote emotional support regardless of our time difference. Kuan, I look forward to our freediving adventures soon. Red Bird, thank you for being a guiding light in the acoustic field and for all the positive encouragement. Vanessa and Shiao-Yu, thank you for the blue rice cooker that nourished me well! Mandy, thank you for the comforting conversations, concerts, and afternoon tea. Sam and Zoe, you will always hold a special place in my heart. Philippa, I will always treasure the fun times with you and Felicitas. Stella and Derek, thank you for giving me a home, literally, and for our cherished Saturday coffee breaks. And Peter! My best pre-neighbor and now a dear bird sound recording friend. I still remember meeting you by your garden, and it is hard to believe you have witnessed me complete both my MSc and my PhD!

Last but not least, I extend my deepest gratitude to my family back in Taiwan: my mom, dad, brother, grandma, and all my extended family. Establishing myself in a new country while staying connected to my roots has not always been easy, but your constant support made it possible. Thank you, Gintas. It has been a long journey. Although waking up early, birding and acoustic monitoring may sound unfamiliar to you, thank you for always understanding my passion. I cherish the fun we had together in the field setting up acoustic recorders in Nida. It brings me great joy to share my understanding of nature, to learn from your perspective, and to explore new knowledge together. Thank you for being on this journey with me, as we continue to learn more about nature and about ourselves.

Dedication

To my grandma, Yu-Ying Lan (1935–2024), a guiding light in our family, who taught me to
move through the world with gentleness.

Preface

This thesis, specifically Chapter 2 to Chapter 5 presented original research and were structured as individual journal manuscripts.

Portions of this thesis have been published in the following peer-reviewed articles:

- Chapter 2 : Tseng, S., D. P. Hodder, and K. A. Otter. 2025. Setting BirdNET confidence thresholds: Species-specific vs. universal approaches. *Journal of Ornithology* 166:1123-1135. <https://doi.org/10.1007/s10336-025-02260-w>
- Chapter 3 : Tseng, S., D. P. Hodder, and K. A. Otter. 2024. Using autonomous recording units for vocal individuality: Insights from Barred Owl identification. *Avian Conservation and Ecology* 19(1):23. <https://doi.org/10.5751/ACE-02680-190123>

Two chapters are expected to be published as peer-reviewed articles:

- Chapter 4: Passive acoustic monitoring reveals seasonal and habitat-related patterns in bird vocal activity, in *Journal of Wildlife Management*.
- Chapter 5: Integrating passive acoustic monitoring and LiDAR for breeding bird habitat modelling in a working forest, in *Methods in Ecology and Evolution*.

In addition, the following publication is related to the methodological developments of this thesis but is not directly included as a chapter:

- Tseng, S., and S. Kahl. 2026. birdnetTools: An R package for working with BirdNET output. *Journal of Open Source Software*. (Under Review)

Chapter 1 General Introduction

1-1 Passive acoustic monitoring

Detecting birds using their vocalizations has long been a fundamental component of avian surveys, including traditional point counts and transect surveys (Bibby et al. 2000). Historically, these surveys relied on human observers to identify birds by sound, an approach that is inherently subject to observer bias arising from differences in identification skill or experience (Cunningham et al. 1999). The development of autonomous recording units (ARUs) has fundamentally transformed passive acoustic monitoring (PAM) by enabling standardized and continuous audio collection while reducing observer-related biases (Gibb et al. 2019; Shonfield and Bayne 2017; Sugai et al. 2019). Recent technological advances have made ARUs increasingly scalable and cost-effective, allowing for the accumulation of large volumes of acoustic data across broad spatial and temporal scales. As a result, PAM has become a widely recognized and commonly applied method in bird monitoring (Shonfield and Bayne 2017).

Despite the rapid expansion of passive acoustic monitoring, a major bottleneck remains due to the needs to transcribe large volumes of audio data into ecologically meaningful information (Gibb et al. 2019). Completing manual annotation of all recordings becomes increasingly infeasible with expanded spatiotemporal scales enabled by ARUs. To speed up the efficiency in analyzing acoustic data, proxy approaches have been developed, such as bioacoustic indices (Alcocer et al. 2022). These indices summarize acoustic properties and have been shown, in some well-characterized systems, to correlate with ecological metrics such as species richness or abundance (Alcocer et al. 2022). However, their interpretation is highly context dependent and often requires detailed knowledge of local soundscapes, limiting their general applicability

(Sueur et al. 2014). As the demand for species-level inference continues to grow in ecological research and conservation, approaches that can reliably derive accurate species detections from acoustic data are increasingly needed.

To address this gap, researchers have increasingly turned to machines to automate the analysis of acoustic data (Pollock et al. 2025). Methods have been developed for sound event detection, which aims to locate acoustic events within recordings, as well as for sound event classification or identification, which assigns a biological label to detected sounds; some approaches integrate both tasks within a single framework (Stowell 2022). These methods typically rely on transforming audio signals into spectrograms, that can be processed using image-based pattern recognition techniques (Stowell 2022). Advancements have led to the creation of commercial software, such as Wildlife Acoustics' Kaleidoscope Pro and open-source tools such as the R package warbleR (Araya-Salas and Smith-Vidaurre 2017), which automate species identification for both researchers and the broader public. Among these methods, one of the most popular platforms for analyzing such data is BirdNET Analyzer (Kahl et al. 2021), whose algorithm is available not only as a free phone app (<https://doi.org/10.1371/journal.pbio.3001670>) but also accessible through a publicly available repository on GitHub (github.com/kahst/BirdNET-Analyzer).

Since its release in 2021, BirdNET Analyzer has emerged as one of the most widely adopted platforms for automated bird sound detection and classification. BirdNET was developed through a collaboration between the K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology and Chemnitz University of Technology (Kahl et al. 2021). Released as an open-source algorithm, BirdNET has expanded rapidly, with current versions (v2.4) supporting automated detection of more than 6,000 bird species worldwide

(BirdNET Analyzer Github, accessed on 2026 Jan. 15th). BirdNET analyzes audio recordings by segmenting sound files into short temporal windows and applying a deep learning model trained on large, curated bird sound libraries, including Xeno-canto (<https://www.xeno-canto.org>) and the Macaulay Library (<https://www.macaulaylibrary.org/>). The algorithm produces species-level detections accompanied by confidence values ranging from zero to one, which reflect the relative certainty of each detection. While this automated approach enables rapid and scalable analysis of large acoustic datasets, challenges remain in interpreting confidence values, accounting for variation in species-specific performance, and incorporating detections into ecological analyses, all of which can influence ecological inference (Pérez-Granados 2023). Addressing these challenges is critical for the reliable application of automated acoustic detections and forms a central motivation for the research presented in this thesis.

1-2 Motivation and research context

This thesis was motivated by the establishment of a long-term biodiversity monitoring program at the John Prince Research Forest (54°27'N, 124°10'W, 700 m a.s.l., ~16,000 hectares), a community-tenure, actively managed forest located in the dry sub-boreal spruce biogeoclimatic zone of British Columbia, Canada (Biogeoclimatic Zones of British Columbia 2024). One of the aims of this initiative was to document avian community dynamics within a managed forest landscape, examining how species respond to different forest management regimes in a community-based forest striving to implement a long-term sustainable and balanced forestry model. The first three years of data collection (2020–2022) provide a unique opportunity to examine patterns of species occurrence, seasonal activity, and monitoring effort.

Coinciding with the launch of this PhD project in 2021, the BirdNET automated sound recognition algorithm became publicly available, offering unprecedented capacity to process large volumes of acoustic data using machine learning. The emergence of this tool fundamentally changed what was feasible for passive acoustic monitoring, enabling rapid and scalable extraction of species-level information from extensive audio datasets. Together, the availability of a growing long-term monitoring dataset and the development of BirdNET created a timely opportunity to explore how automated acoustic detection can be effectively integrated into ecological research and monitoring within a working forest context.

This thesis was therefore developed at the intersection of long-term biodiversity monitoring and emerging machine learning approaches. Its overarching motivation is to evaluate and advance passive acoustic monitoring for ecological inference, while addressing key challenges related to detection reliability, effort standardization, and integration with ecological models. By doing so, this work aims to contribute to the broader methodological framework for applying automated acoustic tools in ecological research.

1-3 Study system monitoring design

The data used in this thesis were collected as part of a long-term biodiversity monitoring program at the John Prince Research Forest. The monitoring program employed passive acoustic methods to survey avian communities across 66 sites using autonomous recording units (ARUs; AudioMoth v1.1.0; Hill et al. 2019), with adjacent sites spaced at least 2 km apart (**Figure 1-1**).

Two seasonal monitoring protocols were implemented to target distinct components of the avian community. Summer monitoring (**Figure 1-1A**) focused on the breeding season of most songbirds and was conducted from May to July over three years (2020–2022), between

daily dawn chorus (4am to 7am). ARUs were set to record for 1 minute, followed by a 4 minutes pause, throughout the three hours recording period, resulting in 36 one-minute recordings per day per site. This effort resulted in approximately 3,984 hours of audio data across three years, with detections of 122 validated bird species (See Appendix **Table A5-1**). In contrast, winter monitoring (**Figure 1-1B**) targeted the breeding period of owls and was conducted from February to April over two years (2021–2022), with recordings scheduled during evening hours to coincide with peak owl vocal activity (6pm to 10pm). This resulted in a total of eight 10-minute recordings collected per day, per site. The winter monitoring program generated approximately 9,266 hours of audio data.

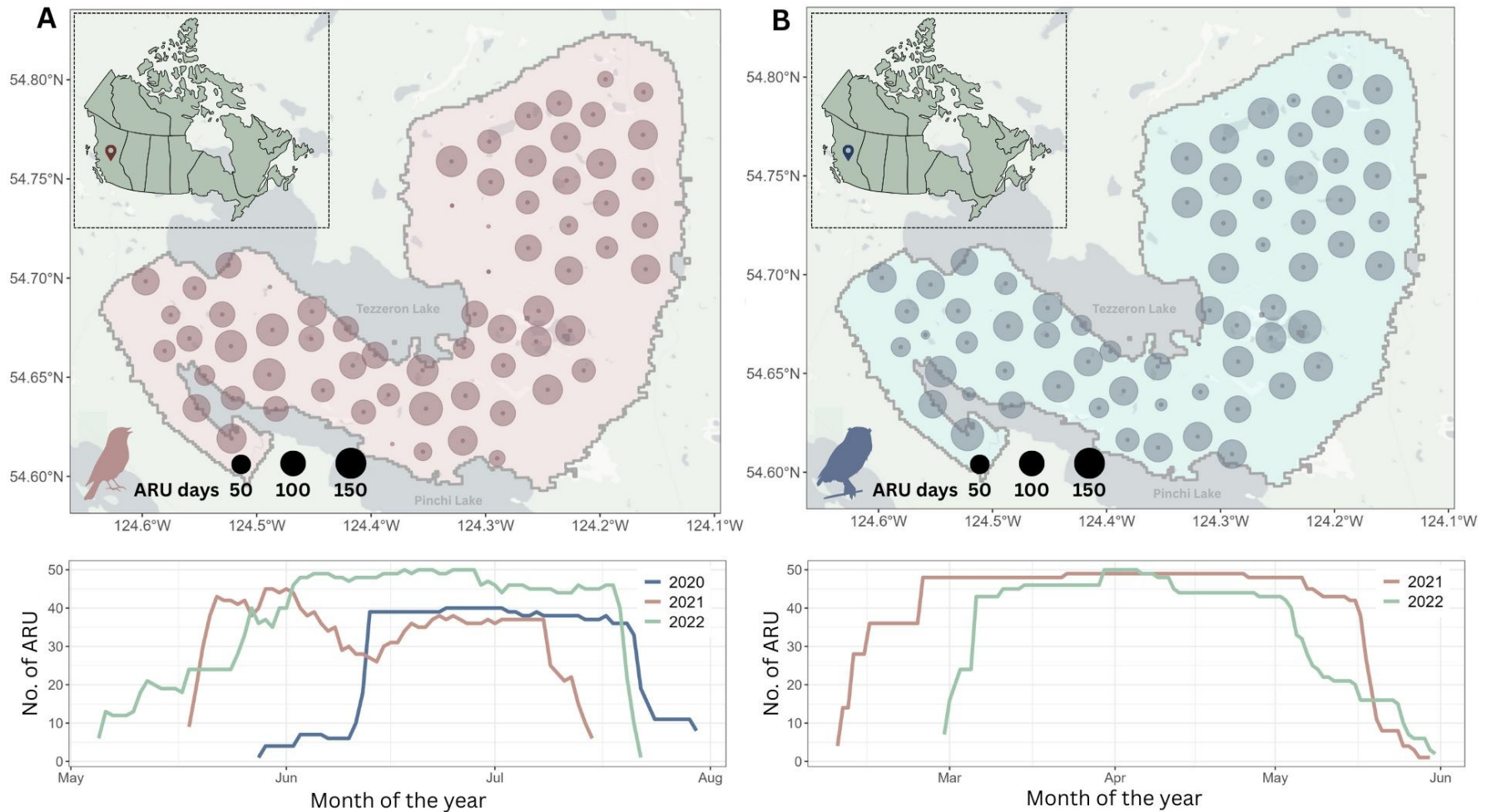


Figure 1-1 Spatial distribution of the 66 Autonomous Recording Units (ARUs) monitoring sites within the 16,000 ha John Prince Research Forest. Circle sizes represent total ARU days per site, and time series showing the number of active ARUs across years during the (A) summer and (B) winter monitoring seasons, illustrating seasonal deployment and retrieval patterns.

1-4 Thesis scope and structure

This thesis integrates ecological applications and methodological development to advance the use of passive acoustic monitoring and automated detection tools. Using data from the John Prince Research Forest long-term monitoring program and detections generated by BirdNET, the thesis addresses key challenges related to detection reliability and ecological inference from large acoustic datasets (**Figure 1-2**).

Chapter 2 compares two approaches for setting BirdNET detection thresholds: a universal threshold applied to all species, and species-specific thresholds tailored to achieve high precision (the proportion of true positives among predicted detections) for each species. Using multi-species songbird recordings, I show that species-specific thresholds achieve high precision and retain more detections in the final dataset compared with a universal threshold. This chapter provides practical guidance for applying BirdNET in biodiversity monitoring and contributes to best practices for automated acoustic analyses. Chapter 3 explores the potential for individual-level monitoring using passive acoustic recordings, focusing on Barred Owls (*Strix varia*). I show that recordings collected by ARUs contain sufficient quality to distinguish individual vocal signatures, demonstrating that passive acoustic monitoring can extend beyond species-level

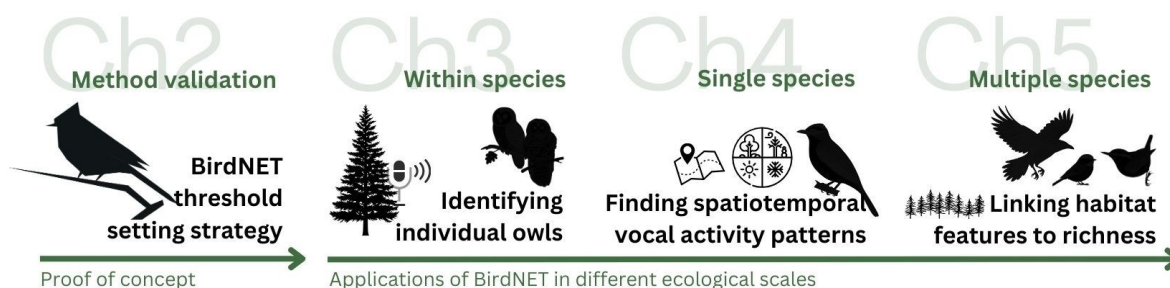


Figure 1-2 Conceptual structure of the thesis, linking methodological development and ecological applications using BirdNET detections.

presence–absence data. This chapter highlights how automated acoustic detections can support finer-scale ecological analyses, such as individual behavior and movement patterns, providing a framework for integrating detection with individual-level inference. Chapter 4 applies BirdNET detections to population-level modelling, integrating detections with occupancy models and generalized additive models to investigate seasonal vocal activity patterns and habitat associations of a targeted species: the Olive-sided Flycatcher (*Contopus cooperi*). This chapter demonstrates how automated detections can be incorporated into established ecological frameworks. Chapter 5 applies BirdNET detections to community-level analyses, using rarefaction-based methods to estimate species richness from large acoustic datasets. This chapter addresses challenges related to uneven sampling effort and demonstrates a scalable approach for comparing biodiversity across sites. By integrating acoustic richness estimates with habitat features derived from remote sensing, it provides insights into how environmental variation influences bird communities and highlights the potential of automated monitoring for large-scale ecological assessments.

Collectively, these chapters contribute both ecologically, by improving inference on avian communities and vocal behaviour in a managed forest landscape, and methodologically, by developing, evaluating, and implementing tools for working with automated acoustic detections. The analytical approaches developed throughout the thesis are consolidated in the *birdnetTools* R package, promoting transparent, reproducible, and accessible workflows for the broader bioacoustics research community.

Chapter 2 Setting BirdNET Confidence Thresholds: Species-specific vs. Universal Approaches

2-1 Abstract

BirdNET is widely used in avian acoustic research, providing species predictions alongside confidence values that represent the algorithm's certainty in species identification. Setting thresholds for these confidence values can increase precision (i.e., the percentage of true positives out of all the identified predictions) but may decrease the predictions retained and exclude true positives that fall below the threshold. This study evaluates two methods for setting confidence thresholds using a two-year audio dataset from western Canada, focusing on 19 target species: (1) a universal threshold of 0.7 across all species and (2) species-specific thresholds defined as the minimum confidence required to achieve a precision of at least 0.9. The universal threshold yielded precision ranging from 0.7 to 1.0 across species but retained only $17 \pm 14\%$ (SE) of BirdNET predictions. In contrast, species-specific thresholds ensured precision above 0.9 while retaining $70 \pm 37\%$ (SE) of predictions. Species-specific thresholds varied across species but were generally lower than 0.35. Confidence values associated with BirdNET predictions were found to be species-specific, but no clear link was observed between BirdNET's performance and species' song/call complexity, defined as song duration, bandwidth, and number of inflections. Our results confirm that species-specific thresholds offer higher precision and retain more predictions compared to a universal threshold. We provide a step-by-step workflow, including R code, to help researchers define species-specific thresholds that ensure reliable interpretation of BirdNET outputs. Additionally, we discuss how our workflow aligns with and complements previously proposed approaches for setting BirdNET thresholds.

2-2 Introduction

Algorithms for bird sound recognition have advanced significantly over recent decades, resulting from collaborations between ornithologists and computer scientists. Key developments include the application of convolutional neural networks for bird sound identification (e.g., Incze et al. 2018, Koh et al. 2019, Tseng et al. 2021), real-time recognition tools (e.g., Kojima et al. 2018, Küçüktopcu et al. 2019), and international competitions in bird sound identification (e.g., Kahl et al. 2023, Stowell et al. 2019). These advancements have led to the creation of commercial software like Wildlife Acoustics' Kaleidoscope Pro and open-source tools like the R package warbleR (Araya-Salas and Smith-Vidaurre 2017), which automate species identification for both researchers and the broader public. The Cornell Lab of Ornithology's free phone app, Merlin Bird ID, enables real-time species identification and has gained widespread adoption among bird enthusiasts globally. Among these methods, one of the most popular platforms for analyzing such data is BirdNET Analyzer, whose algorithm is available not only as a free phone app (<https://doi.org/10.1371/journal.pbio.3001670>) but also as open-source code, accessible through a publicly available repository on GitHub (github.com/kahst/BirdNET-Analyzer).

BirdNET Analyzer has emerged as a leading platform for scientific research due to its open-source nature, flexibility, and user-friendly interface. Developed by the K. Lisa Yang Center for Conservation Bioacoustics at Cornell Lab of Ornithology in collaboration with Chemnitz University of Technology, BirdNET was initially released in 2021 with the capacity to identify 984 North American and European bird species (Kahl et al. 2021). Its latest version, BirdNET Analyzer v2.4, extends its capabilities to over 6,000 species worldwide. The algorithm analyzes audio recordings in 3-second segments, assigning a 'confidence value' (ranging from 0.1 to 1) to each prediction, where higher values indicate greater certainty. While confidence values

reflect the likelihood of accurate predictions, they do not directly translate into probabilities of true predictions (Pérez-Granados 2023, Wood and Kahl 2024). To address this, the BirdNET team provided guidelines for using confidence values appropriately (Wood and Kahl 2024).

Confidence values can be influenced by various factors, including background noise, signal quality, overlapping vocalizations, and the acoustic characteristics of the species' signals (Symes et al. 2023). For some species, true predictions may occur even at lower confidence values, while for others, high-confidence predictions might still include false positives. Previous studies have noted that the identification performance of BirdNET varies across species (Cole et al. 2022). Such variation can be explored by finding the relationship between BirdNET performance and species vocalization characteristics, such as duration, bandwidth, repertoire size or mean frequency of songs (Pérez-Granados 2023). This exploration could enhance our understanding of how and why BirdNET performance varies among bird species.

To enhance the reliability of BirdNET predictions, users can set a confidence threshold, which establishes the minimum confidence value required for reporting predictions. While innovative methods have been proposed to bypass thresholds for call density estimation (see Navine et al. 2024), higher thresholds generally increase precision (i.e., the proportion of true positives among identified predictions) but may also exclude true positives with lower confidence values, thereby reducing recall. Retaining true positives (i.e., improving recall) is particularly important in contexts where predictions are used to analyze species' vocal activity patterns over time. Determining an optimal threshold requires balancing high precision with maximizing the number of predictions retained in the dataset. While Wood and Kahl (2024) provided a theoretical framework for setting species-specific thresholds based on probabilities, they did not assess how these species-specific thresholds influence the precision of retained data.

Such evaluations are critical for improving the utility and interpretation of BirdNET outputs (Pérez-Granados 2023, Ware et al. 2023).

In this study, we analyzed a two-year audio dataset from 66 recording sites in northcentral British Columbia, Canada, focusing on 19 target species selected for their diversity in song/call complexity and singing rates. Our objectives were to: (1) compare the effects of a universal BirdNET confidence threshold with species-specific thresholds designed to achieve high precision while retaining a substantial number of predictions, and (2) evaluate whether BirdNET's performance correlates with species-specific song/call complexity. This study provides a detailed workflow, including step-by-step R code, for defining species-specific thresholds and discusses how our method builds on previously proposed approaches for setting BirdNET thresholds. Ultimately, it aims to enhance BirdNET's utility for future research applications.

2-3 Methods

2-3-1 Study area

The data were collected at the John Prince Research Forest (54° 27'N, 124° 10'W, 700 m a.s.l), located in the dry sub-boreal spruce biogeoclimatic zone in British Columbia, Canada (Biogeoclimatic Zones of British Columbia 2024). The audio recordings were collected in 66 sites, which were selected to include forest stands of various age classes, tree-species compositions and timber harvesting activities in the region. Adjacent recording sites were at least 2 km apart to ensure independent sampling, accounting for the audio detection radius of the ARUs (**Figure 2-1**).

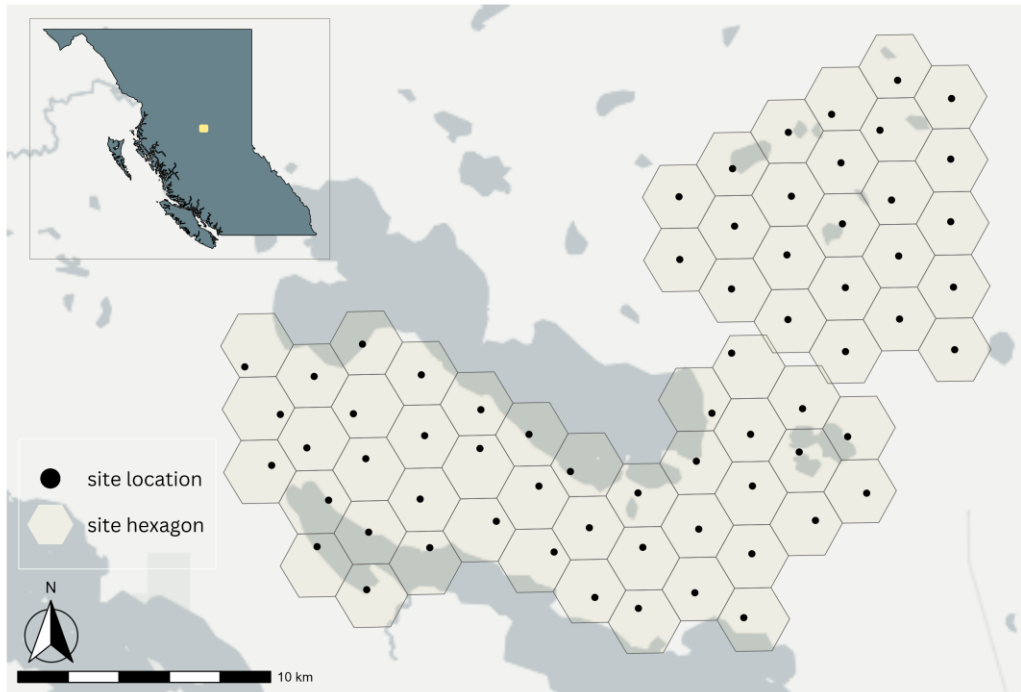


Figure 2-1 Locations of the 66 recording sites. All sites are located within or adjacent to John Prince Research Forest, north of Ft. St. James, BC, Canada, with 42 sites within the research forest (lower contiguous cluster) and 24 sites located northeast of the research forest (cluster to upper right of figure). Dots represent the locations of recorders, and hexagons were superimposed to ensure spatial independence. Greyed areas on map are lakes and water bodies. All ARUs were initially intended to be positioned at the grid centroids; however, certain site locations deviated slightly due to challenges in reaching the grid centroid.

Each site was equipped with one AudioMoth (Hill et al. 2019), and the data were collected during the breeding seasons (May-July) across two years (2020 - 2021). The recording window was set from 4 am to 7 am daily, aligning with the peak vocal activity of most songbirds. Each AudioMoth was programmed to record actively for 1 minute, followed by a 4-minute period of no recording. This recording cycle was repeated throughout the three-hour recording window, resulting in a total of 36 1-minute recordings collected per day, per site. All recordings were digitized at a sampling rate of 48 kHz and stored as 16-bit sound files for subsequent analysis.

2-3-2 Audio data processing

All audio data were processed through BirdNET Lite, a TFLite version of BirdNET. BirdNET enables researchers to specify parameters, enhancing the algorithm's performance. If not explicitly specified, we used the default values for these settings (refer to **Table 2-1**). The BirdNET processing was conducted using the High-Performance Computing lab at the University of Northern British Columbia. Specifically, we used the Intel Xeon Silver 4114 cluster, comprising 16 compute nodes, each equipped with 36 Intel Xeon Gold 6140 core processors. BirdNET's outputs were stored in .csv files.

Table 2-1 Parameters used in this study for the BirdNET analyzer.

Arguments	Definition	Values set in this study
i	Path to input file or directory.	--
o	Path to output directory. If not specified, the input directory will be used.	--
filetype	Filetype of soundscape recordings. Defaults to 'wav'.	'wav'
results	Output format of analysis results. Values in ['audacity', 'raven']. Defaults to 'raven'.	'raven'
lat	Recording location latitude. Set -1 to ignore.	Varies based on site
lon	Recording location longitude. Set -1 to ignore.	Varies based on site
week	Week of the year when the recordings were made. Values in [1, 48]. Set -1 to ignore.	Varies based on recording date
overlap	Overlap in seconds between extracted spectrograms. Values in [0.0, 2.9]. Defaults to 0.	0
spp	Combines probabilities of multiple spectrograms to one prediction. Defaults to 1.	1
sensitivity	Sigmoid sensitivity; Higher values result in lower sensitivity. Values in [0.25, 2.0]. Defaults to 1.0.	1.0
min_conf	Minimum confidence threshold. Values in [0.01, 0.99]. Defaults to 0.1.	0.1

2-3-3 Target species and song/call complexity

We selected 19 species present in our study area (**Table 2-2**), as outlined by one previous study in the same region (Wheelhouse et al. 2022). Among the 19 species, there were 6 from the warbler family (Parulidae), 3 from the thrush family (Turdidae), 4 from the flycatcher family (Tyrannidae), and 6 selected from a variety of families. The selection aimed to encompass a diverse spectrum of song/call complexity and various singing rates.

We quantified the song/call complexity for each target species using signal duration (s), bandwidth (Hz), and number of inflections (defined as the average number of frequency inflections in a song/call signal). The value for each metric was calculated from 50 9-sec recordings for each target species. First, we assessed the song/call signals within a recording using the *energy_detector()* function in the ohun package (Araya-Salas et al. 2023), which detected the timing and the duration of the signal (**Figure 2-2**). Then we used the package warbleR (Araya-Salas and Smith-Vidaurre 2017) to calculate the bandwidth and the number of inflections for each song/call signal. Specifically, the bandwidth was calculated using the *freq_range()* function, and the number of inflections was calculated using the *freq_ts()* and the *inflections()* functions. We selected multiple metrics to quantify complexity because single metrics might not be sufficient when comparing complexity between species (Benedict and Najar 2019).

Table 2-2 Target species in this study.

Family	Common Name	Scientific Name
New World Warblers (Parulidae)	Northern Waterthrush	<i>Parkesia noveboracensis</i>
	Orange-crowned Warbler	<i>Leiothlypis celata</i>
	Tennessee Warbler	<i>Leiothlypis peregrina</i>
	MacGillivray's Warbler	<i>Geothlypis tolmiei</i>
	Wilson's Warbler	<i>Cardellina pusilla</i>
	Myrtle Warbler	<i>Setophaga coronata</i>
Thrush (Turdidae)	American Robin	<i>Turdus migratorius</i>
	Swainson's Thrush	<i>Catharus ustulatus</i>
	Varied Thrush	<i>Ixoreus naevius</i>
Tyrant Flycatcher (Tyrannidae)	Hammond's Flycatcher	<i>Empidonax hammondi</i>
	Olive-sided Flycatcher	<i>Contopus cooperi</i>
	Western Wood-Pewee	<i>Contopus sordidulus</i>
	Western Flycatcher	<i>Empidonax difficilis</i>
Woodpeckers (Picidae)	Northern Flicker	<i>Colaptes auratus</i>
Wrens (Troglodytidae)	Pacific Wren	<i>Troglodytes pacificus</i>
Kinglets (Regulidae)	Ruby-crowned Kinglet	<i>Regulus calendula</i>
Nuthatches (Sittidae)	Red-breasted Nuthatch	<i>Sitta canadensis</i>
Treecreepers (Certhiidae)	Brown Creeper	<i>Certhia americana</i>
New World Sparrows (Passerellidae)	White-throated Sparrow	<i>Zonotrichia albicollis</i>

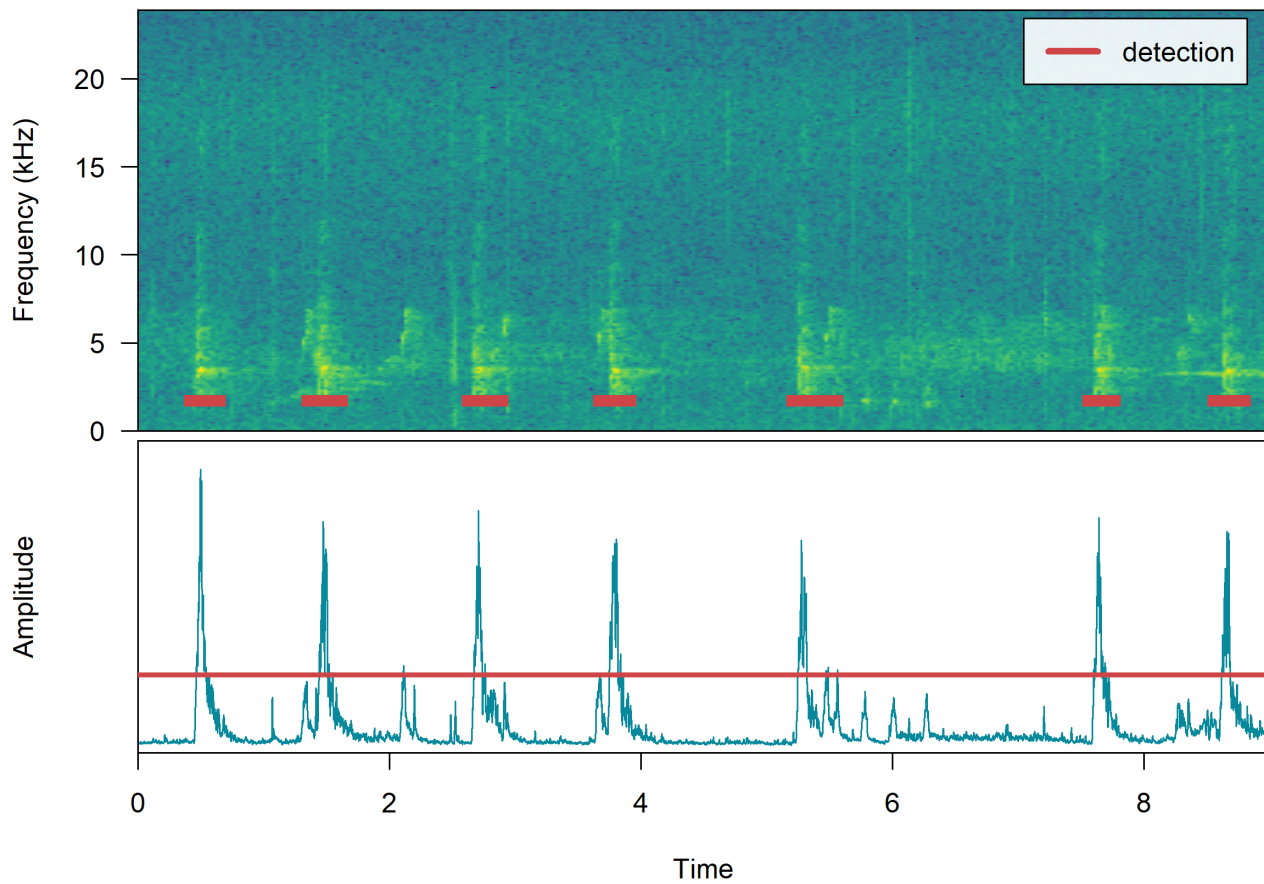


Figure 2-2 Spectrogram illustrating the detection of song/call signals from a sound recording using the *energy_detector()* function in the Ohun package. The example recording is from an American Robin. For each signal, the duration (s), bandwidth (Hz), and number of inflections were calculated and averaged throughout the recording. A total of 50 9-sec recordings were selected and analyzed for each target species to quantify song/call complexity.

2-3-4 Validation dataset

For each of the 19 target species used in the full analysis, we employed a stratified random sampling approach, selecting 10 recording segments from each BirdNET confidence value class with a 0.05 incremental grouping. In other words, for each species, we randomly selected 10 3-sec recording segments which BirdNET had identified as containing the target species with confidence values ranging from 0.1 to 0.15; another 10 recording segments with confidence values ranging from 0.15 to 0.20, and so forth through 0.95-1 confidence values. This process yielded a validation dataset consisting of 180 recording segments for each target species. The application of stratified sampling ensured a balanced validation dataset, mitigating potential bias introduced by the varying number of predictions assessed within each confidence class.

The decision to include 180 predictions per target species was informed by a pilot study on a subset of species, which examined how the number of 3-second recording segments influences BirdNET performance evaluation. In this pilot study, we systematically chose 50, 25, 10, and 5 recording segments from each confidence class for three species: Varied Thrush, Olive-sided Flycatcher, and Myrtle Warbler. While a larger number of segments in the validation dataset intuitively enhances result accuracy, practical constraints, particularly the manual effort required for validating multiple species, must also be considered. Based on this analysis (presented at the beginning of the **Results** section), we selected 10 recording segments per confidence value class (i.e., 180 recording segments for each target species) as achieving a balance between validation effort and quality.

2-3-5 Calibration curves – linking confidence value to true positive rate

All 180 sampled recording segments, were verified through both audio and visual inspection of spectrograms by the first author. Each audio segment was manually reviewed to confirm the presence or absence of the target species as reported by BirdNET. Predictions confirmed as present were classified as true positives (1), while those confirmed as absent were classified as false positives (0). We then fitted a generalized linear model with a binomial distribution to generate a calibration curve, using BirdNET confidence values as the predictor variable and true/false positive classifications as the response variable. Logistic regression, as a method to model the relationship between BirdNET confidence values and the true positive rate, was also proposed by Wood and Kahl (2024). The true positive rate was calculated by back-transforming the model predictions.

2-3-6 Confidence threshold setting

The steps below outline the calculation for precision (i.e., the percentage of true positives out of all the identified predictions) and the number of remaining predictions under varying thresholds for single species. This iterative procedure was applied to all 19 target species within our study:

1. No. of true positives: for each possible confidence value, we multiplied the true positive rate (predicted from the calibration curve) by the number of predictions with that specific confidence value:

$$N_{TP_i} = TPR_i * N_i$$

2. No. of false positives: for each possible confidence value, multiply (1 – true positive rate) by the number of predictions with that specific confidence value:

$$N_{FP_i} = (1 - TPR_i) * N_i$$

3. Number of retained predictions: the sum of true positives and false positives whose confidence value is higher than the confidence threshold:

$$\text{No. of retained detections} = \sum_{i=T}^1 N_{TP_i} + \sum_{i=T}^1 N_{FP_i}$$

4. Precision: the sum of all true positives divided by the number of retained predictions:

$$\text{precision (i.e., proportion of true positives)} = \frac{\sum_{i=T}^1 N_{TP_i}}{\sum_{i=T}^1 N_{TP_i} + \sum_{i=T}^1 N_{FP_i}}$$

where:

i represents all possible confidence values, ranging from 0.1 to 1.

N_{TP_i} is the number of true positives with confidence i .

N_{FP_i} is the number of false positives with confidence i .

TPR_i is the true positive rate with confidence i , derived from calibration curve.

N_i is the total number of predictions with confidence i .

T is the confidence threshold, with potential values ranging from 0.1 to 1.

Based on the calculations, we examined two different scenarios for selecting confidence thresholds and how these scenarios influenced precision and the number of retained predictions:

- 1) universal threshold: a general high threshold of 0.7, as summarized in the Table 2 in Pérez-Granados 2023, to ensure that retained predictions were most likely true positives; and 2)
- species-specific threshold: defined as the minimum confidence threshold the user can apply to achieve a precision of at least 0.9. This method allows us to specify that at least 90% of the predictions retained in the dataset are true positives. This target precision (0.9) is set for demonstration purposes in this study, future users can define their own desired precision as the criteria.

2-4 Results

2-4-1 Linking confidence value to true positive rate

There was a positive relationship between BirdNET confidence and the true positive rate regardless of species, or whether 5 to 50 recording segments were selected (**Figure 2-3**).

Calibration curves derived from datasets comprising 50, 25, and 10 recording segments per confidence class exhibited comparable trends. However, the curve generated from 5 recording segments per confidence class diverged to a greater extent from the remaining calibration curves (see the gray curve in **Figure 2-3C**); therefore, to balance the effort invested in manual validation, we selected 10 segments per confidence class (180 segments per target species) in the validation dataset in this study.

We established calibration curves for our target species, employing a standardized sample size of 10 recording segments per confidence class (180 segments per target species; **Figure 2-4**). Species such as the Myrtle Warbler (green), Red-breasted Nuthatch (yellow), and Pacific Wren (orange) attained high true positive rates even at low BirdNET confidence, suggesting that predictions are likely to be accurate despite low confidence values. Conversely, true positive rates exhibited gradual increase in relation to confidence values for species like Varied Thrush (purple) and Brown Creeper (blue), indicating that only predictions with high confidence were likely to be accurate. Species such as Olive-sided Flycatcher (pink) and Northern Flicker (dark green) fell between these two groups.

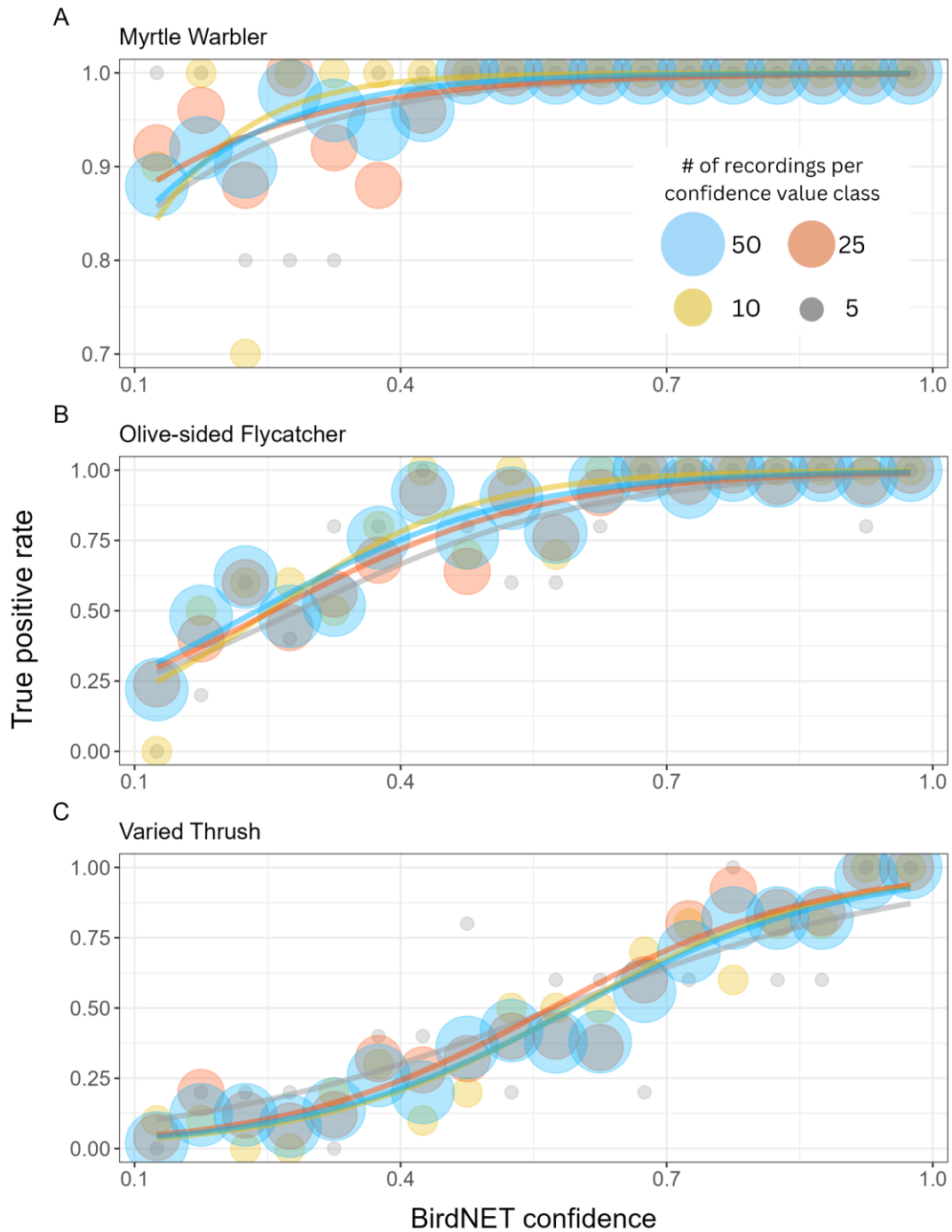


Figure 2-3 Calibration curves to estimate appropriate number of recordings in the validation dataset for (A) Myrtle Warbler, (B) Olive-sided Flycatcher, and (C) Varied Thrush. Curves were generated using 5 (grey), 10 (yellow), 25 (orange), and 50 (blue) recording segments per BirdNET confidence value class.

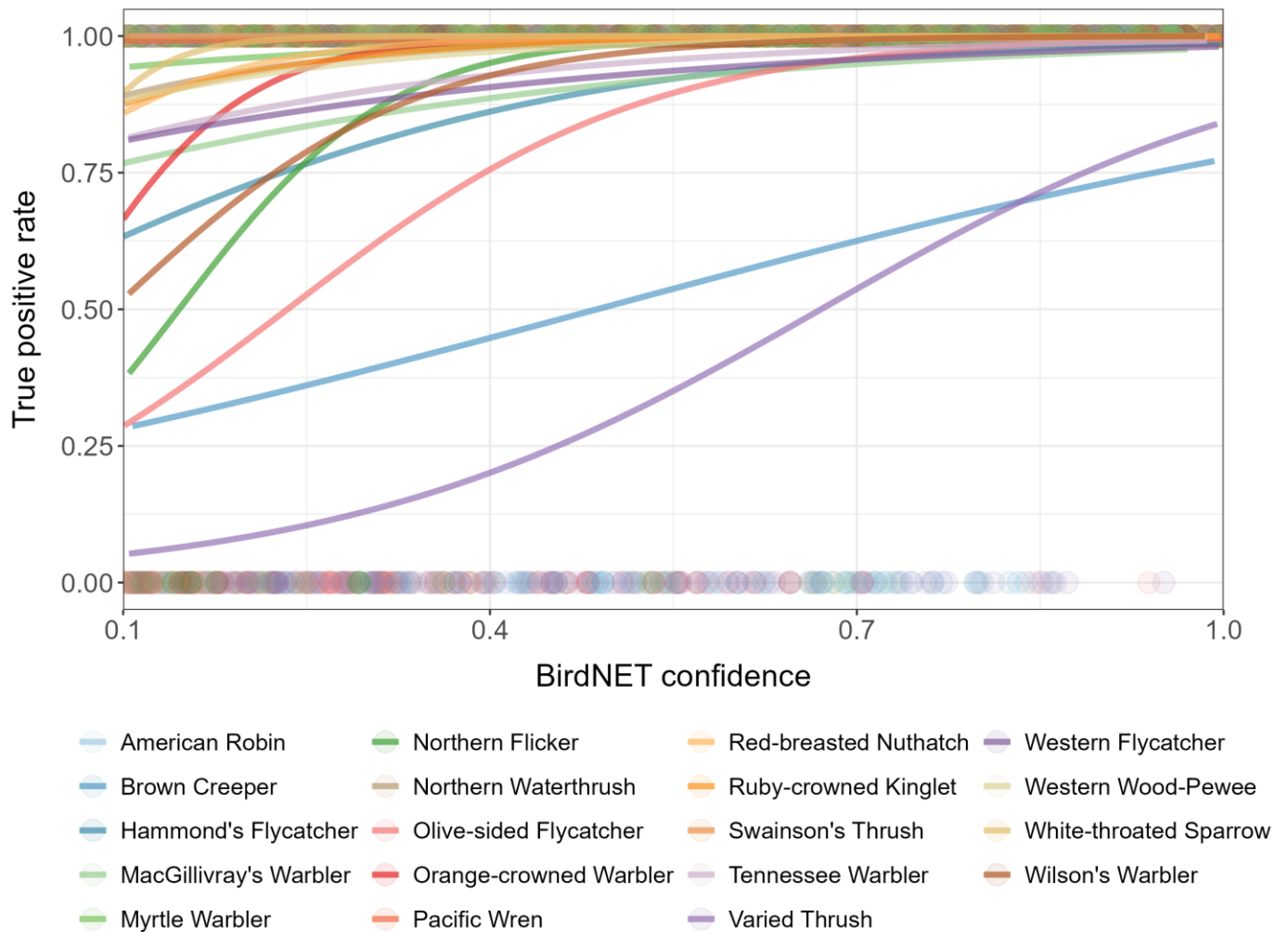


Figure 2-4 Calibration curves for the 19 target species, comparing true positive rate versus BirdNET confidence. The curve was based on logistic regression with true positive (1) or false positives (0) as response variables, and BirdNET confidence values as the predictor variable. The true positive rate was calculated via back-transformation of the model predictions.

2-4-2 Impact of confidence threshold setting

With a universal confidence threshold of 0.7, the precision was consistently high, ranging from 0.7 to 1.0 (**Table 2-3**). However, applying this universal threshold resulted in lower number of BirdNET predictions being retained (blue arrows in **Figure 2-5**). Specifically, only $17 \pm 14\%$ (SE) of BirdNET predictions were retained in the dataset when applying a universal threshold of 0.7.

Conversely, the species-specific threshold achieved precisions higher than 0.9 (**Table 2-3**), while also retain higher number of predictions in the dataset (red arrows in **Figure 2-5**). Overall, $70 \pm 37\%$ (SE) of BirdNET predictions were retained in the dataset when applying species-specific thresholds. These species-specific thresholds were lower than 0.35 for most of our target species, with only two of the 19 target species, Brown Creeper (0.95) and Varied Thrush (0.95), requiring higher thresholds to reach similar level of precision (**Table 2-3**).

Table 2-3 Statistics for each target species, including the total number of detections, thresholds, corresponding precision, and the proportion of detections retained after applying the thresholds. Precision varied across species when applying a universal BirdNET threshold of 0.7, with an average of 17% of detections retained from the unfiltered dataset. In contrast, species-specific thresholds, defined as the minimum threshold to achieve a precision greater than 0.9, retained an average of 70% of detections. This table serves as an example, and researchers can adjust the precision criteria for species-specific thresholds based on their specific objectives.

Common name	Scientific name	No. of detections	Universal			Species-specific		
			Threshold	Precision	Proportion retained (%)	Threshold	Precision	Proportion retained (%)
American Robin	<i>Turdus migratorius</i>	21,284	0.7	1.00	6	0.10	1.00	100
Swainson's Thrush	<i>Catharus ustulatus</i>	376,293	0.7	1.00	32	0.10	1.00	100
Pacific Wren	<i>Troglodytes pacificus</i>	45,232	0.7	1.00	39	0.10	0.99	100
White-throated Sparrow	<i>Zonotrichia albicollis</i>	39,869	0.7	1.00	7	0.10	0.98	100
Northern Waterthrush	<i>Parkesia noveboracensis</i>	89,340	0.7	1.00	31	0.10	0.97	100
Red-breasted Nuthatch	<i>Sitta canadensis</i>	9,007	0.7	1.00	35	0.10	0.97	100
Myrtle Warbler	<i>Setophaga coronata</i>	30,418	0.7	1.00	2	0.10	0.96	100
Ruby-crowned Kinglet	<i>Regulus calendula</i>	13,005	0.7	1.00	13	0.10	0.94	100
Western Wood-Pewee	<i>Contopus sordidulus</i>	4,414	0.7	1.00	13	0.10	0.94	100
Tennessee Warbler	<i>Leiothlypis peregrina</i>	27,162	0.7	0.99	45	0.10	0.94	100
Orange-crowned Warbler	<i>Leiothlypis celata</i>	12,636	0.7	1.00	5	0.15	0.93	61
Northern Flicker	<i>Colaptes auratus</i>	853	0.7	1.00	17	0.18	0.90	48
Hammond's Flycatcher	<i>Empidonax hammondi</i>	92,695	0.7	0.98	31	0.19	0.90	75

Western Flycatcher	<i>Empidonax difficilis</i>	11,698	0.7	0.97	6	0.19	0.90	39
Wilson's Warbler	<i>Cardellina pusilla</i>	3,484	0.7	1.00	4	0.22	0.90	33
MacGillivray's Warbler	<i>Geothlypis tolmiei</i>	24,083	0.7	0.96	7	0.27	0.90	31
Olive-sided Flycatcher	<i>Contopus cooperi</i>	11,388	0.7	0.99	18	0.34	0.90	34
Varied Thrush	<i>Ixoreus naevius</i>	24,525	0.7	0.71	7	0.95*	0.82*	1*
Brown Creeper	<i>Certhia americana</i>	12,403	0.7	0.70	5	0.95*	0.76*	1*

* Varied Thrush and Brown Creeper did not achieve a precision of 0.9, even at highest thresholds. The reported values are based on thresholds set to 0.95.

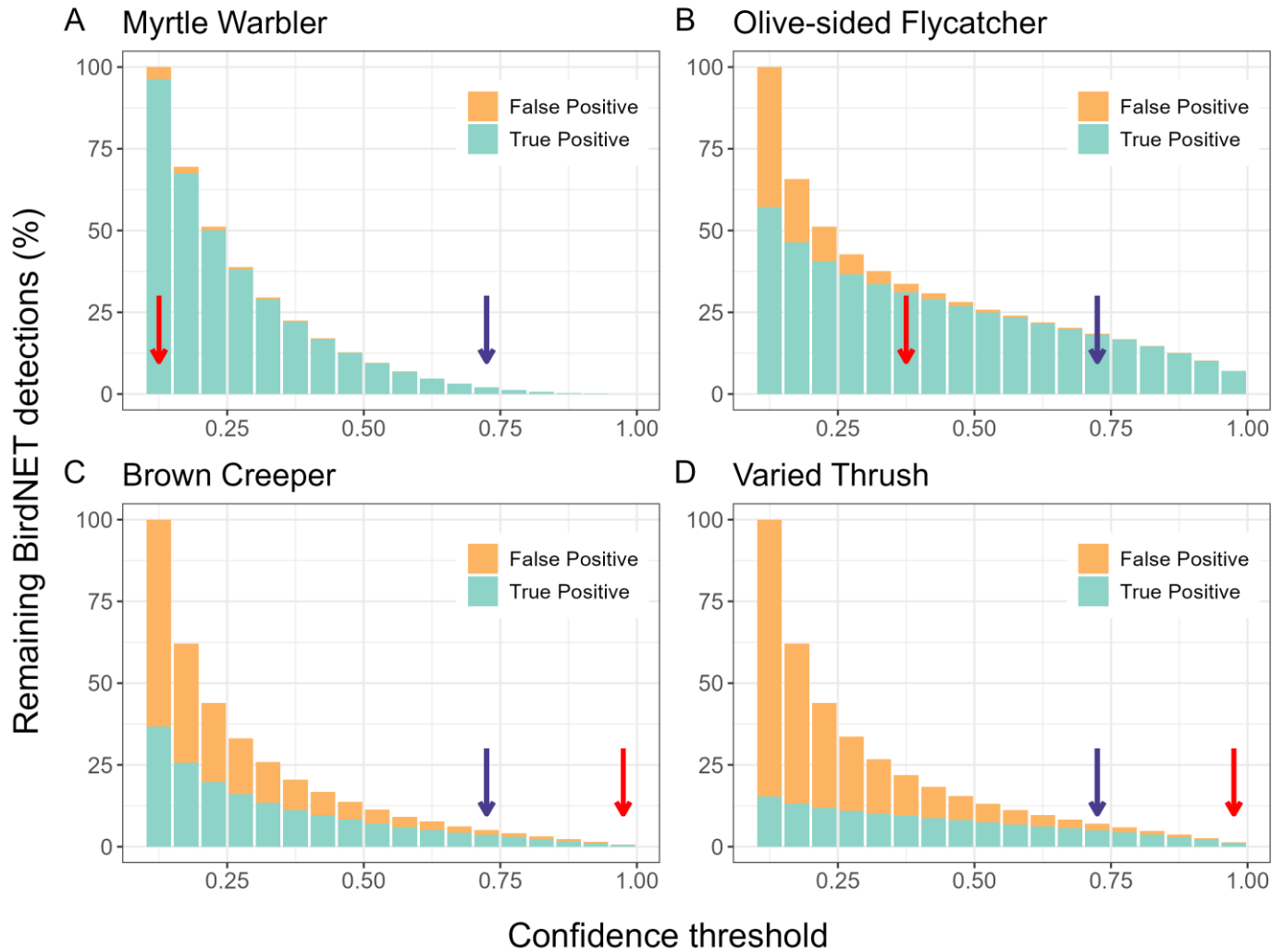


Figure 2-5 The number of detections identified by BirdNET, with true positives (green) or false positives (orange), after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment. We show results for (A) Myrtle Warbler, (B) Olive-sided Flycatcher, (C) Brown Creeper, and (D) Varied Thrush. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9. Note that Brown Creeper and Varied Thrush did not achieve 0.9 precision, even with the highest threshold. For these species, the species-specific threshold was set to 0.95.

2-4-3 Song/call complexity vs BirdNET performance

We did not find a clear link between BirdNET's performance (defined as the threshold needed to achieve at least 0.9 precision, as shown in **Table 2-3**, species-specific threshold column) and song/call complexity (defined as signal duration, bandwidth, and the number of inflections) among the species we tested (**Figure 2-6**). Examples of species for which BirdNET performs poorly (i.e., needing a higher threshold to achieve 0.9 precision) included Varied Thrush and Brown Creeper. Varied Thrush had the narrowest bandwidth among all species measured but has intermediate signal duration and number of inflections (yellow highlight in **Figure 2-6**). Similarly, the Brown Creeper exhibits intermediate values for signal duration, bandwidth, and number of inflections (yellow highlight in **Figure 2-6**). In comparison, the White-throated Sparrow had lower complexity values for all three metrics but required low BirdNET confidence threshold to reach high precision (green highlight in **Figure 2-6**).

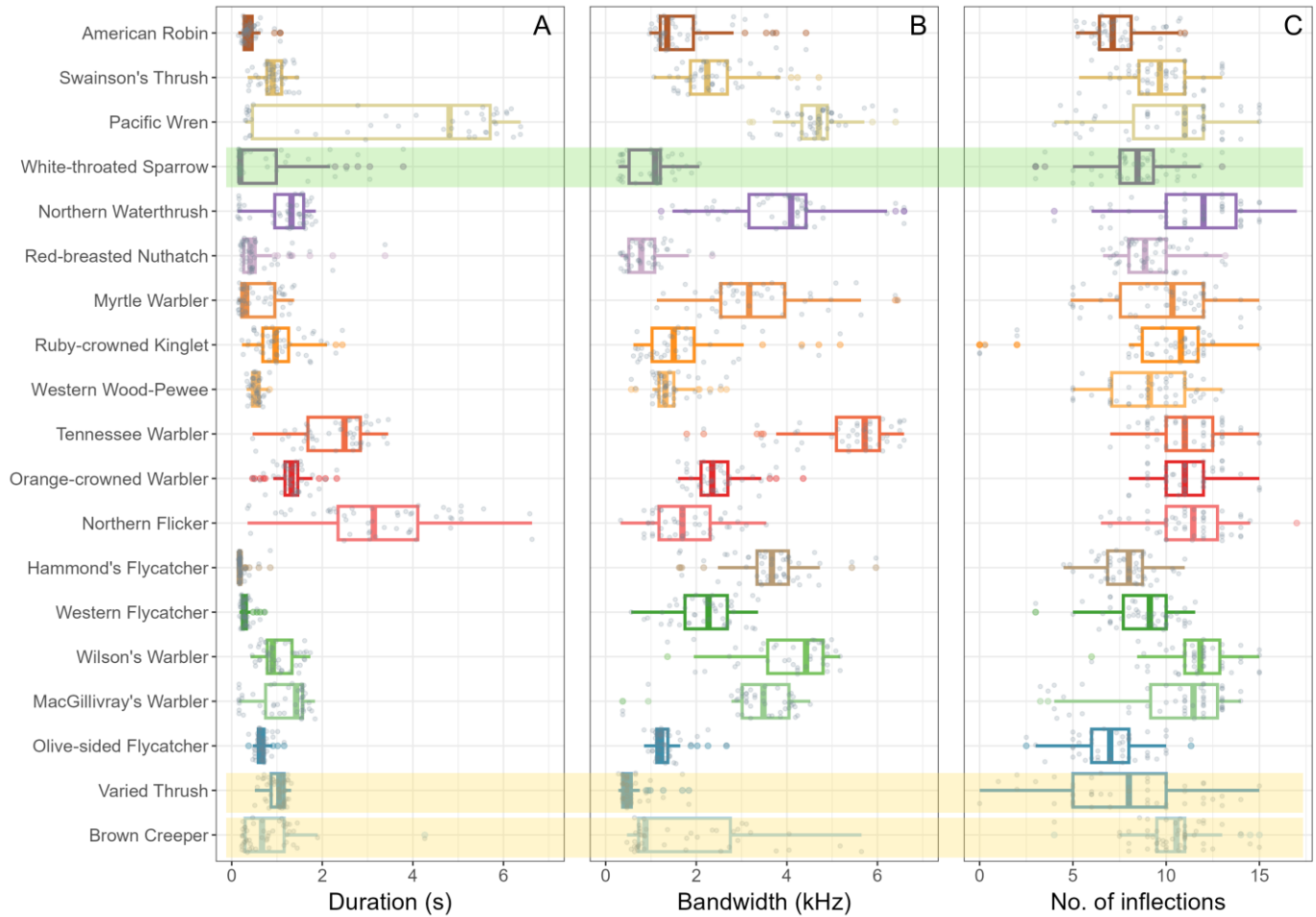


Figure 2-6 Vocal complexity of the 19 target species ranked by the BirdNET's performance (defined as the threshold needed to achieve 0.9 precision, as shown in **Table 2-3**). Vocal complexity is defined by (A) duration of the song/call in seconds, (B) bandwidth of the song/call in kHz, and (C) the number of inflections of the signals. Boxes show quartiles and median of each metrics. Metrics were measured from 50 recordings for each species. BirdNET performed poorly for species highlighted in yellow and performed well for species highlighted in green. The duration of the signals was defined by the *evergy_detector()* function in the *ohun* package. The bandwidth and the number of inflections were extracted by the *freq_range()* and the *inflections()* functions in the *warbleR* package, respectively.

2-5 Discussion

2-5-1 Overview of BirdNET performance and thresholds

Our results suggested setting species-specific confidence thresholds while using BirdNET may be a better option to balance between precision and the number of predictions retained. For species that BirdNET had high success at identifying, such as the Myrtle Warbler (**Figure 2-5A**), users could set a low confidence threshold (e.g., 0.1). Conversely, for species like the Olive-sided Flycatcher (**Figure 2-5B**), a moderate confidence threshold (e.g., 0.34) would be needed to achieve a balance between retaining predictions in the dataset while maintaining a high true-positive rate. In contrast, species like the Brown Creeper (**Figure 2-5C**) and Varied Thrush (**Figure 2-5D**) would require setting a higher confidence threshold (e.g., 0.95) to avoid inclusion of false positives. Although this threshold retains only about 1% of the original predictions for these latter species, it also ensures a higher certainty of true positive predictions in the final dataset.

We did not find a correlation between song/call complexity (defined by duration, bandwidth, and number of inflections) and BirdNET performance (defined as the threshold needed to achieve at least 0.9 precision) among the 19 target species. This suggests that BirdNET may not necessarily perform better at detecting species with higher vocal complexity. However, this study considered only a limited set of song and call metrics, focusing on the 19 target species in our analysis. Many other species present in recordings from our study site remain unexamined. Future research could explore additional complexity metrics, such as repertoire: the number of total song types (e.g., Weiss et al. 2014), delivery: the number of songs recorded over total recording time (e.g., Opaev 2016), switching rate: the proportion of songs which represent a type

switch from the preceding song (e.g., Hill et al. 2018), and variety: short-term song diversity (e.g., dos Santos et al. 2018). Given the low correlation between these metrics (Benedict and Najar 2019), multivariate approaches could help identify the primary factors influencing BirdNET performance. Expanding analyses to include a broader range of species and examining the influence of specific song and call types within species could yield deeper insights into BirdNET's performance.

We did find that misclassifications appeared to be due to overlapping acoustic features between species. During validation, we found that Swainson's Thrush calls were frequently misidentified as Olive-sided Flycatcher. The call of Olive-sided Flycatcher has a similar tonal quality and frequency range as notes in the Swainson's Thrush call. We also found that most Varied Thrush predictions with low BirdNET confidence values were actually White-throated Sparrow songs. This may be due to the similar monotone whistles and overlapping frequency ranges shared by the two species' songs. Interestingly, White-throated Sparrow predictions were mostly accurate and rarely misidentified as Varied Thrush. This might be because BirdNET had a broader training dataset for White-throated Sparrow. For future research, we recommended evaluating how the number, and the variety, of the training dataset used to build the learning algorithm could influence the BirdNET performance for each species.

2-5-2 Applications of two types of thresholds

In practice, while species-specific thresholds have been strongly recommended (Fairbairn et al. 2024, Wood and Kahl 2024, and this study), a universal threshold may still have utility for exploratory analyses, such as preliminary assessments of species richness in large-scale biodiversity surveys (Ware et al. 2023). However, using a universal threshold can result in both

false positives (species predicted by BirdNET but not actually present) and false negatives (species present at the site but not predicted by BirdNET). As a result, it is difficult to recommend a "precise" universal threshold that would reliably reflect species richness. Manual validation of predictions still remains a crucial and necessary step even when employing universal thresholds in species richness studies.

For more detailed investigations, such as analyzing species vocal activity patterns or tracking rare species, species-specific thresholds are essential. These thresholds help maximize true-positive predictions, capturing a more complete vocal census (**Table 2-3**). Using species-specific thresholds prevents biased conclusions that could arise from confidence values (within species) that vary across either locations or time periods, providing a more balanced and realistic representation of vocal activity.

2-5-3 Workflow to define species-specific threshold

The species-specific thresholds calculated in this study (**Table 2-3**) are not transferable to different regions, even for the same species, as they may be influenced by factors such as recording equipment, ambient noise, environmental conditions, and the local species composition. To support future studies, we summarized our proposed steps for defining species-specific BirdNET confidence thresholds (**Figure 2-7**) and provided the corresponding R code and example datasets as supplementary materials. The R code also includes a developed R Shiny App for BirdNET validation, which integrates both audio and spectrographic options for assessing predictions.

While some aspects of our proposed methods align with the approaches outlined by Wood and Kahl (2024) in their “converting scores to probabilities” section, there are several

important distinctions. First, during validation, we recommend stratified sampling across confidence values to ensure representation of both low- and high-confidence predictions, rather than the simple random sampling. The stratified sampling approach is particularly advantageous when confidence value distributions are non-uniform, a common scenario in BirdNET analyses.

The number of validation segments required may vary depending on the spatiotemporal scale and the diversity of vocalizations in the study. In our study, we validated 180 segments per species; however, for projects with greater vocalization variability, we recommend testing how segment numbers influence the calibration curve, as demonstrated in the **Results** section “Linking confidence value to true positive rate”. For large-scale or highly diverse datasets, rather than selecting more segments and determining thresholds per species, users may consider defining thresholds at finer scales, such as by species and site, to better capture variability.

Second, while both methods involve deriving thresholds from logistic regression models, our approach emphasizes precision, assessing the proportion of retained predictions that are true positives. In contrast, Wood and Kahl’s (2024) method focuses on the probability of individual predictions being true positives. These differences lead to different threshold values: our method identifies the confidence threshold at which retained predictions achieve a desired precision, whereas theirs determines the confidence value where the predicted probability equals a given value. We encourage users to evaluate the objectives and select the appropriate method for defining thresholds to enhance the utility of BirdNET.

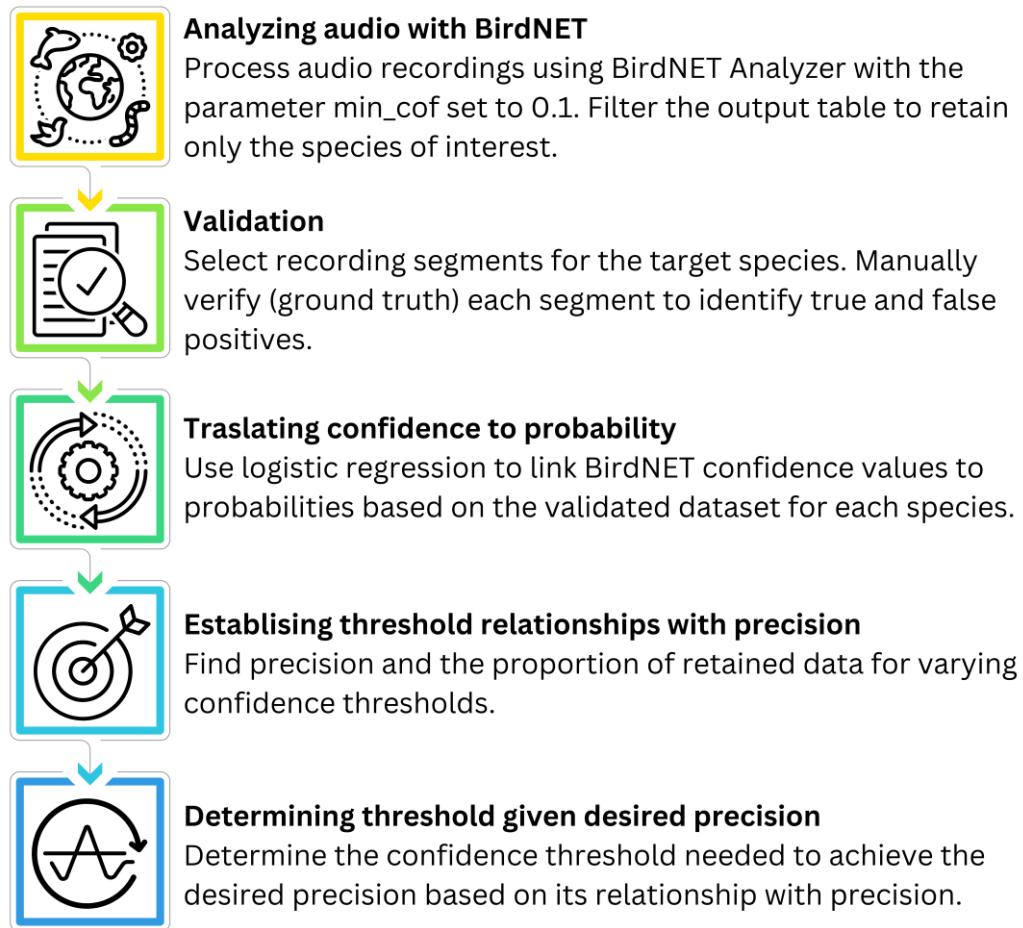


Figure 2-7 Workflow for defining species-specific thresholds. For detailed method descriptions and implementation steps, refer to the R code provided in the supplemental material.

Chapter 3 Using Autonomous Recording Units for Vocal Individuality: Insights from Barred Owl Identification

3-1 Abstract

Recent advances in acoustic recording equipment enable autonomous monitoring with extended spatial and temporal scales, which may allow the censusing of species with individually distinct vocalizations, like owls. We assessed the potential of identifying individual Barred Owls (*Strix varia*) through detections of their vocalizations using passive acoustic monitoring. We placed autonomous recording units throughout the John Prince Research Forest (54° 27'N, 124° 10'W, 700 m a.s.l) and surrounding area, in northern British Columbia, Canada, from February to April 2021. The study area is 357 km² with a minimum of 2 km between the 66 recording stations. During this period, we collected 454 Barred Owl calls, specifically *Two-Phrase Hoot*, from 10 recording stations that were of sufficient quality for spectrographic analysis. From each call we measured 30 features: 12 temporal and 18 frequency features. Using forward stepwise discriminant function analysis, the model correctly categorized 83.2% of the calls to their true recording location based on a 5-fold cross validation. The model showed substantial agreement between which recording station the call was classified to originate from, and where the call was actually recorded. The most important features of the calls that enabled discrimination include call length, interval between the 4th and the 5th notes, interval between the 6th and 7th notes, and the duration of the 8th note. Our results suggest that passive acoustic monitoring can not only be used to detect presence/absence of species, but where individually distinct feature of vocalizations exist, such as in the Barred Owl could also be used in population censusing.

3-2 Introduction

The ability to identify individual animals has many conservation applications (McGregor and Peake 1998), including: tracking individual movements and habitat use; estimating population sizes; studying site/mate fidelity; and analyzing behavior and social interactions. Traditionally, individual identification involves marking animals with physical tags, such as GPS units, PIT tags, leg bands, ear tags, and transmitters. While these methods are well-established for monitoring individuals, they require capturing animals, leading to time-consuming and somewhat invasive data collection methods. Alternatively, some species can be individuated by observing distinctive features on animals, such as scars (Langtimm et al. 2004), fur spot patterns (Jackson et al. 2006), tail marks (Swanepoel 1996), stripe patterns (Karanth 1995), and body colouration patterns (Monteiro et al. 2014). As these features are natural-occurring variation, they can be used for tracking individuals without the requirement of physical capture. The use of natural variation often can be applied to a larger dataset than when individual tracking requires physical capture.

One promising application of tracking using naturally-occurring individual variation is acoustic monitoring (Gilbert et al. 2002; Hartwig 2005). Vocal individuality has been observed in many animal taxa, such as wolves (Sadhukhan et al. 2021), lions (Trapanotto et al. 2022), dolphins (Janik 2009), and owls (Ural Owls *Strix uralensis*; Zhou et al. 2020, European Eagle Owls *Bubo bubo*; Grava et al. 2008, African Wood Owls *Strix woodfordii*; Delport et al. 2002, Tawny Owls *Strix aluco*; Galeotti and Pavan 1991, Barred Owls *Strix varia*; Freeman 2000, Western Screech-Owls *Megascops kennicottii*; Tripp and Otter 2006, Ryukyu Scops Owls *Otus elegans*; Takagi 2020, Eastern Screech-Owls *Megascops asio*; Nagy and Rockwell 2012, Northern Saw-whet Owls *Aegolius acadicus*; Otter 1996, Holschuh and Otter 2005, Great

Horned Owls *Bubo virginianus*; Odom et al. 2013, Pygmy Owls *Glaucidium passerinum*; Galeotti and Pavan 1991, Sunda Scops-Owl *Otus lempiji*; Yee et al. 2016, and Great Gray Owls *Strix nebulosa*; Rognan et al. 2009). Decades of research have demonstrated the effectiveness of acoustic monitoring as a tool for identifying individual owls, given the stable and distinctive features present in their vocalizations. Almost all these studies, however, involved researchers tracking and manually recording single birds. Recent advances in the use of autonomous recording units (ARUs) for surveying bird communities may offer the potential to census populations using individually distinct vocal markers.

The application of ARUs in avian monitoring is currently experiencing significant growth, given the advantages they offer: the ability to subject audio-based data to quantitative analysis and reduced environmental impact. ARU surveys only require labour for deployment, retrieval and analysis, thus enabling lower labor costs compared to traditional point count surveys for coverage of the same number of surveying periods. Furthermore, the development of analytical tools, such as algorithms for automatic species identification based on sounds, has expanded the potential applications of ARUs in large-scale acoustic monitoring. One of the most advanced algorithms is BirdNET, a deep neural network-based bird sound classifier capable of identifying North American and European bird species by their sounds (Kahl et al. 2021). The development of ARU combined with BirdNET may also provide a potentially efficient method for acoustically monitoring individual owls.

Despite being acknowledged and proposed to offer an effective passive alternative, relative to capture/tagging, for censusing owls (Dale et al. 2022), applying ARUs to owl individual identification could present several challenges. The signals recorded by ARUs are often of poor quality compared to manual recordings, as the static location of audio recorders

means many recordings are of distant birds. Owl calls can propagate over long distances but suffer from signal degradation and interference from background noise the farther they are recorded from the source of the sound (Baptista and Gaunt 1994). This might reduce the feasibility of using ARU recordings for individual owl identification. To enable individual identification, signals must exhibit clear "features" that can be accurately measured to distinguish subtle differences between birds. Temporal features from owl calls have been found to be effective in individual identification, including internote interval (Odom et al. 2013, Zhou et al. 2020), call length (Odom et al. 2013, Tripp and Otter 2006), note duration (Odom et al. 2013, Tripp and Otter 2006, Zhou et al. 2020), and the number of notes (Zhou et al. 2020). Other studies have also demonstrated the efficiency of frequency features in individual owl identification, such as mean frequency (Otter 1996), minimum frequency and maximum frequency (Odom et al. 2013). The temporal features, though, are subject to degradation through scattering when subjects are recorded at greater distances (Yip et al. 2017). Similarly, frequency-dependent attenuation with distance could compromise the ability to distinguish among individuals. However, whether these impede the ability of ARU-collected sounds to differentiate owls for population monitoring has not been tested. We deployed ARUs as part of a project assessing landscape biodiversity, and detected sufficient numbers of Barred Owls during these surveys to test the potential of ARU-based individual identification.

Barred Owls are found in mature mixed-wood forests across much of Canada, and this association with habitat means they are often used as an indicator species in forest management. Furthermore, the expansion of the Barred Owl's range has led to overlap with the closely related Spotted Owl (*Strix occidentalis*) in the Pacific Northwest, and been the focus of a number of studies on potential conflict between the two species (Long and Wolfe 2019). Developing an

effective tool for monitoring Barred Owls could help contribute to evidence-based management plans. The aim of this study is to assess the feasibility of using ARU recordings not just for presence/absence detection, but potentially for individually distinguishing recorded individuals for more accurate censusing. When reviewing our landscape-level detections, we found a significant number of stations detecting Barred Owls, which have previously been shown to exhibit individual vocal variation (Freeman 2000), so we chose this species as the subject of analysis. Our specific objectives were to: (1) apply the advanced bird sound classifier, BirdNET, to screen for Barred Owl signals detected in our long-term monitoring ARU project, (2) identify the temporal and/or frequency features of Barred Owl calls from ARU recordings that provide the best individual distinction, and (3) calculate the accuracy of identifying Barred Owls using acoustic features and compare it to studies on other owl species that used targeted, individual recordings. Ultimately, this serves as a baseline study to demonstrate the framework of using ARUs as a long-term monitoring tool for individual species, such as Barred Owls.

3-3 Methods

3-3-1 Target species and study area

Barred Owls are a non-migratory species that is widely distributed across North America. They are predominately associated with large tracts of mixed mature forests, often near water. Barred Owls are strongly territorial, a behavior proposed to result from limited nest sites (Mazur et al. 1998). Data from 158 banded Barred Owls revealed that the annual home range of Barred Owls is typically less than 10 km² (Johnsgard 1988); however, movement is much more spatially restricted during the breeding season (Feb-April), when the average territory size is between 250 to 300 hectares (Livezey 2007), representing a circular area with around a 1 km radius. The boundaries of home ranges are generally well-maintained by Barred Owls and remain stable from year-to-year and even generation to generation (Nicholls and Warner 1972). Barred Owl vocal activity peaks during the breeding season, with highest rates of spontaneous calling occurring during late March to late April, prior to egg laying (Elderkin 1987). Another peak time is in the fall, potentially corresponding to the dispersal of the young.

The audio data were collected at the John Prince Research Forest (54° 27'N, 124° 10'W, 700 m a.s.l), BC, Canada, from February to April 2021, corresponding with the peak time of Barred Owl vocal activity. ARUs (AudioMoths; Hill et al. 2019) were deployed at 66 sites throughout the study area as part of a larger biodiversity monitoring project, with adjacent sites at least 2 km apart. These sites were selected to include various forest age classes, compositions of forest and harvesting activities in the region (**Figure 3-1**), but many of these sites included habitat conducive to Barred Owls, which are known to occupy the region. The majority of sites with Barred Owl detections are concentrated in the west, possibly attributed to the favorable

habitat characterized by the presence of mixed-wood forests. Conversely, the absence of detections in the east grid may be linked to recent forest harvesting (< 20 years), reducing Barred Owl settlement. However, a definitive explanation would require more detailed occupancy analyses.

The biodiversity monitoring project deployed ARUs in a hexagonal grid, as the design is straightforward when considering the nearest neighbourhood (Birch et al. 2007). This also facilitated sound analysis and spatial separation of sites from the perspective of monitoring Barred Owls. Our deployment distance (i.e., 2 km) ensures that owls detected at one recording site would not be simultaneously recorded at a neighbouring site, and the recording sites were separated by approximately twice the average radius of a Barred Owl pair's breeding home range (Livezey 2007). The ARUs were placed at the centroid of these hexagons, unless there were logistic constraints (e.g., waterbodies, inaccessible terrain) in reaching these locations; in this case the ARUs were placed as close to the centroids as possible.

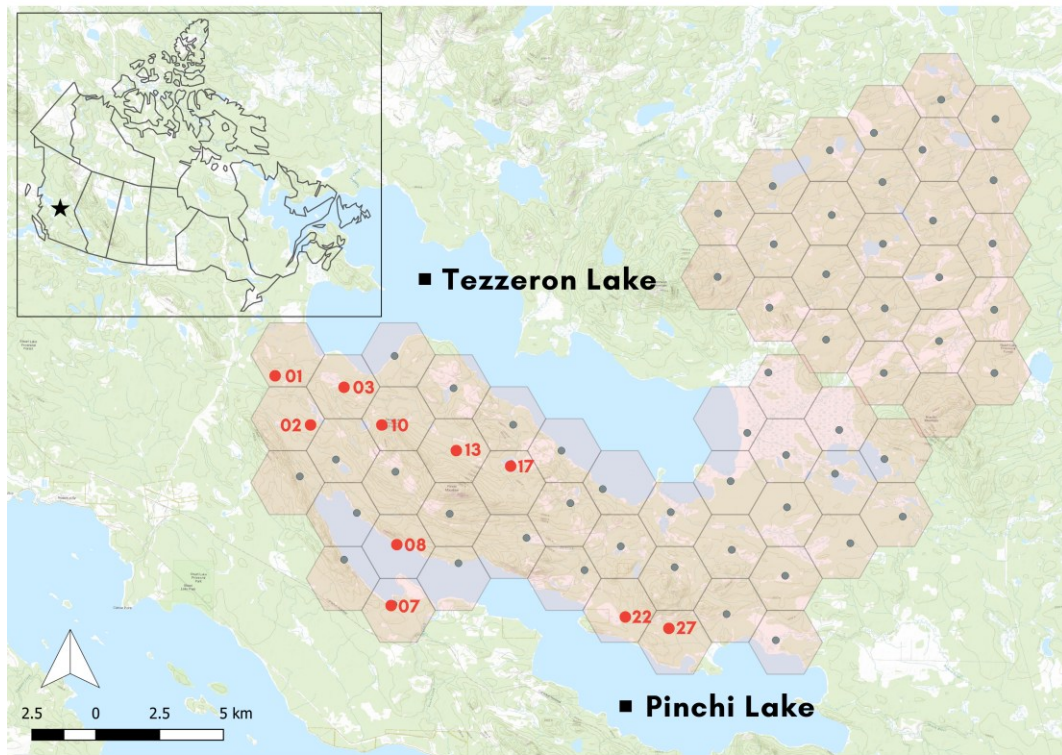


Figure 3-1 Locations of the 66 recording sites. All sites are located within or just outside the John Prince Research Forest, north of Ft. St. James, BC, Canada, with 42 sites within the research forest and 24 sites locate northeast of the research forest. Dots represent the locations of recorders, and hexagons were superimposed to ensure spatial independence. Sites that detected Barred Owls were marked in red with site ID annotated.

3-3-2 Audio data collection

We utilized AudioMoth (v1.1.0) ARUs for audio recordings in this study. The recording window was set from 6 pm to 10 pm daily (Pacific Standard Time), with the sunset time varying between 5:30 – 6:30 pm during the study period. The recording window aligned with the known peak in vocal activity of Barred Owls during the breeding season (0 to 2 hours after sunset - Clement et al. 2021). Each AudioMoth was programmed to record actively for 10 minutes, followed by a 20-minute period of no recording. This recording cycle was repeated throughout the four-hour recording window, resulting in a total of eight 10-minute recordings collected per day, per site. At the end of the project, around 20,000 10-minute recordings were collected, representing around three thousand hours of monitoring effort. A more extensive recording effort was needed during nocturnal surveys for owls compared to typical songbird monitoring programs, due to the lower population densities of most large owls and vocal bouts occurring more infrequently than songbird vocalizations. All recordings were digitized at a sampling rate of 48 kHz and stored as 16-bit sound files (wav format) for subsequent analysis. The high sampling rate ensured accurate representation of the owl vocalizations and preserved the necessary frequency range for detailed examination.

3-3-3 Sound feature processing

Following data collection, we employed BirdNET (Kahl et al. 2021) to identify recordings that contain Barred Owl vocalizations. Specifically, we obtained the source code of BirdNET through the Github repository (<https://github.com/kahst/BirdNET-Analyzer>) and systematically processed all of our data through BirdNET on the High-Performance Computing lab at the University of Northern British Columbia. We provided BirdNET optional parameters to enhance the species detection. For this, we specified the recording longitude and latitude based on the site location, and the week of the year based on recording date. BirdNET analyzes sound recordings by 3-second segments and produces outputs as .csv files, with columns indicating the detected species and a 'detection confidence' ranging from 0.1 to 1. Higher detection confidence values indicate a greater certainty associated with a species detection. The raw output from BirdNET indicated there were around 2,500 3-second segments in our ARU recordings that had detection confidence greater than 0.1 of Barred Owl vocalizations (i.e., more than 2,500 3-seconds segments potentially include Barred Owl). We then manually inspected all the recordings segments which BirdNET indicated as potentially including Barred Owl vocalizations; from this pool, most incidences were true-positive Barred Owl detections, but included a variety of call types and some were too faint for spectrographic analysis.

Our field recordings captured various types of Barred Owl vocalizations, including *Ascending Hoot* and *Two-Phrase Hoot* (Mazur and James 2021). The *Ascending Hoot* has 6–9 regularly spaced notes followed by a downwardly inflected *hoo-aw* ([Audio 1](#)). However, the most frequently recorded vocalization is the *Two-Phrase Hoot*, consisting of 2 groups of 4 or 5 notes each, and commonly being phoneticized as "*Who cooks for you, who cooks for you all?*" ([Audio 2](#)). The *Two-Phrase Hoot* was suggested to be a territorial call (Johnsgard 1988), and a

contact call (Brewster and Chapman 1891) of Barred Owls. This vocalization was used throughout the night and produced during both spontaneous bouts of vocal activity and when an owl confronts conspecifics (Odom and Mennill 2010a). This hoot can be heard from both sexes and occasionally from juveniles, with the juveniles sometimes omitting a few notes at the end. Further, the *Two-Phrase Hoot* routinely occurred both inside and outside of duets (Odom and Mennill 2010b). Given that the *Two-Phrase Hoot* represented the dominant call type among all the detections in our study area, and was a vocal signal in Barred Owls previously shown to allow individual identification (Freeman 2000), we selected recordings of this call type (hereafter “Barred Owl calls”) as our target for the subsequent analysis. A total of 454 calls from 10 sites were of sufficient amplitude and clarity to allow spectrographic analysis and were included in the subsequent analyses, with each site yielding 9 - 217 calls (**Figure 3-2**). Calls with sufficient quality are defined as those with a signal-to-noise ratio above 4 dB.

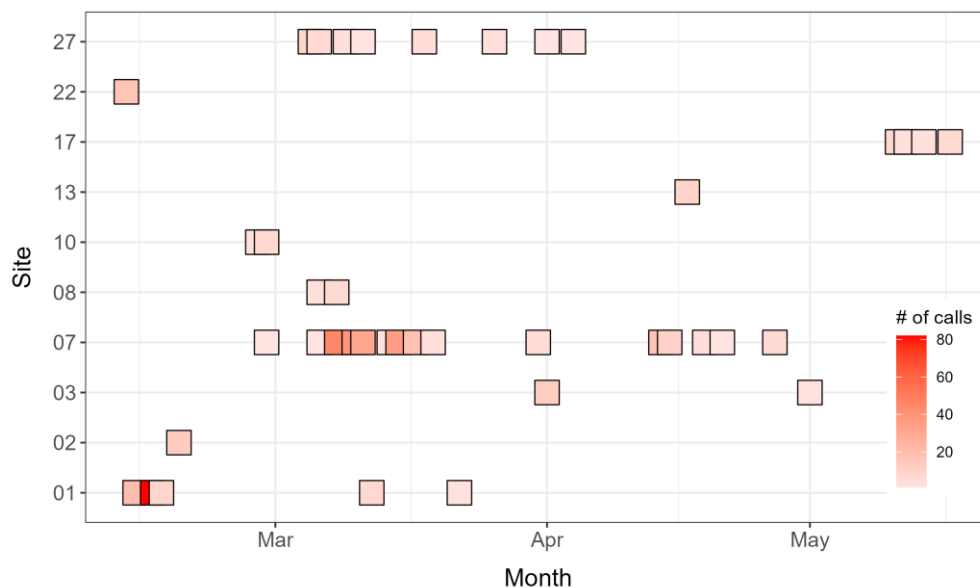


Figure 3-2 Barred Owl calls collected during the 2021 breeding season. A total of 454 calls were selected and analyzed from 10 sites.

The recordings were processed using Raven Lite (K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology 2023), a free software for spectrogram notations. Specifically, we imported the recordings that contained Barred Owl calls into Raven Lite software, manually inspected and created the selection table (i.e., boundaries for signals of interest) of each Barred Owl call. For this, spectrograms were generated with Raven Lite default setting: Hanning Window with window size 256 samples, resulting in a frequency resolution of 90Hz and a temporal resolution of 10ms. Selection tables produced by Raven Lite defined the boundary of each Barred Owl call (i.e., start time and end time of the call, maximum frequency and minimum frequency of the call).

To measure the sound features within each selection table, such as dominant frequency and duration of the call, we used the R (version 4.3.2; R Core Team 2024) package *warbleR* (Araya-Salas and Smith-Vidaurre 2017), a free modular tool for sound analysis and synthesis, which uses Raven Lite selection tables as input. Specifically, we used the *spectro_analysis()* function to extract 12 temporal features and 18 frequency features from each Barred Owl call (**Figure 3-3**). These 30 features were selected as they are standard measurements employed in vocal individuality studies of non-passerines (e.g., Holschuh and Otter 2005, Tripp and Otter 2006). The 12 temporal features, measured in seconds, included the total length of the call (cl), the duration of individual notes within the call (d1 - d8), the interval between the 2nd and 3rd notes (i1), the interval between the 4th and 5th notes (i2), and the interval between the 6th and 7th notes (i3). The 18 frequency features, measured in Hz, encompassed the range of dominant frequency in individual notes (dfr1 - dfr8), the minimum dominant frequency of the call (min_d), the maximum dominant frequency of the call (max_d), and the mean dominant frequency in notes (md1 - md8).

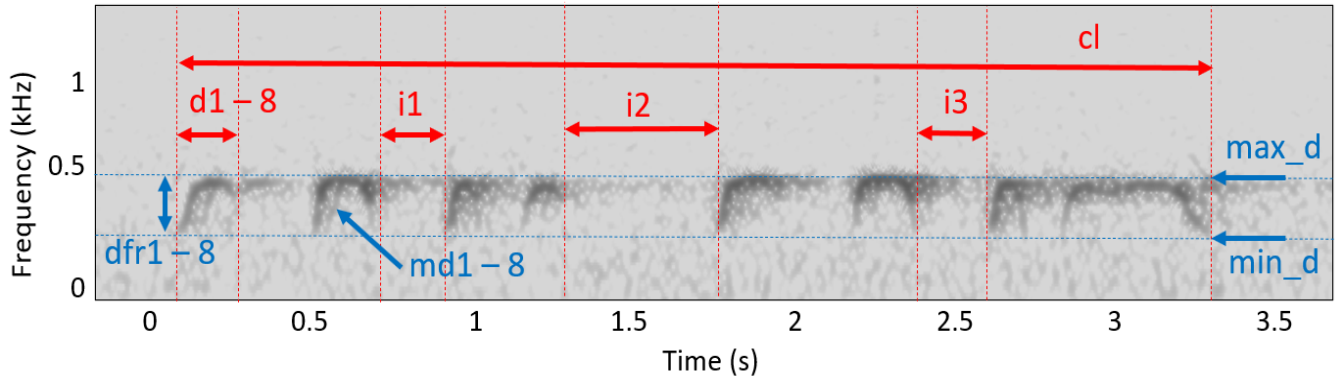


Figure 3-3 Spectrogram of a Barred Owl call, consisting of eight notes. In this study, we measured 12 temporal features (marked in red) and 18 frequency features (marked in blue). *cl* is the total length of the call, *d* is the duration of individual notes within the call, *i* is interval between notes, *dfr* is the range of dominant frequency in individual notes, *md* is the mean dominant frequency in notes. *max_d* and *min_d* are max and min dominant frequency of the call, respectively.

3-3-4 Statistical modelling

We employed linear discriminant analysis (LDA) to classify sounds collected from individual sites. LDA is a widely employed statistical technique for classification of unique groups, using a linear combination of features that maximizes the discrimination between different groups within a dataset (Xanthopoulos et al. 2013). We utilized sound features, specifically temporal and frequency features, as predictor variables, and individual site ID as the response variable in the LDA. The LDA models were fitted using *lda()* function in R package MASS (Venables and Ripley 2002).

For variable selection, we employed a stepwise forward approach based on Wilks' lambda statistic, which is a statistical measure commonly used in LDA to quantify the importance of variables in a model. This was done by *greedy.wilks()* function in R package klaR (Weihs et al. 2005). Initially, the model was defined by selecting one single variable that exhibited the greatest separation between groups. Subsequently, additional variables were incorporated based on their Wilks' lambda statistic. This iterative process of adding variables continued until there was no further improvement in the model's performance, as assessed by overall accuracy and kappa statistics. The overall accuracy and kappa statistics were derived by the average performance a 5-fold cross-validation process, where the dataset was randomly divided into 80% training dataset and 20% test dataset. We used a bootstrap approach with 100 iterations of the 5-fold cross-validation process to compute the standard error for both overall accuracy and kappa statistics.

3-3-5 Spatial distance and vocal similarity

Even though the recording sites were at least 2km apart, there was still potential the same Barred Owl may be detected at two adjacent sites on sequential nights if individual owls moved extensively within or between nights (e.g., movement distances exceeded the inter-recorder spacing such that the bird could be recorded at neighbouring stations at different time in the recording season). This could lead to vocalizations recorded at neighbouring sites appearing to overlap in temporal/frequency characteristics (i.e., have similar multivariate centroids in the LDA model) and lead to instances of the model misclassifying calls recorded at one location to having originating at a neighbouring location. If Barred Owls were moving extensively between recording sites in our study, we would predict that sites with closer spatial distance should detect Barred Owl calls with more similar vocal features. To examine this effect, we measured the correlation between spatial distance and the vocal similarity of Barred Owl calls in each site. Spatial distances were determined by measuring physical distances between the recording sites where the Barred Owls were detected. Vocal similarity was assessed by calculating the Euclidean distance between the multi-dimensional centroids of linear discriminant components of the recorded owls. Measuring the Euclidean distance between the centroid is a means of determine how similar two vocalizations are in multivariate clusters.

3-4 Results

3-4-1 Temporal features outperform frequency features

Descriptive statistics of the 30 sound features extracted from the 454 Barred Owl calls were reported in **Table 3-1**. The focus of this examination was on the coefficient of variation (CV) for these features, considering two perspectives: CV across all sites (CV_all) and CV within each specific recording site (CV_site). Features indicative of individual variation should have lower CV_site relative to CV_all. Based on paired t-tests, both frequency and temporal features showed significant differences between CV_all and CV_site ($\alpha = 0.05$), suggesting the potential for these features to contribute in distinguishing sounds collected from different sites.

The stepwise forward variable selection method based on the Wilk's lambda criterion determined the most influential variables for distinguishing the calls recorded from individual locations. The top ten variables, in descending order, were as follows: total length of the call (cl), interval between the 4th and 5th notes (i2), interval between the 6th and 7th notes (i3), duration of the 8th note (d8), mean dominant frequency of the 6th note (md6), duration of the 3rd note (d3), mean dominant frequency of the 8th note (md8), duration of the 2nd note (d2), mean dominant frequency of the 5th note (md5), and mean dominant frequency of the 1st note (md1). It is noteworthy that the initial four variables selected in the model were temporal features, highlighting the higher discriminative capability in distinguishing individual Barred Owl vocalizations when compared to frequency features.

Table 3-1 Statistics, including mean, standard deviation (SD), and coefficient of variation (CV) for 30 sound features: 12 temporal features and 18 frequency features.

Feature type	Feature	Code	mean	SD	CV group (%)	CV individual (%)
Frequency	Range of dominant frequency in 1st to 8th note (Hz)	dfr1	229	76	33	31
		dfr2	222	70	31	29
		dfr3	217	70	32	32
		dfr4	189	69	37	37
		dfr5	232	74	32	26
		dfr6	221	80	36	34
		dfr7	219	67	30	32
		dfr8	232	72	31	28
Frequency	Maximum dominant frequency of the call (Hz)	max_d	627	40	6	5
Frequency	Minimum dominant frequency of the call (Hz)	min_d	322	58	18	15
Frequency	Mean dominant frequency of the 1st to 8th note (Hz)	md1	532	41	8	6
		md2	564	30	5	4
		md3	533	30	6	4
		md4	552	29	5	4
		md5	561	32	6	4
		md6	572	27	5	3
		md7	540	26	5	4
		md8	555	28	5	3
Temporal	Duration of the 1st to 8th note (sec)	d1	0.17	0.04	25	21
		d2	0.19	0.03	16	12
		d3	0.12	0.03	25	18

Feature type	Feature	Code	mean	SD	CV group (%)	CV individual (%)
		d4	0.19	0.09	46	28
		d5	0.19	0.04	19	17
		d6	0.2	0.03	14	12
		d7	0.13	0.03	23	18
		d8	0.36	0.09	26	24
Temporal	Interval between the 2nd and 3rd notes (sec)	i1	0.41	0.02	5	3
Temporal	Interval between the 4th and 5th note (sec)	i2	0.64	0.06	9	5
Temporal	Interval between the 6th and 7th note (sec)	i3	0.42	0.02	5	4
Temporal	Total length of the call (sec)	cl	3.06	0.14	5	3

3-4-2 LDA model

Variables were added incrementally to the cross-validated LDA model based on their discriminatory power in distinguishing between sites. As the number of variables increased in the model, we observed an increase in both the overall accuracy and the Kappa statistics (**Figure 3-4**). However, the performance of the model reached a point of saturation when six variables were included. Beyond this point, the incremental improvement in model performance by adding more variables was marginal. Consequently, we selected the first six features with the highest discriminatory power (i.e., c1, i2, i3, d8, md6, and d3) as our final model.

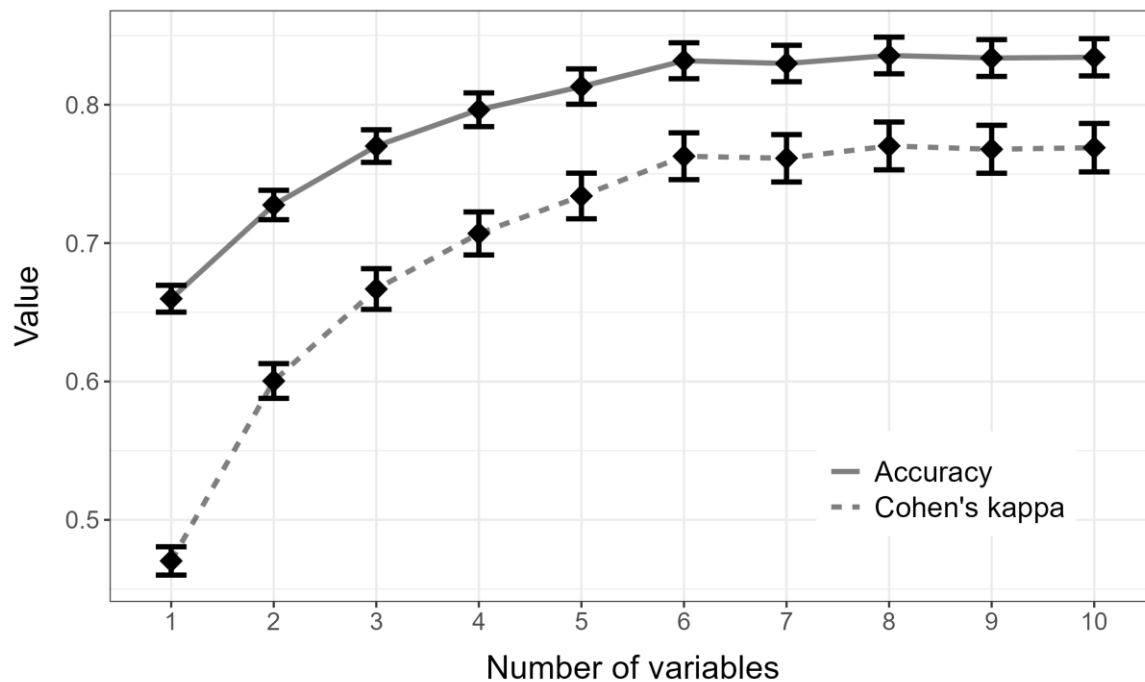


Figure 3-4 The change of overall accuracy and Kappa statistic when adding variables incrementally, with standard error derived by bootstrapping of 100 iterations. The performance of the model increases with the number of variables, reaching a point of saturation when six variables were included.

The correlations between discriminant function and the predictor variables (i.e., loadings) are presented in **Table 3-2**. The first two discriminant functions exhibit the highest correlation with the total length of the call (cl) and the interval between interval between the 6th and 7th notes (i3), respectively. Additionally, these top two discriminant functions collectively account for 80% of the total variability in the dataset (57% and 23%, respectively). We additionally conducted a principal component analysis and received comparable results to the discriminant function analysis (please refer to the Appendix for details).

Table 3-2 Correlations between discriminant function and predictor variables (i.e., loadings). High correlation (absolute value ≥ 0.5) was marked in bold. The trace value reflects the among-group variation captured by that specific discriminant function. The number of functions is either N-1, where N is the number of groups, or the number of predictors, which ever is smaller. Here we have 10 groups and 6 parameters, thus we have 6 linear discriminant functions from the LDA model.

LD functions	Sound features						Proportion of trace
	cl	i2	i3	d8	md6	d3	
LD1	0.81	-0.48	0.33	0.63	0.67	-0.34	57%
LD2	-0.26	-0.71	-0.75	0.42	-0.10	0.23	23%
LD3	0.23	0.22	-0.43	-0.27	-0.23	0.46	12%
LD4	-0.26	-0.34	-0.08	-0.71	0.50	0.47	7%
LD5	-0.36	-0.14	-0.31	-0.45	0.15	-0.64	1%
LD6	-0.18	-0.29	0.21	-0.39	-0.46	-0.01	0%

3-4-3 Identification performance

The final model achieved an overall accuracy of 83% and a Kappa statistic as 0.76, which implies substantial agreement between classification of calls into the predicted site (which recording site the model determined the call had originated) versus observed site (the actual recording site where the call was recorded). The ordination plot of the final LDA model showed the distribution of each of the 454 Barred Owl calls in a reduced dimensional space (Figure 3-5). The calls from the same site generally cluster closely together, indicating distinct groups, such as calls from site 01 (grey), site 07 (light blue), and site 17 (yellow). However, we also found some overlap of the clusters, indicating insufficient discrimination, such as between site 22 (dark blue) and all other sites.

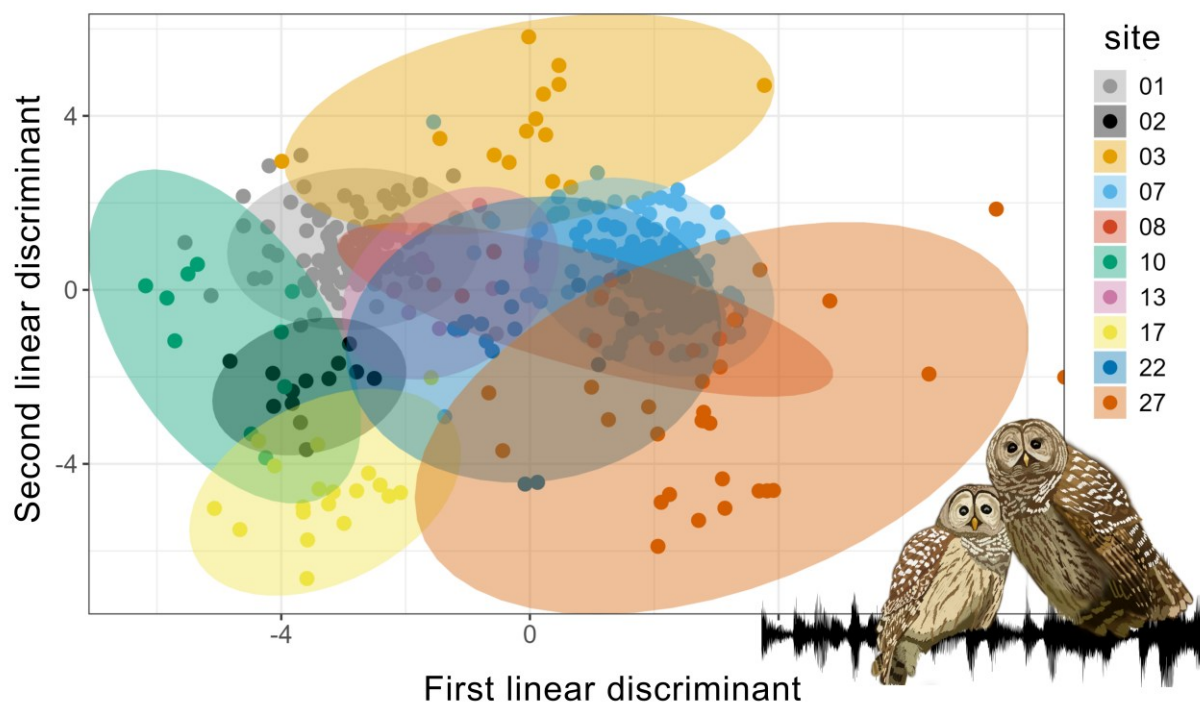


Figure 3-5 The ordination plot of the final LDA model. Only the first two discriminant functions were shown. Some sites have overlapping sound features, indicating insufficient discrimination.

The confusion matrix provides a concise overview of the performance achieved by our final model (**Figure 3-6A**). This tabular representation offers a comprehensive count of calls being assigned by the model to their predicted site of origin (predicted site), as compared to the sites where recordings are known to originate from (observed site). Most of the calls were correctly classified by the model – the predicted site aligning with the observed site (cells along the diagonal). However, there were instances of misclassification. Specifically, we found that 21 calls from site 03 were mis-classified as belonging to site 07, while 6 calls from site 08 were misclassified as 07. Additionally, 6 calls from site 13 were misclassified as 17, and 6 calls from site 22 were misclassified as site 27. We computed the sensitivity (proportion of correctly identified positive instances) and specificity (proportion of correctly identified negative instances) for each site (**Figure 3-6B**). The model exhibited high specificity (> 0.85) across all sites, while sensitivity varied from 0 (site 08) to 0.98 (site 01).

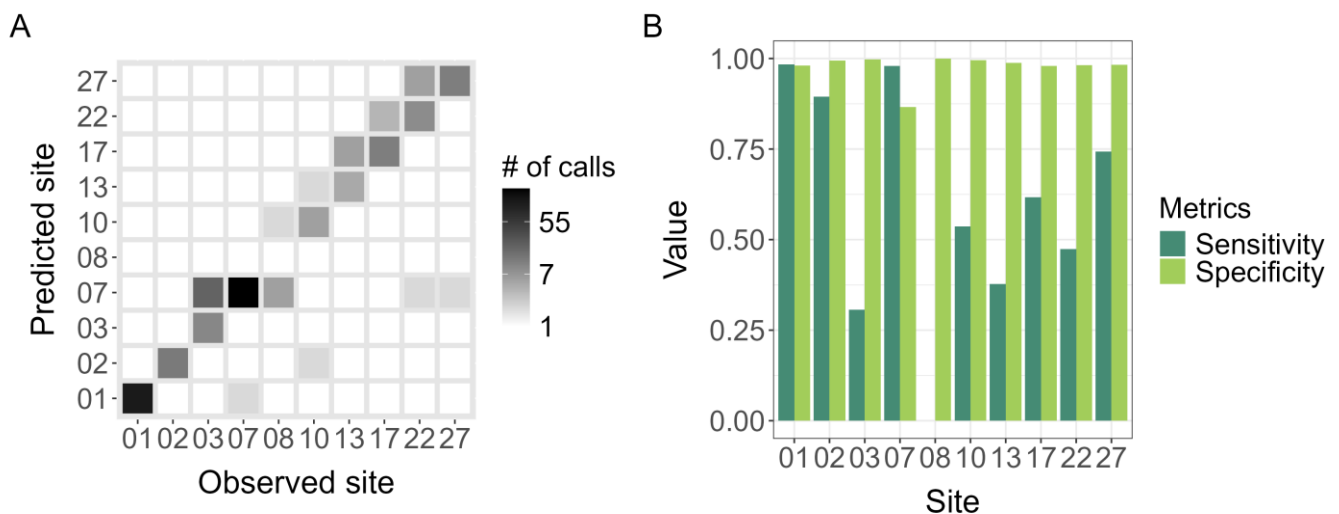


Figure 3-6 (A) The confusion matrix summarizes the final linear discriminant model's outcomes, with call counts averaged from 100 bootstrap iterations. (B) The sensitivity and specificity were calculated for each site.

3-4-4 Low correlation between spatial distance and vocal similarity

We examined the relationship between spatial distance (distance between any two recording sites) and vocal similarity (measure of Euclidean Distance between the multidimensional centroids of the ordination plots as represented in **Figure 3-5**) and found a correlation coefficient of 0.28 ($p=0.06$), indicating a relatively low degree of association between spatial distance and vocal similarity (**Figure 3-7**).

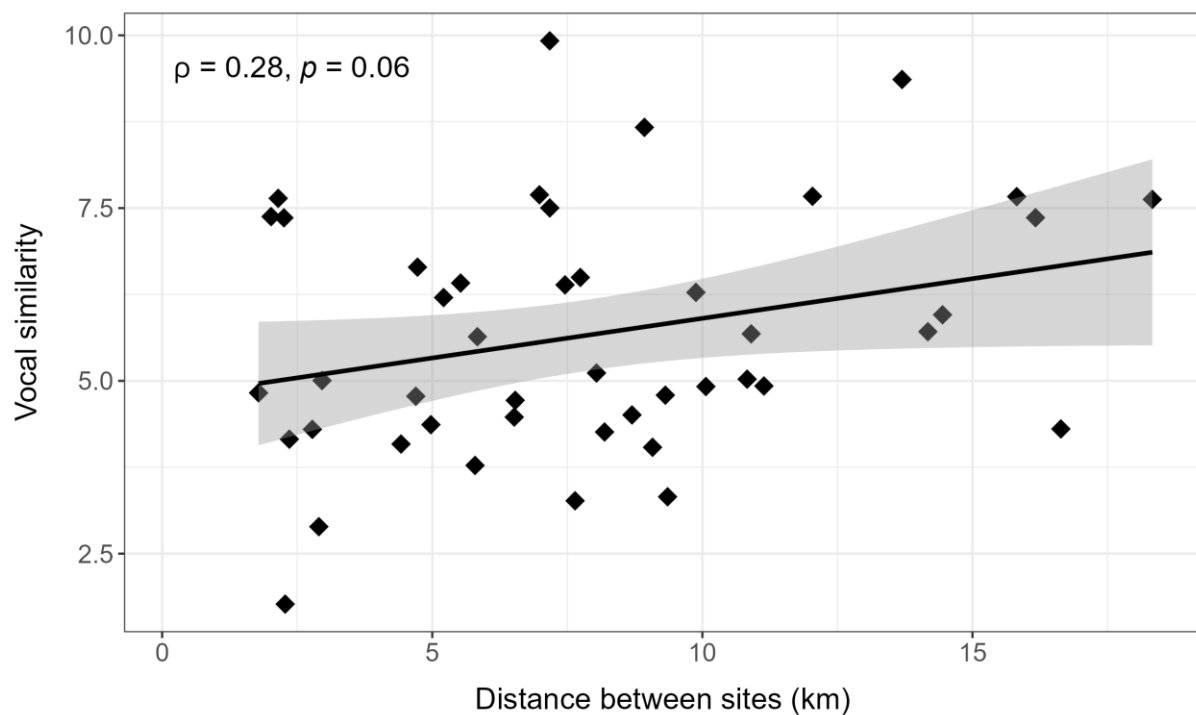


Figure 3-7 The correlation analysis showed there is low correlation (correlation coefficient = 0.28) between the distance and the vocal similarity of Barred Owl calls.

3-5 Discussion

Our study offers evidence supporting the feasibility of individual acoustic monitoring using ARUs, particularly in the context of our focal species, the Barred Owl. Despite inherent variability in audio recordings due to fixed ARU placement, our analysis suggests that the discriminant ability of our models was not significantly compromised, demonstrating that ARU recordings can capture sounds of sufficient quality for individual identification. While our research did not involve a marked population of Barred Owls or directional recordings from individuals, our findings indicate promise for discriminating among individual birds based on vocalizations. We acknowledge the importance of replicating this work on populations of radio or GPS-tracked individuals to validate the effectiveness of ARU-based acoustic monitoring in discriminating among individual birds.

Our LDA model achieved an accuracy of 83% of assigning calls to the sites, which is slightly lower, but still comparable to other individual identification studies using manual sound recording, where the quality of the recording is better controlled (e.g., 89% for female European Eagle Owls (Grava et al. 2008), 99.1% for Tawny Owls (Galeotti and Pavan 1991), 97.4% for Ryukyu Scops Owls (Takagi 2020), 89% for Eastern Screech-Owls (Nagy and Rockwell 2012), and 92% for Western Screech-Owls (Tripp and Otter 2006).

Our study found temporal features were the most powerful distinguisher for Barred Owl vocal individual identification. This was unexpected, given that temporal features are mostly likely to suffer degradation with distance between the bird and recorder (Bradbury and Vehrencamp 2011). Unlike manual recordings, ARUs introduce a challenge in establishing a standard distance-to-recording subject framework. Consequently, this discrepancy could complicate the precise measurement of start and end of notes on spectrograms, owing to

scattering effects. We predicted this might compromise the precision of temporal feature measurements, potentially decreasing the ability to show subtle differences across individuals. However, our LDA model demonstrates that, despite using static passive recorders, temporal measures still retain the greatest discriminant ability in individually distinguishing calls emanating from different sites. And this finding corresponds well with one previous study for Barred Owl individual identification where Freeman (2000) found temporal features, duration and interval of notes, were powerful in distinguishing individuals.

We found a low correlation between the spatial distance between recordings sites where birds were detected and the vocal similarity of the Barred Owl calls recorded at each of these pair-wise ARU stations. This suggests it is unlikely that Barred Owls are moving between, and being recorded at, adjacent sites, which would have predicted that sites that were closer together would have greater vocal similarity in their associated recordings. As a result, the misclassification of calls by the model between observed and predicted site is likely not due to birds moving between and being detected at adjacent recording sites. This suggests that most of the Barred Owls were somewhat sedentary, at least within 2 – 3km from the recording center, within a single breeding season. This makes species with restricted home ranges, like the Barred Owl, good target subjects for using acoustic monitoring to census populations. With broad spacing of the recorders, large areas can be covered and monitored simultaneously, yet our data suggest that owls remain within their individual territories throughout the breeding season. Some of the model misclassification that was seen may be due to more than one individual (e.g., both male and female within a pair) being recorded in a single site (Odom and Mennill 2010a). We observed instances of duet activity in specific study sites, indicating the presence of owl pairs within these areas. Duet calls were documented at site 15, site 27 (Audio 3), and site 28 (Audio

4). Previous research has demonstrated that male and female Barred Owls can be distinguished based on their *Two-Phrase Hoots* (Odom and Mennill, 2010a), with females typically producing higher frequency calls compared to males. The identification of the sex of owls within the same site could potentially enhance the accuracy of individual identification models. Therefore, future research could explore whether ARU recordings within a site cluster sufficiently to differentiate between calls made by a single bird versus counter-singing pairs.

Given that the dataset used in this study is constrained to a single breeding season, an assessment of vocal individuality across multiple years has not been investigated. Nonetheless, prior studies of owl vocalizations (e.g., Galeotti and Pavan 1991, Nagy and Rockwell 2012, Takagi 2020) have consistently found strong stability of sound features over years through manual recordings. To expand the applications of passive acoustic monitoring, we suggest future studies investigate the constancy of owl vocalizations over extended periods using ARUs. This can be done by collecting multi-year acoustic data at the same sites and integrating tracking techniques like MOTUS or GPS to verify the individuals. Upon method validation, passive acoustic monitoring enables prolonged observation of individual owls, facilitating the study of behaviors such as year-to-year fidelity and territorial dynamics. The ultimate objective is to use passive acoustic monitoring as a tool for comprehensive investigations of individual owls.

3-6 Conclusion

This study establishes that the *Two-Phrase Hoot* of Barred Owls, as captured through autonomous recording units (ARUs), exhibits sufficient natural variation in sound features for potential individual identification. While direct proof of individual identification by ARU is not provided, our research lays the groundwork for employing passive recordings in owl census and individual monitoring. Our findings highlight the discriminatory power of temporal features in Barred Owl identification, despite their susceptibility to degradation with increasing distance from the recorder. Notably, the low correlation between spatial distance and vocal similarity revealed the stability of Barred Owl's vocalization both temporally (during the breeding season) and spatially (within a 2 – 3 km radius), making them well-suited for acoustic census using monitoring techniques. Future studies could further explore the generalizability of this technique across other owl species and investigate inter-year comparisons at the same sites, paired with tracking methods like MOTUS or GPS, to validate individuality. This approach would expand the scope of passive monitoring applications in studying individual owl behaviors.

Chapter 4 Quantifying Seasonal and Habitat-related Bird Vocal Activity Using Passive Acoustic Monitoring with GAMs and Occupancy Models

4-1 Abstract

Vocalizations are a primary cue for detecting many bird species; however, species-specific spatiotemporal patterns in vocal activity can bias survey results, particularly when community monitoring programs rely on a single, standardized protocol across multiple species. Effective species-specific survey protocols require an understanding of fine resolution spatiotemporal patterns in vocal activity. In this study, we used three years of data collected by passive acoustic monitoring (PAM), combined with BirdNET automated detections, to quantify seasonal and habitat-related vocal activity of Olive-sided Flycatchers (*Contopus cooperi*). Generalized additive models revealed a pronounced peak in detections in mid-June, followed by a smaller secondary peak in mid-July, with modest interannual variation in peak timing and magnitude; this potentially reveals post-breeding vocal behaviour of Olive-sided Flycatchers. Occupancy modelling suggested that Olive-sided Flycatchers were less likely to occupy stands with dense upper-canopy (i.e., canopy cover >10 m), and that the probability of detecting Olive-sided Flycatchers when they occupied a site was reduced in areas with dense lower-canopy (i.e., canopy cover 1–3 m). These findings identify optimal periods (mid-June and mid-July) and habitats (open areas with sparse canopy cover) for conducting Olive-sided Flycatcher surveys. Overall, our study demonstrates that PAM can effectively capture species-specific spatiotemporal patterns in vocal activity, providing data-driven insights to refine survey design, improve detection, and support informed conservation decisions.

4-2 Introduction

Bird vocalizations play a central role in inter- and intra-sexual communication, facilitating territory defense, mate attraction, and social interactions. This prevalence of acoustic signalling enables the detection of many species during audio-based surveys (Dawson and Efford 2009). Because vocal activity varies across both seasons and habitat types, the probability of observing a species is closely tied to the underlying spatiotemporal patterns of their vocal activity. These patterns can differ substantially among species, some begin breeding and vocalizing early in the year, while others peak later in the season (Singer et al. 2025). Some species prefer open forests, whereas others prefer mature forest habitats (Billerman et al. 2025). Consequently, understanding when and where species vocalize is fundamental for effective species-specific survey protocols that allow accurate monitoring of populations over long-term and large spatial scales.

North America benefits from several long-term, large-scale bird monitoring initiatives that combine the efforts of professional biologists and community scientists. Structured programs such as the Breeding Bird Survey (<https://www.birdscanada.org/bird-science/breeding-bird-survey>) and Christmas Bird Count (<https://www.birdscanada.org/bird-science/christmas-bird-count>) follow standardized protocols that define survey windows, routes, and sampling effort, ensuring consistency and comparability across years. In parallel, flexible platforms like eBird (<https://ebird.org>) and FeederWatch (www.feederwatch.org) allow participants to record birds opportunistically across diverse locations and times, generating extensive datasets. Together, these initiatives have transformed our understanding of bird distributions and population trends, providing broad temporal and geographic coverage of observations.

Although these monitoring programs provide essential long-term and large-scale information, their survey windows and protocols are typically standardized across broad regions, aimed at detecting a wide breadth of species rather than specific target species. Conventionally, these designs are informed by expert knowledge and logistical constraints, and do not always align with species-specific peaks in vocal activity (Singer et al. 2025). Mismatches between survey timing and species-specific vocal activity patterns can potentially introduce bias into trend estimates (Massimino et al. 2021). Even when survey protocols cannot be modified, integrating independent, data-driven information on when and where species are most vocally active can greatly improve the interpretation of existing datasets by clarifying periods of high and low probability of observing a species and identifying potential sources of bias (Best 1981).

Recent advances in passive acoustic monitoring (PAM), combined with automated identification tools such as BirdNET (Kahl et al. 2021), make it possible to quantify vocal activity at fine resolutions across large scales. Continuous or semi-continuous acoustic recordings provide objective measures of the seasonal timing of when species vocalize, as well as the habitat-related variation in vocal activity (e.g., Chronister et al. 2025; Czarnecki et al. 2024; Larkin et al. 2024), which were logistically infeasible to document through traditional field surveys (Shonfield and Bayne 2017). These data-driven activity profiles can refine survey timing, inform modelling approaches, and reveal behavioural patterns not possible through conventional methods.

In this study, we used three years of passive acoustic monitoring (PAM) data collected in the John Prince Research Forest in central British Columbia, Canada, and processed with BirdNET to quantify spatiotemporal patterns of vocal activity, using the Olive-sided Flycatcher (*Contopus cooperi*) as a case study species. The Olive-sided Flycatcher is a late-breeding species

listed as Special Concern in Canada, making it well suited for testing the feasibility of this methodology and evaluating species-specific activity patterns. Our objectives were: (1) to characterize the seasonal vocal activity of the Olive-sided Flycatcher and identify species-specific peak periods, which we expected to occur later than those of most songbirds due to its late-breeding behaviour; and (2) to determine habitat features associated with Olive-sided Flycatcher vocal activity, thereby revealing spatial patterns of occupancy. We expected the species to occupy areas characterized by sparse vegetation and young forest. By outlining a workflow that integrates PAM detections with modelling, this study provides a detailed examination of Olive-sided Flycatcher spatiotemporal vocal activity to inform survey design for this species, while also presenting a transferable approach that can be applied to other species to define data-driven spatiotemporal activity patterns. Such information has the potential to refine survey design, improve interpretation of conventional monitoring datasets, and help identify sources of analytical bias.

4-3 Methods

4-3-1 Target species

The Olive-sided Flycatcher (*Contopus cooperi*) is a long-distance migratory songbird currently listed as Least Concern by the IUCN (BirdLife International 2025). However, the species is designated as Special Concern by the Committee on the Status of Endangered Wildlife in Canada (Environment and Climate Change Canada 2019) given the significant population declines across its range in recent decades (Billerman et al. 2025). Olive-sided Flycatchers breed in coniferous and mixed forests across western and boreal North America, typically near forest edges with tall trees or snags for perching, adjacent to open areas containing standing trees and snags. Optimal breeding habitat is generally considered to be late-successional forest with relatively open canopy structure (0–39% canopy cover; Verner 1980).

Male Olive-sided Flycatchers produce loud, far-carrying vocalizations, most notably the characteristic “quick-three-beers” song ([Audio](#)). Vocal activity varies across the breeding cycle, as documented in a study from Alaska (Wright 1997): males sing frequently upon arrival on the breeding grounds and prior to pairing, reduce singing during nest-building, and increase vocal activity again during incubation. Singing is uncommon during the nestling stage, when males are primarily engaged in parental care. As a result, the probability of observing Olive-sided Flycatcher based on song was highest prior to pairing (94%), declined following pair formation (34%), increased during incubation (53%), and was lowest after hatching (12%; Wright 1997). Only one brood is raised per season.

4-3-2 Study area

The study was conducted in the John Prince Research Forest, which spans approximately 16,000 ha and is located in central British Columbia, Canada (54°27'N, 124°10'W; 700 m a.s.l.) within the dry sub-boreal spruce biogeoclimatic zone. Acoustic data were collected using 66 autonomous recording units (ARUs; AudioMoth; Hill et al. 2018), each deployed at least 2 km apart to minimize spatial autocorrelation. Acoustic data was collected from 2020 to 2022, during May to July, between 4am and 7am. The ARUs were scheduled to record for 1-minute followed by 4 minutes of inactivity, resulting in 12 recordings per hour and 36 recordings per day. In total, 67,301 one-minute recordings were obtained across the three years at the 66 sites. All recordings were standardized to a 48 kHz sampling rate and stored as mono Pulse Code Modulation (PCM) WAV files.

4-3-3 Audio processing

Acoustic data were analyzed using the BirdNET Analyzer v2.4 model (<https://github.com/BirdNET-Team/BirdNET-Analyzer>), implemented via the Python module running in a local terminal. To retain as many true positive detections as possible initially, we set the min_conf parameter to 0.1. This parameter filters out any detections with a confidence score below the specified value. Processing the entire dataset, which comprised 1.5 terabytes of audio, required approximately 72 consecutive hours using four-core CPUs.

A BirdNET species-specific threshold of 0.35 was applied for Olive-sided Flycatcher detections, which was previously determined for our study area to ensure a precision of 95% (Tseng et al. 2025). Tseng et al. (2025) provides a detailed framework on how these species-specific thresholds were defined. At sites with at least one Olive-sided Flycatcher detection, the

number of active ARU days ranged from 1 to 54 per year, with a mean of 6.84 ± 10.89 days per ARU per year (**Figure 4-1**).

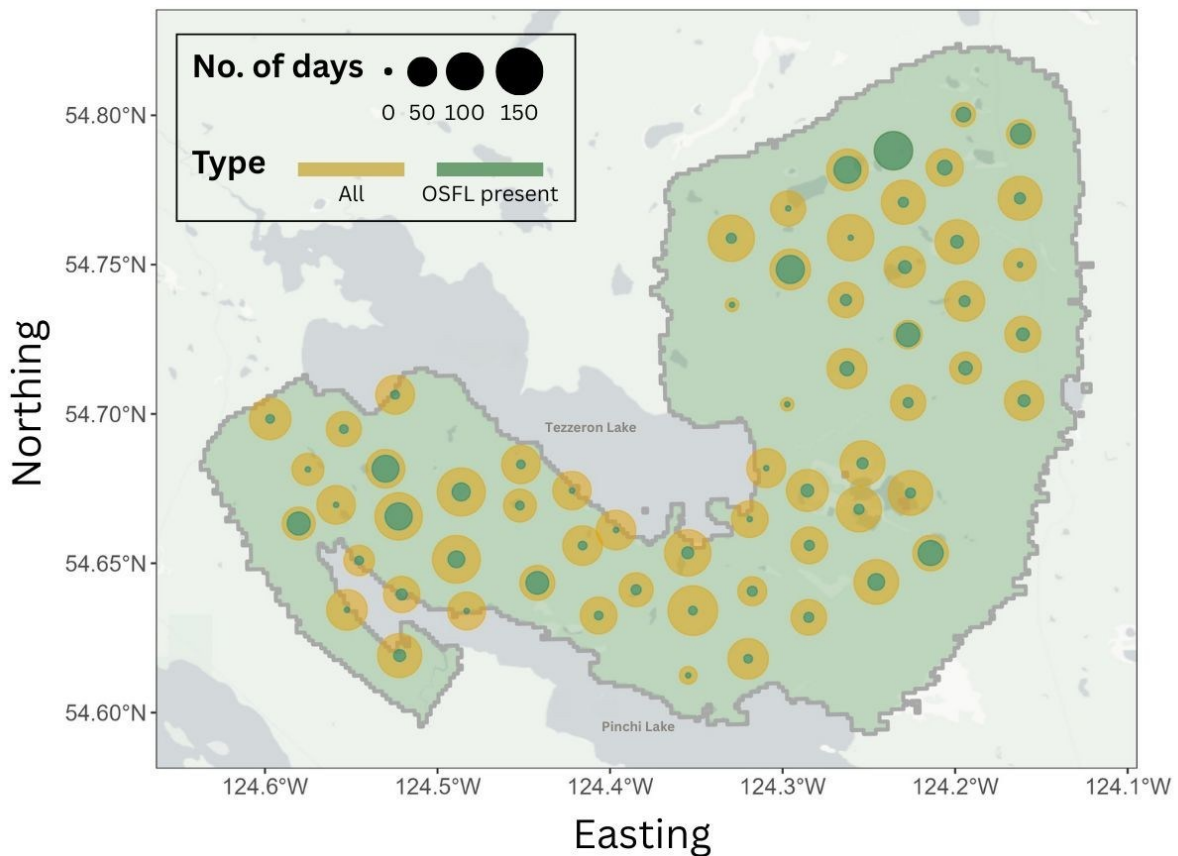


Figure 4-1 Site map of the John Prince Research Forest in British Columbia, Canada, illustrating survey effort across three breeding seasons (2020–2022; May–July) at 66 monitoring sites, each equipped with one autonomous recording unit (AudioMoth). Dot size represents total ARU deployment days (yellow) and the number of days with Olive-sided Flycatcher detections (green). Northeastern sites exhibited particularly high detection frequency, with Olive-sided Flycatcher detected on most recording days.

4-3-4 Modelling temporal pattern – generalized additive models

To characterize seasonal vocal activity of Olive-sided Flycatcher, we first refined the dataset by retaining only site–year combinations with at least two consecutive days of Olive-sided Flycatcher detections. This filter excluded sites with very few or sporadic detections, as single-day detections may represent opportunistic vocalizations or transient individuals rather than meaningful patterns of local activity. Using this criterion, we identified 9, 10, and 13 sites in 2020, 2021, and 2022, respectively, which had consistent flycatcher detections.

We used generalized additive models (GAMs) to model seasonal activity patterns, leveraging their ability to capture nonlinear and cyclic temporal trends. For all candidate models, the response variable was the daily number of Olive-sided Flycatcher detections (i.e., the total detections across all 36 one-minute recordings per day). Julian day was included as a cyclic smooth term to represent seasonal activity patterns, and site and year were included to account for spatial and interannual variation. We constructed a suite of models of increasing complexity (**Table 4-1**; see also **Figure A4-1** for model ecological meaning and formula).

All models were fitted using the *gam()* function in the *mgcv* R package (Wood 2001). Maximum likelihood was used to enable valid comparison of Akaike Information Criterion (AIC) values among models with differing fixed-effects structures (Zuur et al. 2009). Models with the lowest AIC values, including those with $\Delta\text{AIC} \leq 3$, were considered supported by the data. These top models were then refitted using restricted maximum likelihood to obtain stable smoothing-parameter estimates and to generate the final inference and visualizations (Zuur et al. 2009).

Table 4-1 Candidate generalized additive models (GAMs) of daily Olive-sided Flycatcher detections. Models represent different seasonal phenology (smooth terms), the dependency of phenology on year (interaction types), and annual or spatial variation (random intercepts). Smooth terms for day of year (yday) utilized cyclic cubic splines.

Model ID	Smooth term	Interaction type	Random intercepts	Interpretation
model 0 (m0)	--	--	site	Null model. A constant detection rate with a random site intercept.
model 1 (m1)	yday	--	site	A single seasonal curve with a random site intercept.
model 2 (m2)	yday	--	year + site	A single seasonal curve with random site and year intercepts (no interaction).
model 3 (m3)	yday	Varying coefficient (by = year)	site	Year-specific seasonal curves with a random site intercept.
model 4 (m4)	yday	--	year * site	A single seasonal curve with a random site $\times$ year intercept.
model 5 (m5)	yday	Random smooth (fs = year)	site	A single seasonal curve with smooth year-specific deviations and a random site intercept.
model 6 (m6)	yday	Varying coefficient (by = year)	year * site	Year-specific seasonal curves with a random site $\times$ year intercept.

4-3-5 Modelling spatial pattern – occupancy models

To satisfy the closure assumption, that occupancy status remains constant across repeated surveys, we restricted detections to the period from 1–30 June in each of three years, when Olive-sided Flycatchers are considered settled on individual breeding territories (Billerman et al. 2025; and **Figure 4-4**). To minimize false positives, we applied the species-specific BirdNET confidence thresholds (see **Methods - Audio processing**), ensuring 95% precision in identifying Olive-sided Flycatcher vocalizations. To meet the assumption of independence across sampling occasions, we aggregated ARU detections into 5-day intervals to construct presence–absence detection histories for Olive-sided Flycatchers. This resulted in a detection history comprising 66 sites (i.e., 66 samples) and 18 sampling periods (six periods per year over three years; **Figure 4-2**). The selected window length follows recommendations for continuous-monitoring sensors, including autonomous recording units and camera traps (Chambert et al. 2018; Goldstein et al. 2024).

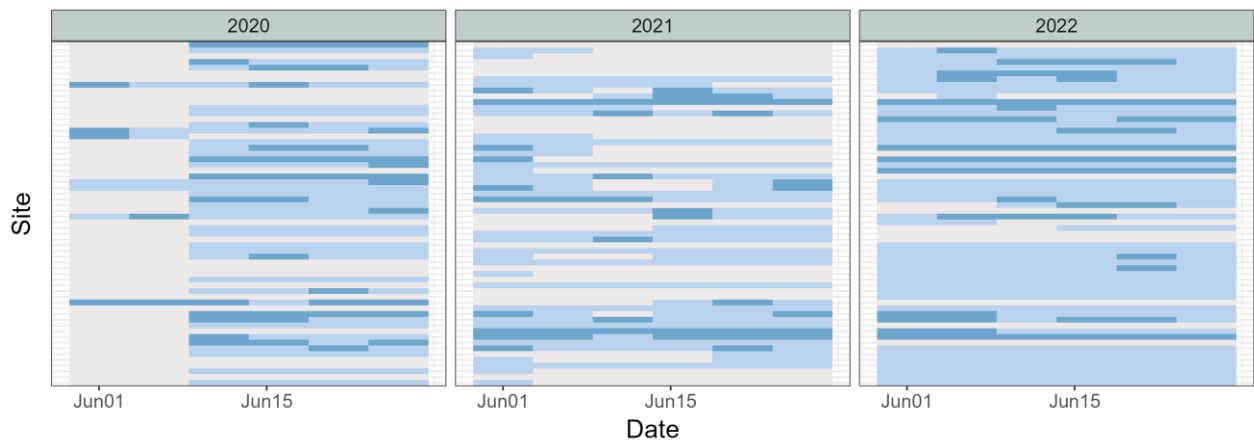


Figure 4-2 Detection history for Olive-sided Flycatcher across sites in June of 2020, 2021, and 2022, using a 5-day detection window. Grey cells indicate periods when ARUs were not functioning, light blue cells indicate Olive-sided Flycatcher absence (0), and dark blue cells indicate Olive-sided Flycatcher presence (1). This detection history was used as the input for the occupancy model.

Occupancy models estimate two parameters: (1) occupancy probability (ψ) - the probability that a site is truly occupied, and (2) detection probability (p) - the probability of detecting the species given it is present. For the occupancy component, we included seven LiDAR-derived habitat covariates summarized within a 100-m radius around each site, representing the approximate area a bird may occupy when detected (Pérez-Granados 2025). Covariates were selected based on the documented habitat preferences of Olive-sided Flycatchers, which primarily breed along forest edges and openings and rely on tall, prominent trees or snags as singing and foraging perches, as well as unobstructed airspace for aerial foraging (Billerman et al. 2025). These occupancy covariates were: (1) distance to stream, wetland, or lake edge; (2) distance to polygon stand edge; (3) canopy cover at 1–3 m; (4) canopy cover at 3–10 m; (5) canopy cover >10 m; (6) the proportion of the surrounding polygon with forest age >80 years; and (7) the proportion of deciduous trees with height >10 m.

For the detection component, we included two covariates expected to influence acoustic detectability: (1) Julian day and (2) LiDAR-derived canopy cover at 1–3 m. These covariates were selected because sound transmission and signal attenuation are likely affected by lower-canopy density at the height of the recording units, and because detection probability may vary over the breeding season.

Occupancy models were fitted using the *PGOcc()* function in the *spOccupancy* R package (Doser et al. 2022). To examine landscape patterns in occupancy and detection probability, we generated spatial predictions using the covariates identified as significant in the occupancy model. Raster layers for these covariates were aggregated to a 15-m analysis resolution and used as inputs to the fitted occupancy model to obtain posterior predictive samples for every grid cell across the study area.

4-4 Results

4-4-1 Temporal vocal activity pattern

Sites with at least two consecutive days of Olive-sided Flycatcher detections exhibited vocal activity concentrated in June and July (**Figure 4-3**). Among the GAMs fitted to quantify seasonal activity patterns, models m4 and m6 had the lowest AICc values and received substantially greater support ($wAIC = 0.79$ and 0.21 , respectively) than all alternative models, indicating that both models captured the underlying data structure reasonably well compared with other models in the model set (**Table 4-2**).

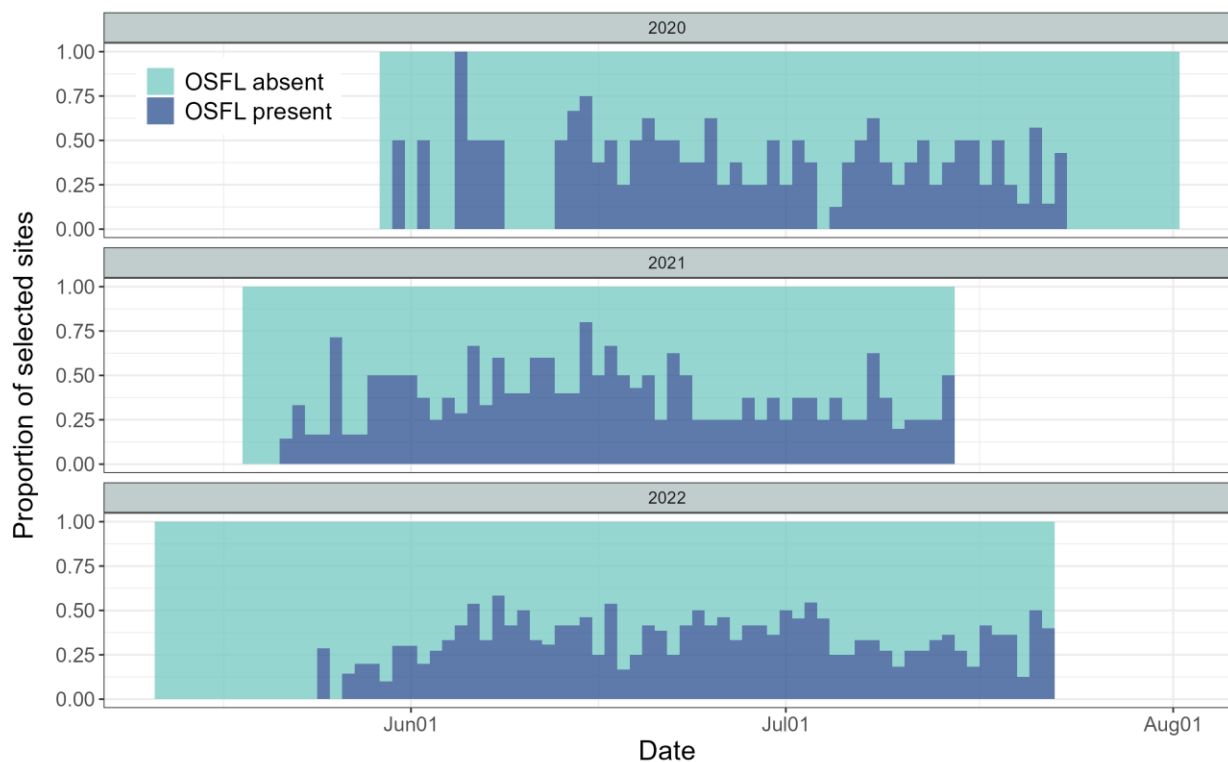


Figure 4-3 Seasonal activity of sites with at least two consecutive days of Olive-sided Flycatcher within a site-year. This included 9, 10, and 13 selected sites in 2020, 2021, and 2022, respectively. Sites with at least one Olive-sided Flycatcher detection that day are shown in dark blue, and those without detections are shown in light blue.

Table 4-2 AICc-based comparison of candidate GAM models. Models are listed in order of increasing AICc. The two highest-ranked models (bolded) exhibited the strongest empirical support. Full model specifications are provided in the Methods section and illustrated in **Figure A4-1**.

Model	Model interpretation	DF	AICc	Delta AICc	AICc weight
m4	One seasonal curve, with random site*year intercept	40	5017	0	79%
m6	Seasonal curve by year, with random site*year intercept	52	5020	3	21%
m5	One seasonal curve and smooth seasonal deviation by year, with random site intercept	36	5082	65	0%
m3	Seasonal curve by year, with random site intercept	42	5085	68	0%
m2	One seasonal curve, with random site + year intercept	31	5089	72	0%
m1	One seasonal curve, with random site intercept	29	5103	86	0%

In the model with a single seasonal smooth (m4), there was strong evidence of seasonal variation in detections ($\chi^2 = 249.6$, $p < 0.05$). Population-level predictions showed a pronounced seasonal pattern, with detections peaking in mid-June, and exhibiting a smaller secondary peak in mid-July (**Figure 4-4A1**). Site-year random effects contributed substantially to model fit ($\chi^2 = 525.9$, $p < 0.05$), with site-years differing widely in baseline detection rates. Some locations showed predicted counts several-fold higher or lower than the population-level mean (**Figure 4-4A2**). Overall, m4 explained approximately 40% of the deviance (**Figure 4-4A3**), that 40% of the variation in detection counts across all observations is captured by the model predictors and smooths.

The model with year-specific seasonal smooths (m6) produced a similar seasonal pattern but allowed peak timing and magnitude to vary among years. Seasonal effects remained significant across all three years (2020: $\chi^2 = 124.5$, $p < 0.05$; 2021: $\chi^2 = 83.5$, $p < 0.05$; 2022: $\chi^2 = 144.0$, $p < 0.05$). All years exhibited a primary peak in mid-June followed by a secondary peak in

mid-July, although the smoothness and height of these peaks differed among years (**Figure 4-4B1**). Site-year random effects again explained considerable variation ($\chi^2 = 519.9$, $p < 0.001$), reflecting persistent differences in baseline detection rates among site-years (**Figure 4-4B2**). Model m6 explained slightly more deviance (41.4%) than m4 (**Figure 4-4B3**).

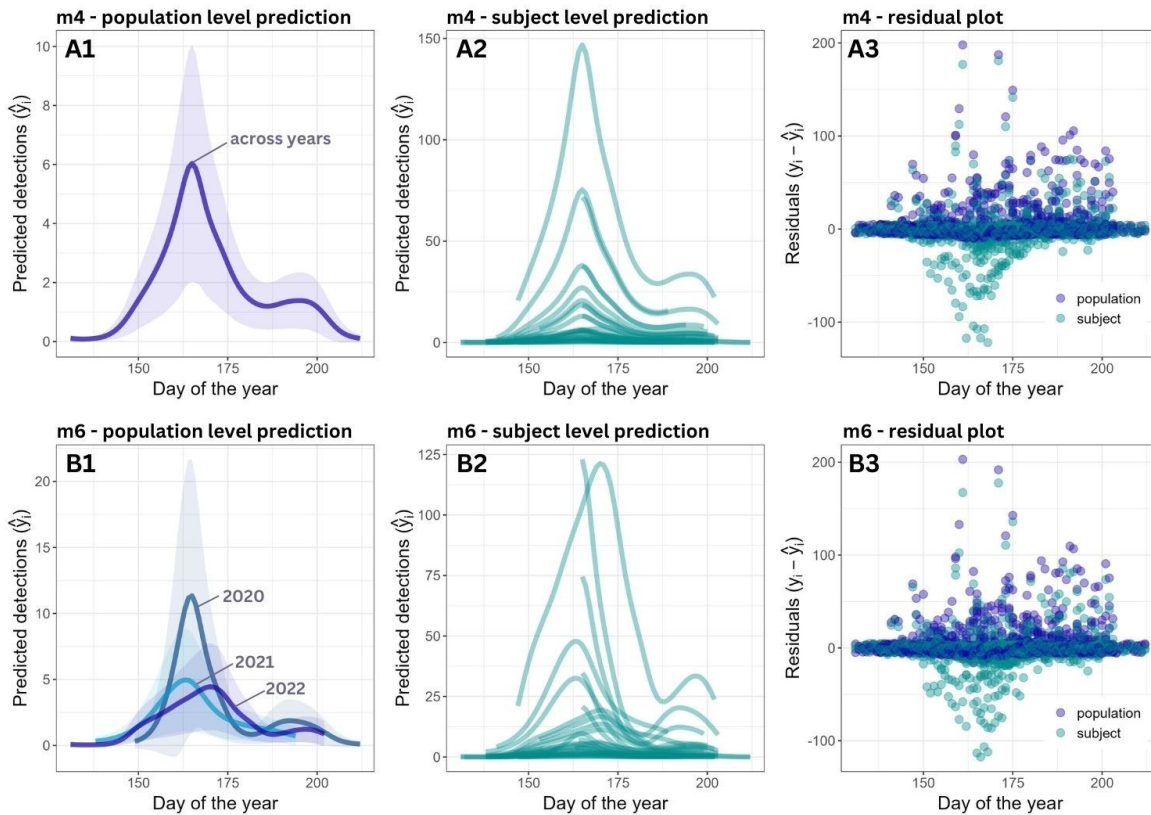


Figure 4-4 Model predictions at the (1) population level, (2) site-year (subject) level, and (3) model residuals for (A) model 4 and (B) model 6. Both models showed a bimodal seasonal pattern in Olive-sided Flycatcher vocal activity and substantial variation among site-year combinations.

4-4-2 Spatial vocal activity pattern

The naïve occupancy of Olive-sided Flycatchers, defined as the proportion of sites with at least one detection, was 57% in 2020, 51% in 2021, and 41% in 2022. The full model, which included all candidate occupancy and detection covariates, showed improved model fit relative to a null model containing only an intercept for occupancy, as evidenced by a lower WAIC (643.8 vs. 650.1). Among the occupancy covariates, only canopy cover >10 m had a significant effect on site use (posterior effect size mean: -2.02 ; 95% credible interval: -3.60 to -0.52), indicating that Olive-sided Flycatcher were less likely to occupy sites with dense upper-canopy cover compared with more open sites (**Figure 4-5A**). For detection covariates, canopy cover at 1–3 m had a small but significant negative effect (posterior mean = -0.023 , 95% CI: -0.04 to -0.006), suggesting that dense lower-canopy cover slightly reduced the likelihood of detecting singing Olive-sided Flycatchers when the bird is actually occupying the site (**Figure 4-5B**).

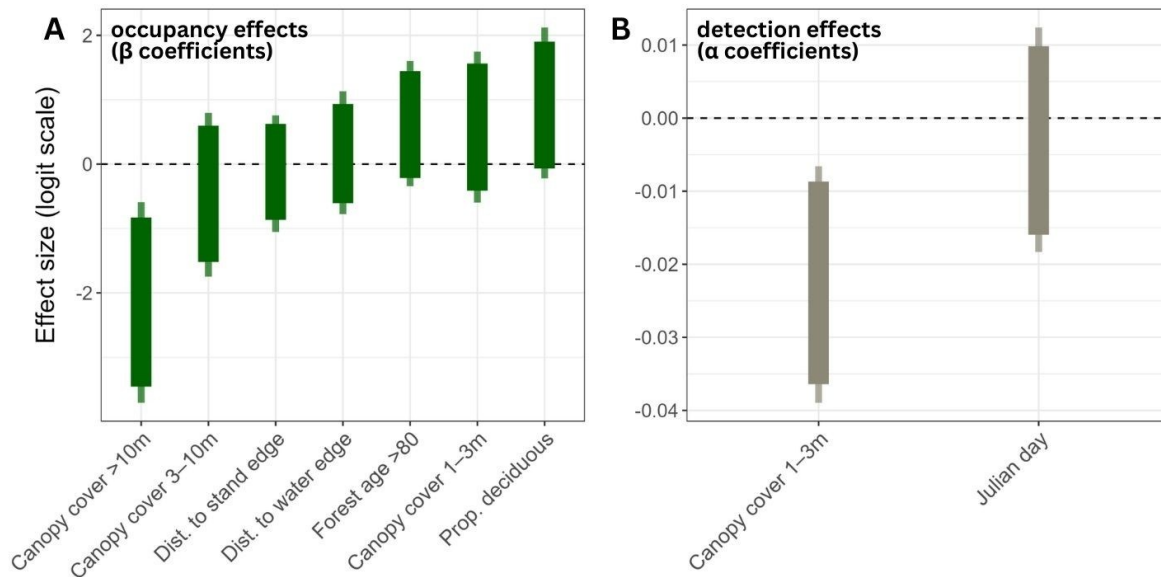


Figure 4-5 Posterior effect sizes of covariates on (A) occupancy and (B) detection probabilities for Olive-sided Flycatcher. Thick bars represent 90% credible intervals, and thin bars represent 95% credible intervals. All covariates were standardized prior to modeling.

Landscape-level predictions indicated that estimated occupancy probability was highest in the northeastern portion of the study area and along the eastern shoreline of Tezzeron Lake, where young forest stands were most common (**Figure 4-6A**). Predicted detection probabilities were below 0.5 throughout most of the study area, with the lowest values occurring in dense shrub habitats, particularly near forest edges where low-height vegetation cover (1–3 m) was greatest (**Figure 4-6B**).

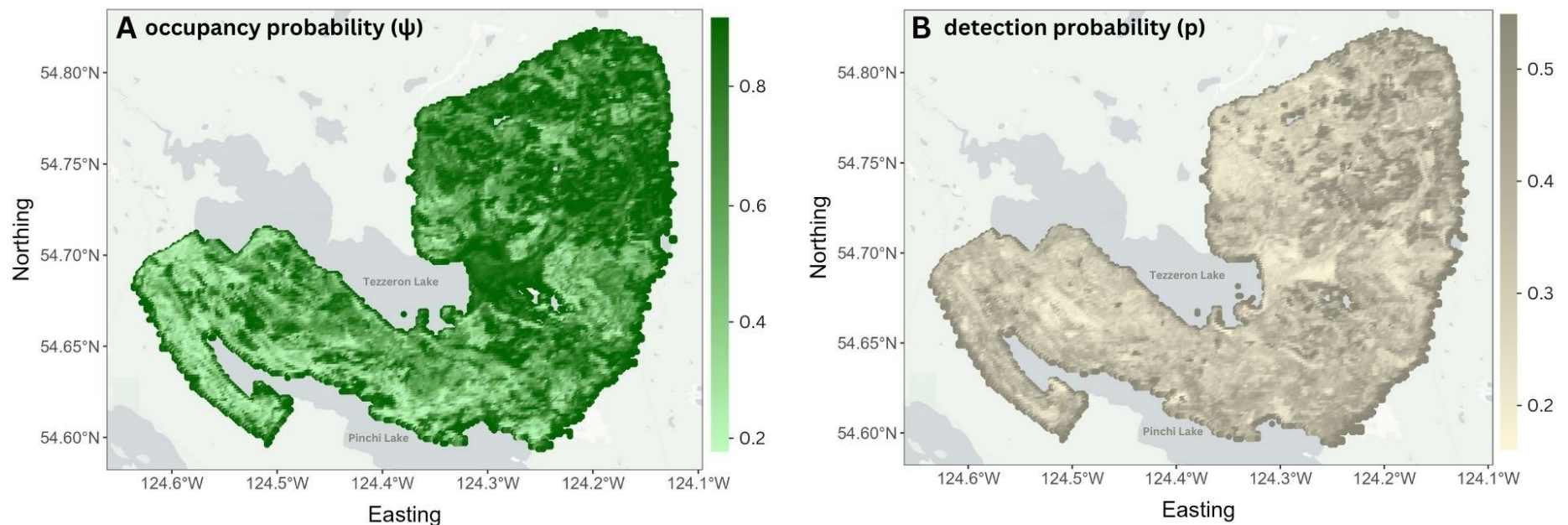


Figure 4-6 Spatial predictions from the occupancy model for Olive-sided Flycatcher in the study area. (A) Predicted occupancy probability (ψ), showing the likelihood that a site is occupied by Olive-sided Flycatchers. (B) Predicted detection probability (p), indicating the probability of detecting Olive-sided Flycatchers at a site if they are present.

4-5 Discussion

4-5-1 Temporal vocal activity pattern

We used three years of acoustic monitoring data combined with generalized additive modelling to identify the seasonal vocal activity patterns of Olive-sided Flycatchers, addressing the need for fine-resolution activity information to inform species-specific survey protocols. Our results revealed a clear seasonal pattern, characterized by a primary peak in mid-June, followed by a smaller secondary peak in mid-July (**Figure 4-4**). This pattern corresponds well with previous observations of Olive-sided Flycatcher activity, which also reported two activity peaks (Wright 1997).

In our results, the primary peak in mid-June likely corresponds to territory establishment and mate attraction during female's arrival, when males sing most intensively to advertise occupancy and secure breeding opportunities. This timing broadly aligns with previous studies showing high song output early in the breeding season, particularly during territory defense and pair formation (Wright 1997).

Our results further showed a secondary peak in mid-July, which is most plausibly explained by post-breeding vocal activity. Elevated singing after successful fledging has been documented in other passerines as a mechanism for territory maintenance, mate guarding, or insurance against late breeding failure (Fig. 6 in Beckett and Ritchison 2010). The temporal spacing between pre-breeding and post-breeding song peaks typically corresponds to the combined duration of incubation and nestling development. In Olive-sided Flycatchers, incubation lasts approximately 14–16 days and the nestling period approximately 16–18 days (Wright 1997), yielding a total of 4–5 weeks. This closely aligns with the ~30-day separation

between the two vocal activity peaks identified in our model (**Figure 4-4**). Alternatively, this late-season surge in vocal activity may be attributed to nesting reattempts by males following an initial failure. Notably, this secondary peak differs slightly from previous observations, which reported increased vocal activity primarily during incubation, followed by a decline after hatching (Wright 1997). This discrepancy warrants further investigation and could be explored through additional acoustic data from other study sites or targeted field observations, which were previously limited, to gain a better understanding of Olive-sided Flycatcher vocal activity in relation to their breeding behaviour.

Our results also showed that the year-specific smooths revealed modest interannual variation in peak timing and magnitude (**Figure 4-4**), likely reflecting environmentally driven shifts in breeding phenology. Factors such as warmer spring temperature and earlier snowmelt timing are known to advance arrival dates in long-distance migratory birds (Gordo 2007), such as Olive-sided Flycatchers, and may contribute to the observed variability among years. Multi-year datasets could further help identify relationships between phenological changes and climate, including potential mismatches between migration timing and vegetation green-up or the influence of migration timing on population trends. Acoustic monitoring provides a feasible approach for such investigations.

4-5-2 Spatial vocal activity pattern

We used three years of acoustic monitoring data, combined with occupancy modelling and LiDAR-derived habitat variables, to identify habitat associations of Olive-sided Flycatchers, addressing the need for fine-resolution information on their spatial activity. Our occupancy models identified forest structure as a key determinant of Olive-sided Flycatcher occupancy and

detection probability across the study area. Upper-canopy cover (>10 m) emerged as the strongest predictor of occupancy, with dense overstory substantially reducing the probability of site use (**Figure 4-5**). This finding is consistent with previous studies showing that Olive-sided Flycatchers preferentially breed in open or early-successional forests, where reduced canopy cover facilitates aerial foraging and provides exposed perches for singing and hunting (Billerman et al. 2025). Higher occupancy probabilities concentrated in areas dominated by younger forest stands, particularly in the northeastern portion of the study area (**Figure 4-5**). Together, these results reinforce the importance of canopy openness and forest age structure in shaping Olive-sided Flycatcher breeding habitat at local and landscape scales.

At a broader scale, however, the apparent association between Olive-sided Flycatchers and disturbed or edge habitats should not be interpreted as evidence that increasing forest harvest, for example, in the northeast of the study area, necessarily benefits the species. Although anthropogenic forest openings may resemble natural post-fire habitats structurally, accumulating evidence suggests that reproductive success is often lower in harvested landscapes, likely due to increased nest predation and altered predator communities (Robertson and Hutto 2007). As a result, such habitats may function as ecological traps, attracting individuals without supporting long-term population viability (Robertson and Hutto 2007). This distinction between habitat use (i.e., occupancy) and habitat quality may help reconcile our spatial results with long-term population declines documented across much of the species' range (BirdLife International 2025). In heavily logged areas of the study region (i.e., northeast part of our study area), suitable structural conditions for occupancy may exist, yet these sites may not contribute positively to population growth.

Detection probability was generally low across the landscape, with predicted values rarely exceeding 0.5 (**Figure 4-6**). Several factors likely contribute to this pattern. Most notably, our recording window (4 – 7am) may not have fully captured the Olive-sided Flycatcher’s peak vocal activity, which declines rapidly after dawn (Wright 1997). Second, the Olive-sided Flycatcher’s large territories, often 10–20 ha (Altman 1997), result in naturally low densities. This reduces detection probability even at occupied sites, as fewer individuals are present to be detected. Furthermore, our results indicate that dense lower-canopy vegetation (1–3 m) slightly but significantly reduced detection probability (**Figure 4-5**), likely due to acoustic attenuation or obstruction within shrub layers. Finally, our workflow of defining species-specific thresholds prioritized high precision (95%) to minimize false positives; this inherently reduced recall, thereby potentially lowering the estimated detection probability.

4-6 Conclusion

This study demonstrates that passive acoustic monitoring (PAM), combined with generalized additive modelling and occupancy modelling, can effectively characterize species-specific temporospatial patterns in Olive-sided Flycatcher vocal activity. Aligning surveys with the mid-June peak, and focusing efforts in young or open forest stands, is likely to increase both detection probability and monitoring efficiency. The modest interannual variation observed in seasonal activity underscores the value of multi-year datasets for capturing environmentally driven shifts in vocal behaviour. More broadly, our findings highlight how integrating full-season acoustic data with modelling approaches can generate species-specific spatiotemporal activity profiles, offering an evidence-based foundation to potentially refine survey protocols, reduce detectability-driven bias, and improve population assessments.

Chapter 5 Integrating Passive Acoustic Monitoring and LiDAR for Breeding Bird Habitat Modelling in A Working Forest

5-1 Abstract

Accurately modelling bird species richness is essential for guiding conservation and forest management decisions; however, traditional ground-based surveys are often limited in spatial and temporal coverage. Here, we integrated passive acoustic monitoring (PAM) and Light Detection and Ranging (LiDAR)-derived habitat covariates to model breeding songbird richness (composed of 40 species) within the John Prince Research Forest, a 16,000 ha managed working forest in central British Columbia, Canada. Acoustic data was collected from 59 sites over three breeding seasons using autonomous recording units to inform site-level richness, which was then compared to 17 LiDAR covariates calculated using a 100-m buffer around each site. Our results showed that none of the LiDAR covariates strongly predicted richness of breeding songbirds, and only one covariate, the proportion of deciduous trees, showed a weak positive association for a species sub-group. Vertical canopy structure and other commonly used LiDAR metrics had limited explanatory power, likely reflecting the heterogeneous habitat preferences among breeding songbirds and the generally similar richness across sites. Overall, this study demonstrates the potential of integrating PAM and LiDAR to quantify biodiversity in working forests and highlights the importance of standardized approaches for large-scale acoustic monitoring.

5-2 Introduction

Identifying bird species richness and its association with habitat is crucial for guiding conservation priorities for different species (Francis and Currie 2003; Hawkins et al. 2003). Traditionally, estimates of richness relied on ground-based bird surveys that provided direct observations of species presence. To extend beyond surveyed locations, models have been developed to predict bird richness using habitat covariates, thereby linking biodiversity patterns to environmental drivers and supporting conservation and management decisions (Rahbek et al. 2006). In working forests, such models are particularly valuable for identifying priority areas for conservation, informing harvest planning, and evaluating the ecological impacts of forest management strategies (e.g., Maes et al., 2005).

Obtaining accurate and comprehensive data on bird richness, however, remains challenging. Ground-based surveys, such as point counts and transects, are labor-intensive, time-consuming, and often limited in spatial and temporal coverage, making it difficult to monitor large or remote areas effectively. Moreover, habitat covariates derived from conventional remote sensing methods, such as satellite imagery or aerial photography, generally emphasize horizontal landscape features. However, vertical habitat structure, such as canopy height, foliage density, and understory complexity, plays a critical role in shaping bird communities (Culbert et al. 2013). These vertical attributes are hard to capture with traditional remote sensing products, limiting the ecological relevance of bird–habitat models (Gottschalk et al. 2005).

Passive acoustic monitoring has emerged as a powerful and scalable tool for biodiversity assessment (Ross et al. 2023; Sugai et al. 2019). Autonomous recording units (ARUs) can collect continuous or repeated audio data over large spatial and temporal scales with relatively low human effort (Sugai et al. 2019). This approach enables detection of vocalizing bird species even

in remote or difficult-to-access areas, thereby greatly expanding survey coverage. Numerous studies have demonstrated the effectiveness of acoustic monitoring for estimating species presence (e.g., Picciulin et al. 2019), vocal activity patterns (e.g., Pérez-Granados et al. 2021), density (Pérez-Granados and Traba 2021), and community composition (e.g., Mattmüller et al. 2022).

Light Detection and Ranging (LiDAR) technology provides a complementary perspective by offering detailed three-dimensional information on vegetation structure (Vierling et al. 2008). Unlike conventional remote sensing, LiDAR captures both horizontal and vertical habitat attributes, making it particularly well-suited for studying bird–habitat relationships (Bradbury et al. 2005; Tattoni et al. 2012). Previous research has demonstrated the utility of LiDAR-derived metrics in modeling bird diversity and habitat suitability across diverse ecosystems (Bakx et al. 2019). However, the integration of LiDAR with passive acoustic monitoring remains relatively unexplored, despite its potential to improve our understanding of the structural drivers of bird species richness.

In this study, we integrate passive acoustic monitoring and LiDAR remote sensing to model breeding bird richness within the John Prince Research Forest, a managed working forest in central British Columbia, Canada. Acoustic data were collected using autonomous recording units (ARUs) during three breeding seasons. LiDAR data provided structural habitat covariates characterizing vertical and horizontal forest complexity. Our objectives were to: (1) compile a comprehensive species list and estimate asymptotic site-level breeding bird richness, and (2) evaluate the associations between LiDAR-derived habitat covariates and bird asymptotic richness.

5-3 Methods

5-3-1 Study area and acoustic recording sites

The study was conducted in the John Prince Research Forest (~16,000 hectares in area), located in central British Columbia, Canada, within the dry sub-boreal spruce biogeoclimatic zone (**Figure 5-1A**). Audio data were collected from 2020 to 2022 throughout the breeding season (early May to late July) between daily dawn chorus (4 am to 7am), using Audio Moth ARUs (Hill et al. 2019). ARUs were set to record for 1 minute, followed by 4 minutes pause, throughout the three hours recording period, resulting in 36 one-minute recordings per day per site. Each ARU was placed at least two kilometers apart to minimize spatial correlation. Variability in the number of active ARUs at each site/date (**Figure 5-1B**) occurred due to setup logistics (e.g., accessibility of sites during early season) and field/equipment challenges (e.g., battery depletion, firmware issues, disturbances by wildlife). We excluded data from sites which had less than 15 days of ARU data to avoid potential bias due to low surveying effort. This resulted in data from 59 sites, with each site having an average of 106 ARU days across three years, ranging from 32 to 182 ARU days.

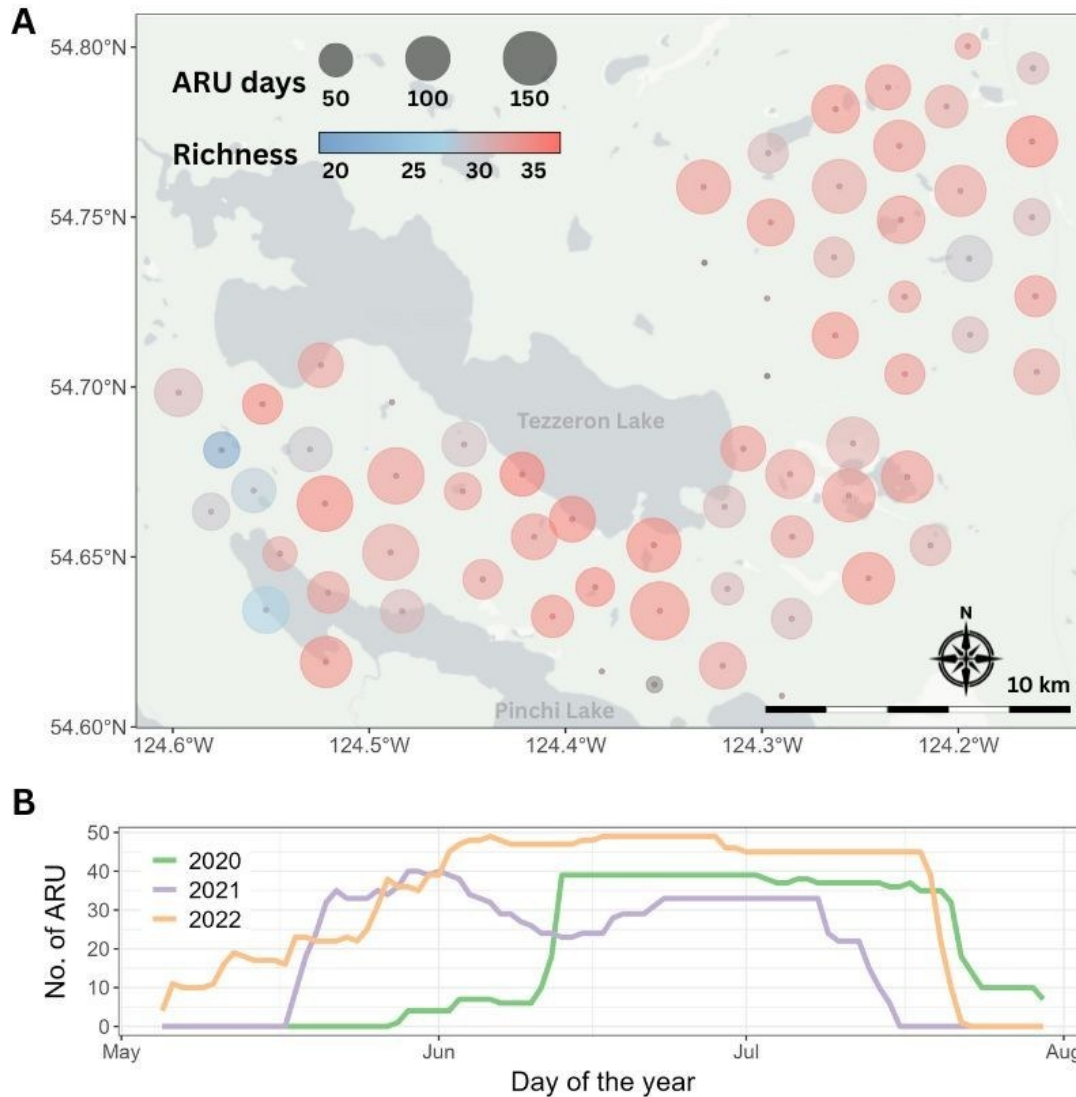


Figure 5-1 Spatiotemporal distribution of survey effort for data collected in the John Prince Research Forest, British Columbia, Canada. (A) Spatial distribution of the 66 monitoring sites; solid central dots indicate the locations of autonomous recording units (ARUs), and circle size represents the number of days sampled during the survey period. (B) Annual number of active ARUs across the survey period, showing a gradual increase during deployment and a decline as units were retrieved.

5-3-2 Acoustic data processing and target species

We analyzed collected acoustic data with the BirdNET Analyzer v2.4 model (BirdNET Team 2025), using the parameters listed in **Table 5-1**. The entire dataset, comprising 1.5 terabytes of audio, required approximately 72 consecutive hours of processing. An overall bird community list was generated in four steps: (1) BirdNET detections were initially filtered using a confidence threshold of 0.8. For each detected category, five recording segments with the highest confidence scores were manually reviewed. Sound categories with at least one confirmed vocalization were retained, resulting in 136 sound categories. (2) Non-bird categories, such as Car Engine, Red Squirrel, Wood Frog, and Slender Meadow Katydid, were removed, reducing the list to 129 sound categories. (3) Species not listed in the British Columbia Breeding Bird Atlas (Davidson et al. 2015), which documents species recorded in the Prince George area since 2008, were excluded, leaving 123 species. (4) Species detected at fewer than two sites or on fewer than two days in June (i.e., main breeding season) were excluded, resulting in a final list of 122 species (**Appendix Table A5-1**). The overall species list included a diverse range of bird taxa such as raptors, waterfowl, warblers, sparrows, flycatchers, woodpeckers, owls, and other families.

We further refined this overall species list to a target guild of passerine species that have confirmed breeding records in the study region, and which are known to be vocally active during the dawn chorus period. These criteria were used to avoid detection bias given that recordings were scheduled during the dawn chorus of the breeding season. We then assessed individual species-specific BirdNET confidence thresholds for the 40 species, following the methods of Tseng et al. (2025), to ensure the precision of detections. To do this, we used stratified sampling to select 360 recording segments in which BirdNET had detected each of the target species, with 20 segments drawn from each 0.05 detection confidence interval bin (ranging from 0.1 to 1.0).

Each segment was then manually reviewed, by both listening to audio and viewing spectrograms in comparison with known exemplars of the species, to classify detections as true or false positives. For each species, a BirdNET confidence threshold was set that achieved a precision of 0.95, indicating that at least 95% of the retained detections were true positives (see Tseng et al. 2025 for the full workflow). These thresholds were set individually for each of the final list of 40 passerines target species included (**Table 5-2**).

Table 5-1 BirdNET algorithm arguments with default values and the values used in this study.

Argument	Default value	Used value
i	none	--
o	none	--
lat	-1	same as default
lon	-1	same as default
week	-1	same as default
slist	none	same as default
sensitivity	1.0	same as default
min_conf	0.1	same as default
overlap	0	same as default
rtype	table	r
threads	1	4
batchsize	1	4
locale	en	same as default
sf_thresh	0.03	same as default
classifier	none	same as default
fmin	0	same as default
fmax	15000	same as default
output_file	none	--
skip_existing_results	FALSE	TRUE

Table 5-2 The 40 breeding passerine species detected during the dawn chorus and selected as target species in this study. Detections were obtained from three years of songbird breeding-season data collected from 66 sites at the John Prince Research Forest. For each species, the table reports the total number of BirdNET detections and the number of sites where the species was detected. Values were calculated after applying species-specific BirdNET confidence thresholds achieving 95% precision and excluding sites with fewer than 15 ARU recording days.

Common name	Scientific name	Threshold (95%)	No. detections	No. sites
Swainson's Thrush	<i>Catharus ustulatus</i>	0.10	327234	59
Hammond's Flycatcher	<i>Empidonax hammondii</i>	0.10	140580	59
Yellow-rumped Warbler	<i>Setophaga coronata</i>	0.10	110942	59
Warbling Vireo	<i>Vireo gilvus</i>	0.10	86199	59
Golden-crowned Kinglet	<i>Regulus satrapa</i>	0.10	84229	59
Magnolia Warbler	<i>Setophaga magnolia</i>	0.10	74066	59
Dark-eyed Junco	<i>Junco hyemalis</i>	0.10	48329	59
Western Tanager	<i>Piranga ludoviciana</i>	0.10	39421	59
American Robin	<i>Turdus migratorius</i>	0.10	37644	59
MacGillivray's Warbler	<i>Geothlypis tolmiei</i>	0.10	30276	59
Orange-crowned Warbler	<i>Leiothlypis celata</i>	0.10	11907	59
Pine Siskin	<i>Spinus pinus</i>	0.10	6399	59
Northern Waterthrush	<i>Parkesia noveboracensis</i>	0.10	145743	58
American Redstart	<i>Setophaga ruticilla</i>	0.10	74866	58
White-throated Sparrow	<i>Zonotrichia albicollis</i>	0.10	42408	58
Pacific Wren	<i>Troglodytes pacificus</i>	0.10	60245	57
Red-breasted Nuthatch	<i>Sitta canadensis</i>	0.21	15373	57

Common name	Scientific name	Threshold (95%)	No. detections	No. sites
Tennessee Warbler	<i>Leiothlypis peregrina</i>	0.10	36095	56
Ruby-crowned Kinglet	<i>Corthylio calendula</i>	0.10	13718	56
Least Flycatcher	<i>Empidonax minimus</i>	0.16	36919	55
Chipping Sparrow	<i>Spizella passerina</i>	0.10	5162	54
Cedar Waxwing	<i>Bombycilla cedrorum</i>	0.36	1994	53
Olive-sided Flycatcher	<i>Contopus cooperi</i>	0.31	9201	52
Common Raven	<i>Corvus corax</i>	0.10	4668	52
Varied Thrush	<i>Ixoreus naevius</i>	0.61	6693	51
Townsend's Warbler	<i>Setophaga townsendi</i>	0.24	3039	51
Boreal Chickadee	<i>Poecile hudsonicus</i>	0.20	2918	51
Western Wood-Pewee	<i>Contopus sordidulus</i>	0.10	10992	50
Dusky Flycatcher	<i>Empidonax oberholseri</i>	0.26	3495	49
Alder Flycatcher	<i>Empidonax alnorum</i>	0.10	21031	48
Common Yellowthroat	<i>Geothlypis trichas</i>	0.10	4153	46
Western Flycatcher	<i>Empidonax difficilis</i>	0.41	1306	37
Hermit Thrush	<i>Catharus guttatus</i>	0.79	3301	36
Wilson's Warbler	<i>Cardellina pusilla</i>	0.46	603	35
Black-capped Chickadee	<i>Poecile atricapillus</i>	0.26	606	32
American Crow	<i>Corvus brachyrhynchos</i>	0.10	1477	30
Song Sparrow	<i>Melospiza melodia</i>	0.10	3390	23
Red-winged Blackbird	<i>Agelaius phoeniceus</i>	0.10	204	23
Yellow Warbler	<i>Setophaga petechia</i>	0.40	394	21
Blackpoll Warbler	<i>Setophaga striata</i>	0.85	51	10

5-3-3 Asymptotic richness detected by ARUs

We calculated the number of days on which each of the 40 target species was detected at each of the 59 sites (**Figure 5-2**). To characterize similarities in site-use among species, we applied k-means clustering to group species with comparable detection patterns across sites. Prior to clustering, we applied data normalization by dividing the number of species ARU days by the total number of ARU days per site, thereby accounting for variation in sampling effort. To determine the optimal number of clusters, we evaluated two approaches: the total within-cluster sum of squares (“elbow method”) using the *factoextra::fviz_nbclust()* function (Kassambara 2017), and the gap statistic using the *cluster::clusGap()* and *factoextra::fviz_gap_stat()* functions (Kassambara 2017). Final clustering was performed using the *kmeans()* function in R (Lemenkova 2019).

Direct comparisons of observed species richness across sites (i.e., the number of coloured cells in each column of **Figure 5-2**) can be misleading, as survey effort (i.e., number of ARU days) varied among sites. To account for this variation, we used asymptotic richness estimates as a standardized measure of site-level diversity (Chao et al. 2014). We calculated asymptotic richness using the iNEXT (iNterpolation and EXTrapolation) R package (Hsieh et al. 2016), which implements diversity estimation based on Hill numbers. Specifically, we used the *iNEXT::ChaoRichness()* function with incidence data to estimate asymptotic richness and its associated standard errors.

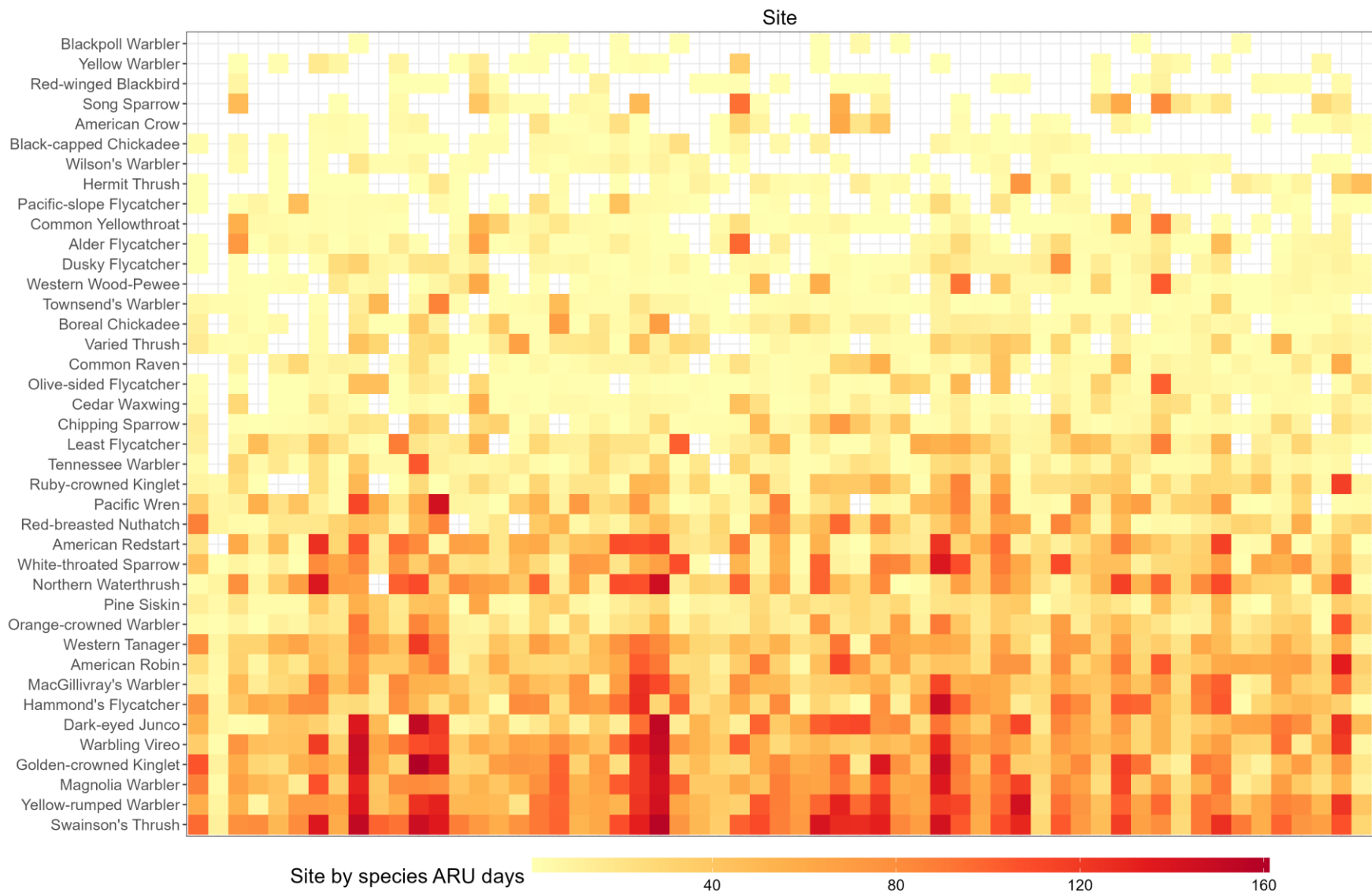


Figure 5-2 Number of days each species was detected at each site using ARUs. Species in the lower part of the graph, such as American Robin, Dark-eyed Junco, and Golden-crowned Kinglet, were detected across most sites and likely represent generalists. In contrast, species toward the top of the graph, including Yellow Warbler and Red-winged Blackbird, were detected at fewer sites, potentially reflecting more specific habitat preferences.

5-3-4 LiDAR covariates and model selection

LiDAR data were collected in 2015 during the leaf-off season. As all sites in the study had not undergone significant disturbance since 2015, the LiDAR covariates reflect similar relative habitat composition/shifts to allow for comparison with the audio data collected in year 2020 to 2022. We applied an area-based approach to rasterize the point clouds at a spatial resolution of 10 meters. For modeling purposes, we aggregated habitat covariates within a 100-meter radius around each ARU site to represent the area a bird might be using when detected (Pérez-Granados 2025). A total of 17 LiDAR-derived variables were considered as candidate covariates (**Table 5-3**). These covariates were classified following the framework outlined in Bakx et al. (2018), which highlights canopy structure and total vegetation metrics as the most commonly associated with bird species richness (see Fig. 4 in Bakx et al. 2019).

To reduce multicollinearity and ensure model stability, we calculated pairwise correlations among the covariates and removed those with correlations greater than 0.8. Furthermore, we computed variance inflation factors (VIFs) to ensure all the covariates having a VIF less than 5. This pre-selection ensured that only relatively independent covariates were included in the subsequent model selection. We then used an information-theoretical approach for model selection. Specifically, we fitted models using all possible linear combinations of the LiDAR covariates, calculating Akaike weights to quantify the relative support for each model. For each covariate, we then computed the sum of model weights by summing the weights across all models in which the covariate appeared. Covariates with high sum of weights (> 0.8) consistently appeared in the best-supported models, representing a high variable importance. This analysis was conducted using the *MuMIn::dredge()* function (Bartoń 2025), with the full model specified as a linear regression.

Table 5-3 LiDAR-derived habitat covariates used in the analysis, including measures of canopy structure, vegetation density, and ground features. Covariates retained after the correlation and multicollinearity screening are shown in bold (see **Results** section for details).

Type	Name	Definition
Topography/ Ground	dem	Average elevation
	slope	Average slope
	aspect	Average aspect
	d_lid_rip_wet_str_le	Distance to LiDAR stream, wetland polygon, lake edge
	d_vri_polyedge	Distance to vegetation resources inventory (VRI) polygon stand edge
Canopy	cc1_3	Average crown closure between 1 – 3 m
	cc3_10	Average crown closure between 3 – 10 m
	cc10	Average crown closure above 10m
	chm	Average canopy height
	vdi_95	Vertical distribution if intercepted canopy elements ranging from 0 and 1 using 95% height
Total vegetation	less_10	Proportion of polygon with forest age less than 10 years
	age_80	Proportion of polygon with forest age greater than 80 years
	prop_decid	Proportion of deciduous tree counts for all trees with height > 10m
	decid_dens	Number of deciduous trees with height > 10m per unit area
	conf_dens	Number of conifer trees with height > 10m per unit area
	tree_dens	Number of total trees with height > 10m per unit area
	ba_dens	Sum of basal area per unit area, with trees having height > 10m

5-4 Results

5-4-1 Asymptotic richness estimates

Estimated asymptotic species richness across sites ranged from 29.7 to 48.8 species. Regardless of the number of autonomous recording unit (ARU) sampling days per site, observed richness was, on average, within the 91% of the asymptotic richness (**Figure 5-3**). Specifically, the mean difference between estimated asymptotic and observed species richness was 3.0 species for low-effort sites (i.e., 1 – 90 ARU days), 4.6 species for medium-effort sites (i.e., 90 – 120 ARU days), and 2.8 species for high-effort sites (i.e., > 120 ARU days). See Appendix **Table A5-2** for the estimated asymptotic richness and associated estimation uncertainty.

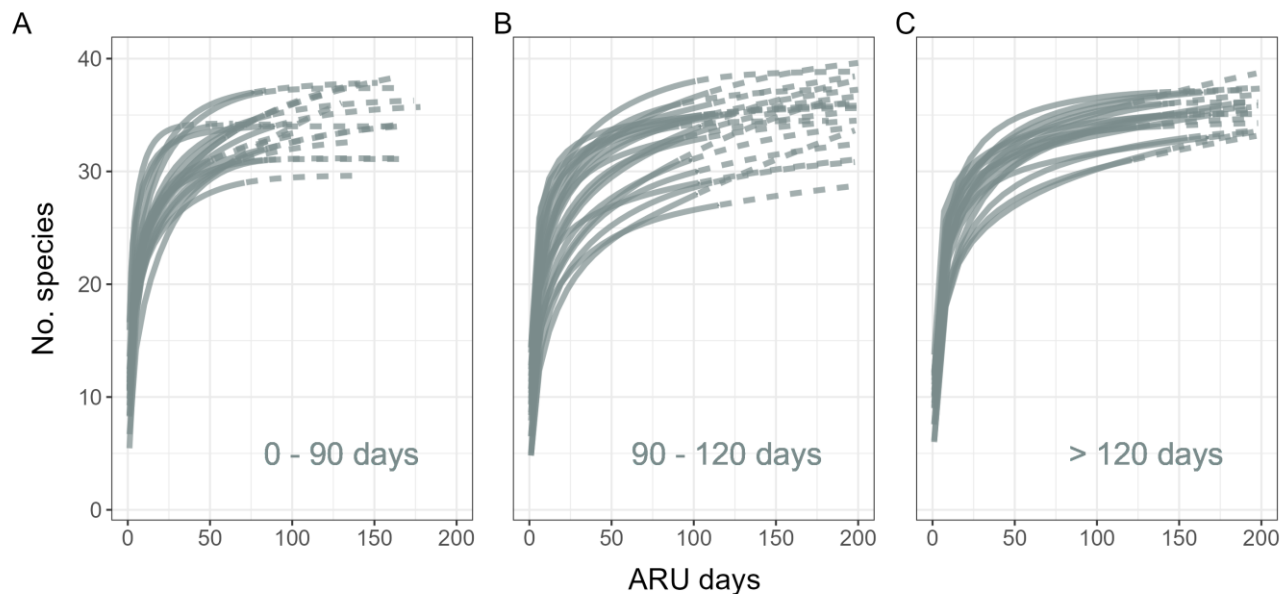


Figure 5-3 Rarefaction curves for sites with different survey efforts (i.e., ARU days): (A) fewer than 90 days, (B) between 90 and 120 days, and (C) greater than 120 days. Each semi-transparent line represents the curve for an individual site. The solid parts of the curve are interpolation of collected data while the dashed parts are extrapolation. If the solid line is already approaching the asymptote, it indicates that current sampling effort is sufficient in surveying richness.

5-4-2 LiDAR covariate pre-selection

Pairwise correlations between the 17 LiDAR-derived covariates revealed several instances of high collinearity ($|r| > 0.8$; **Figure 5-4A**). Specifically, several covariates representing canopy structure (**Figure 5-4A** green highlight) and total vegetation (**Figure 5-4A** yellow highlight) were strongly correlated. To reduce effects of multicollinearity among variables, we retained only the variables that were most interpretable and widely used in ecological studies, reducing the number of covariates from 17 to 10 (**Figure 5-4B**). All retained variables had variance inflation factor (VIF) values below 5, indicating low levels of multicollinearity among variables (**Figure 5-4C**; bolded variables in **Table 5-3**). Because aspect is a circular variable, we further decomposed it into its sine and cosine components, resulting in a total of 11 LiDAR covariates used in the subsequent model selection.

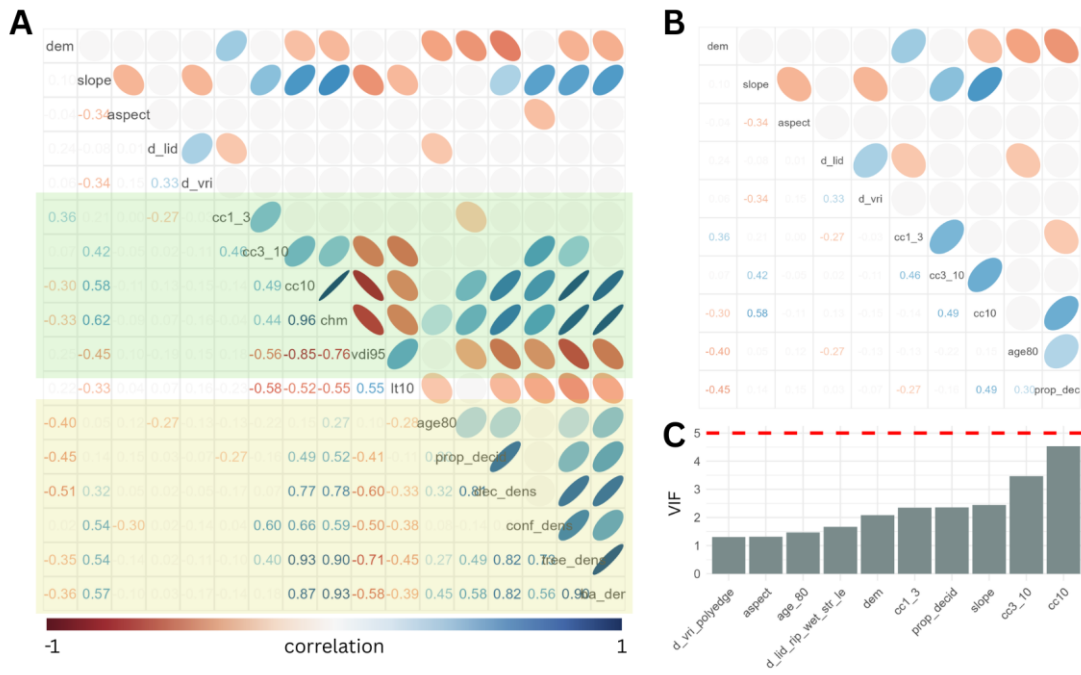


Figure 5-4 Pre-selection of LiDAR covariates. (A) Correlation plot of the original set of 17 covariates. (B) Correlation plot of covariates retained after removing highly correlated variables ($|r| > 0.8$). (C) Final check showing that the variance inflation factors (VIF) for all variables were < 5 .

5-4-3 Asymptotic richness model with LiDAR covariates

Among the 11 pre-selected LiDAR covariates, none had a sum of model weights exceeding 0.8 when predicting richness of all species combined (**Figure 5-5A**). To account for the potential heterogeneity across all species, we grouped species based on their site-level occurrence patterns (**Figure 5-2**) using k-means clustering (**Figure 5-6**). This analysis identified three clusters comprising 13 (purple), 6 (orange), and 21 (green) species, respectively. Species within the same cluster exhibited similar distribution patterns across study sites, suggesting shared habitat associations or frequent co-occurrence.

All pre-selected LiDAR covariates had model weights lower than 0.8 for the purple (**Figure 5-5B**) and green (**Figure 5-5D**) groups. Even the covariates with the highest model weights, average crown closure above 10 m for the purple group and aspect for the green group, showed non-significant relationships with asymptotic richness (**Figure 5-7**). For the orange group, the proportion of deciduous trees (prop_decid) had a model weight greater than 0.8 (**Figure 5-5C**) and showed a positive relationship with asymptotic richness (**Figure 5-7**). However, the association was weak ($p = 0.04$).

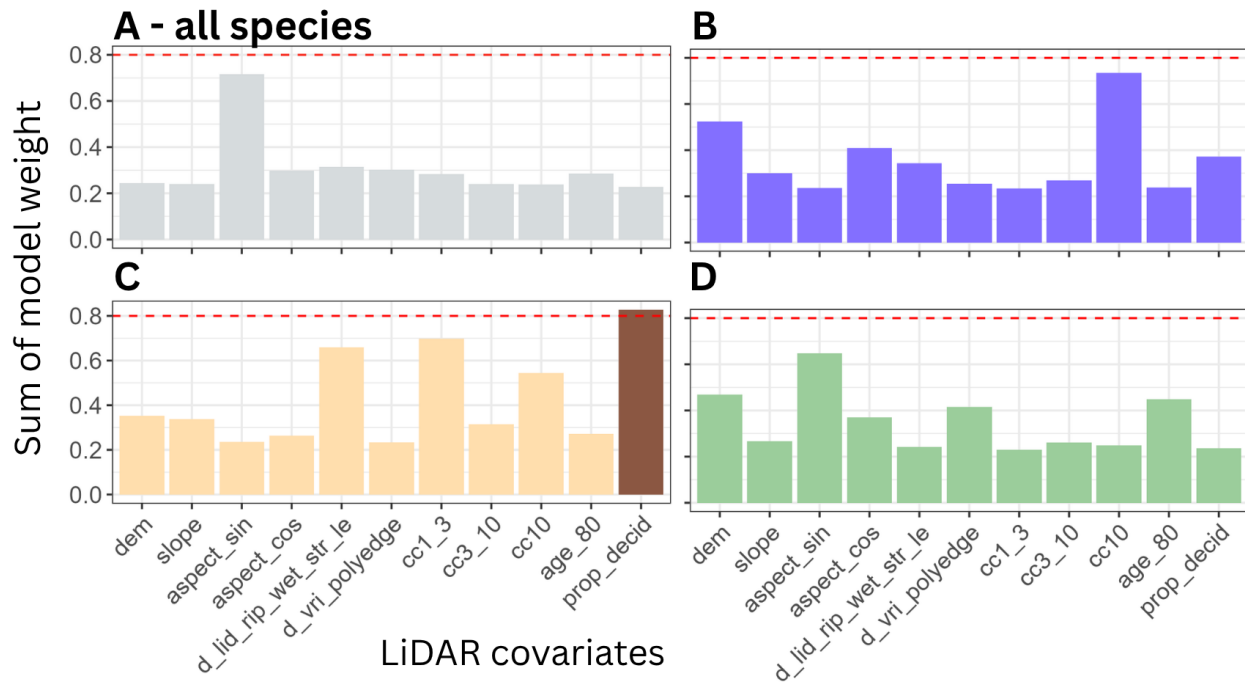


Figure 5-5 Sum of model weights for each LiDAR covariate across different species groups: (A) full species list (40 species of breeding songbirds), (B) purple group (13 *species*), (C) orange group (6 species), and (D) green group (21 species). The red line indicates a sum of model weights of 0.8, a commonly used threshold for identifying influential covariates.

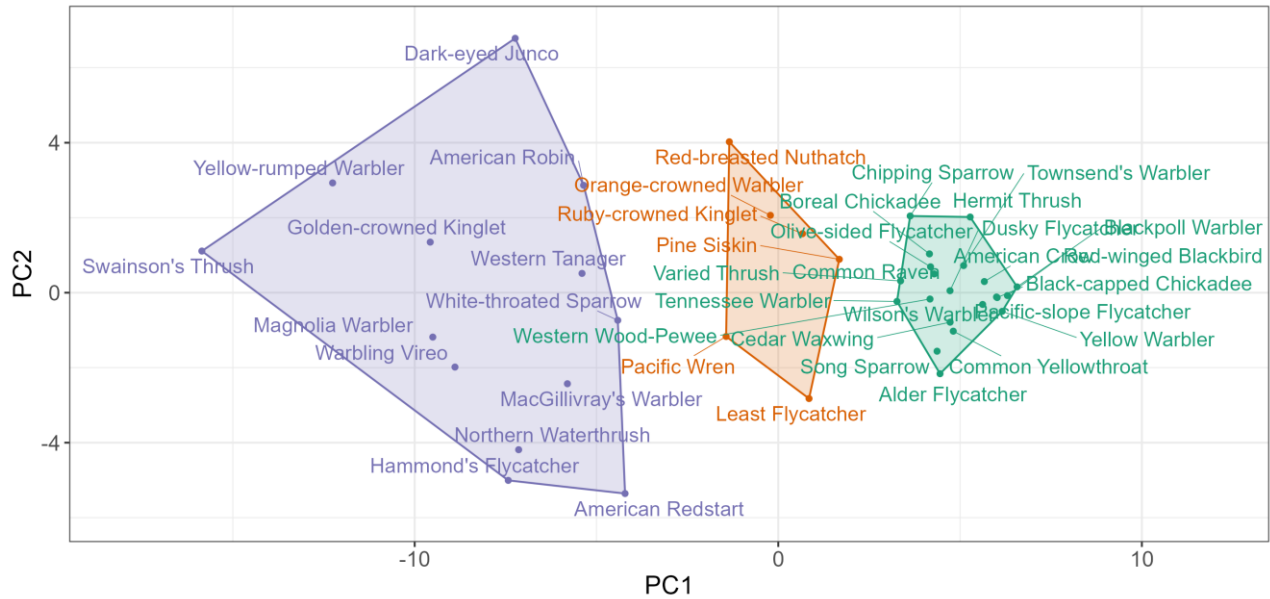


Figure 5-6 K-means clustering of target species based on site-use patterns, visualized in principal component space. Species within the same cluster show similar distributions across study sites. The purple group includes 13 species, the orange group 6 species, and the green group 21 species.

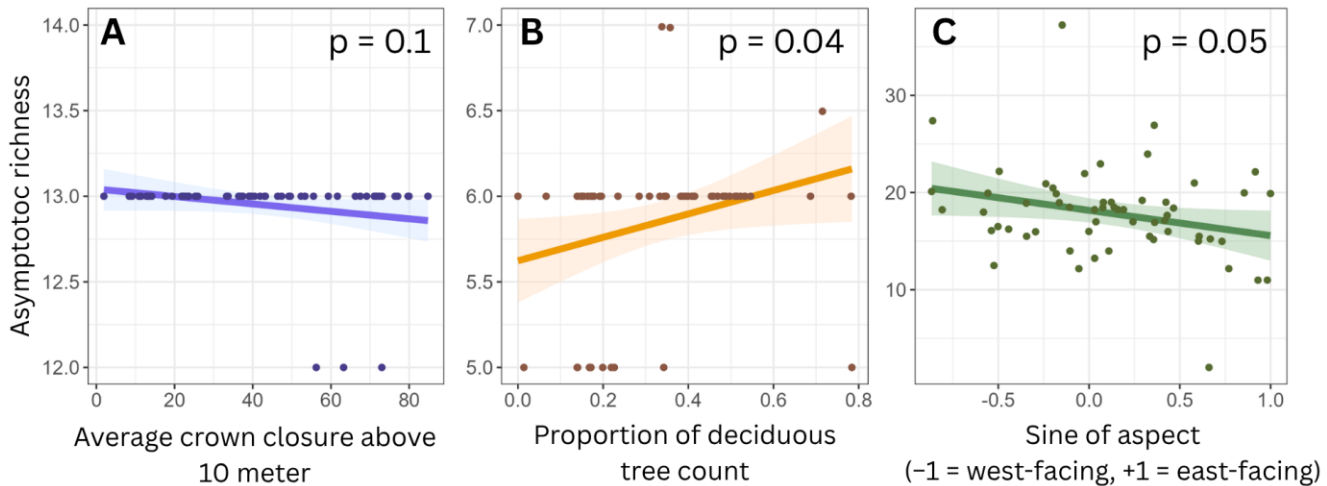


Figure 5-7 Simple linear regression of the most influential LiDAR covariate for each species group in explaining asymptotic richness: (A) purple group – average crown closure above 10 m, (B) orange group – proportion of deciduous trees, and (C) green group – aspect. Note that even the most influential covariates showed weak or no association with richness within each group.

5-5 Discussion

Our results showed that none of the LiDAR covariates effectively predicted songbird asymptotic richness (**Figure 5-5**). The lack of strong covariate importance likely reflects differences in habitat preferences among species, which reduced the overall signal when analyzed together. When species were grouped based on similarities in site-use, only one covariate, the proportion of deciduous trees, showed a weak positive association with richness for a single species group (**Figure 5-7**). Rarefaction analyses indicated that surveying effort across study sites was generally sufficient, as observed richness closely approached estimated asymptotic richness (**Figure 5-3**). This suggests that the number of ARU days at each site provided an adequate sampling of the local breeding songbird richness, which included 40 species in total. Overall, our large-scale acoustic survey provided a general assessment of the breeding bird community within the John Prince Research Forest in central British Columbia (Appendix **Table A5-1**).

Contrary to expectations from previous studies that identified canopy covariates as key drivers of bird richness (Bakx et al. 2019), LiDAR variables related to vertical canopy structure, such as average crown closure at different heights, showed limited explanatory power for asymptotic richness across all species. This discrepancy may reflect the composition of our focal community: the 40 species of dawn-chorus songbirds occupy various habitat types, which may have blurred any clear habitat association in the overall model. When analyzed by species groups, none of the covariates strongly predicted richness, as species richness was relatively similar across sites. Specifically, the 13 species in the purple group (**Figure 5-6**) occurred at nearly all sites (**Figure 5-7A**), and the 6 species in the orange group (**Figure 5-6**) were also widespread across sites (**Figure 5-7B**), resulting in low variation in asymptotic richness across sites. Overall, the weak relationships between asymptotic richness and LiDAR covariates likely

reflect both heterogeneous habitat preferences among target species and the generally uniform species richness across sites.

Expanding analyses to include the full community species list could provide broader ecological insights but presents several practical challenges. First, validating BirdNET detections requires substantial manual effort to establish species-specific thresholds (Tseng et al. 2025; Wood and Kahl 2024). Based on our workflow (see **Methods**), validating 360 song segments per species took approximately 4–8 hours, depending on factors such as vocal repertoire overlap (e.g., Varied Thrush vs. White-throated Sparrow; Alder Flycatcher vs. Hammond’s Flycatcher), frequency characteristics (e.g., the high-pitched songs of Brown Creeper and Golden-crowned Kinglet), and call length (e.g., short warbler calls). Scaling this process to 122 species would require roughly 100 person-days of effort. Second, even when focusing only on breeding songbirds, our results showed that LiDAR covariates were generally non-significant due to heterogeneous habitat preferences among species. Expanding the analysis to the full species list would probably further obscure any trends, requiring more complex models to disentangle covariate associations. A promising alternative for investigating community-level richness and species-specific habitat use is multi-species occupancy modeling (Tingley et al. 2020). Future research comparing our current approach, rarefaction curves integrated with linear models, against multi-species occupancy models (potentially incorporating classifier-guided validation) would offer valuable insights into the most robust framework for deriving richness estimates from machine learning detections.

One potential approach to reduce the validation effort and enable richness analyses across all species is the use of universal thresholds (Tseng et al. 2025), which apply a standardized BirdNET confidence threshold across species and aggregate results over longer temporal

windows. Recent studies suggest that this strategy can yield community-level estimates comparable to expert validation while minimizing manual effort (Fairbairn et al. 2025). However, universal thresholds bypass species-specific error assessment, and their outputs should therefore be interpreted with caution. Without explicit quantification of precision or detection probability, such thresholds may be most appropriate for coarse-resolution analyses, where presence–absence data are aggregated over broad spatial or temporal scales (Fairbairn et al. 2025).

5-6 Conclusion

Our findings showed none of the LiDAR covariate was strongly associated with the breeding songbird richness, resulting from the breeding songbirds in the study area exhibit varied habitat associations, and breeding songbird richness was generally similar across sites. Strategies such as universal thresholds could help facilitate broader community-level analyses, but they must be applied carefully due to the lack of species-specific error assessment. Future research could expand beyond richness to include other biodiversity metrics, such as evenness and abundance, though deriving these metrics from ARU data should be approached with caution. As acoustic monitoring becomes increasingly widespread for large-scale biodiversity assessments, the development of clear guidelines for using ARU-derived data is highly recommended. Such guidelines could include standardized procedures for generating species lists (as in our **Methods**), estimating richness across sites (e.g., using asymptotic richness), and deriving relative abundance for single or multiple species. Implementing these standardized practices will help ensure that acoustic monitoring can reliably inform biodiversity conservation and forest management decisions.

Chapter 6 General Conclusion

6-1 Summary of contributions

Advances in passive acoustic monitoring have transformed the scale at which avian biodiversity can be surveyed, but the rapid growth of audio data has created a major analytical bottleneck (Shonfield and Bayne 2017). In response, BirdNET Analyzer has emerged as one of the most widely adopted platforms for automated bird sound detection and classification since its release in 2021 (Kahl et al. 2021). This thesis advances the application of BirdNET in ecological research by developing practical methods and tools for managing large acoustic datasets, and demonstrating the potential use of these detections generated by BirdNET to address questions in multiple case-studies. I developed a framework for validating BirdNET outputs, which allowed defining species-specific thresholds and thus, derive reliable detections from extensive ARU recordings (Chapter 2). I then used this compiled data to extract acoustic features that allowed for individual identification (Chapter 3), integrate detections into occupancy models, assess signalling patterns using generalized additive models (Chapter 4), and compile BirdNET outputs into rarefaction curve modelling for standardized species richness estimates (Chapter 5).

The next stage of this process was to consolidate these methodological achievements to make them more accessible to other researchers, which I have done in the `birdnetTools` R package (**Figure 6-1**; software submitted to Journal of Open Source Software, currently under review). The `birdnetTools` package supports data import, visualization, and manipulation, to help users validate detection confidence values with precision within their study systems, thereby improving the accessibility, reproducibility, and transparency of BirdNET-based workflows for

the research community. This package is open-access and can be found on Github:

<https://github.com/birdnet-team/birdnetTools>.

birdnetTools::work flow



1. Data Import



Multiple selection tables
stored in **local path**



Single selection table
stored in **local path**



Selection table in R
(i.e., produced by birdnetR)

birdnet_combine()

to combine all selection tables
stored in a local path

readr::read_csv()

to read in a single selection
table into R environment

selection table in R
(a tibble object)

2. Data Exploration

birdnet_filter()

to filter the detections based on
species, threshold, or date/time

birdnet_heatmap()

to view activity pattern, hour by day,
for single species

birdnet_add_datetime()

to auto-detects date/time from file
paths and adds datetime columns

3. Detection Validation

birdnet_subsample()

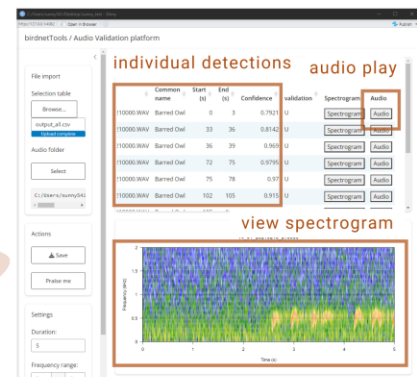
to subsample detections for validation
(random, stratified random, top, etc)

birdnet_launch_validation()

to start a shiny app that assist to view/listen
to the selected audios for validation

birdnet_calc_threshold()

to calculate species-specific thresholds
based on validated detections



birdnet_launch_validation() launches
an R ShinyApp for detections validation

birdnetTools workflow updated 2025 Sep.15th

Figure 6-1 Overview of the workflow implemented in the R package birdnetTools, which consolidates the methods developed in this thesis for processing, validating, and analyzing BirdNET outputs. The package is available on GitHub (<https://github.com/birdnet-team/birdnetTools>).

6-2 Recommendations for the passive acoustic monitoring program

Based on the insights gained from this thesis, I propose two recommendations to enhance the long-term value of the monitoring program at the John Prince Research Forest (JPRF), which may also be applicable to other areas engaged in long-term passive acoustic monitoring.

1. Monitoring schedule: I recommend adopting a 24-hour monitoring schedule, using a 1-minute on / 4-minute off duty cycle for songbird monitoring and a 10-minute on / 20-minute off cycle for owl monitoring. A full diel schedule would reduce biases associated with sunrise and sunset and allow detailed investigation of daily activity patterns. It would also enable finer-scale analyses of species' responses to unexpected extreme events. For example, while the current dataset allowed examination of vocal activity during the late-June 2021 heatwave (Baum et al. 2026), it did not provide enough temporal coverage to assess whether birds shifted their diel activity or how vocal behaviour changed with rapid temperature fluctuations outside the three-hour morning sampling period. Further, extending the monitoring period from late January to late August would ensure coverage of both the owl and songbird breeding seasons, providing a more complete representation of annual vocal dynamics.

2. Monitoring spatial design: I recommend maintaining the current grid of 66 autonomous recording units, which were placed approximately 3 km apart. This placement provides an effective balance between spatial independence and habitat coverage. The spacing is particularly appropriate for owl research, where territories are larger than those of most songbirds. The monitoring effort achieved through this grid appears sufficient for capturing local richness, with observed richness approaching asymptotic estimates. If research objectives extend beyond general richness monitoring, such as examining species–habitat associations or evaluating responses to different forest management practices, alternative sampling designs may

be required. In such cases, deploying ARUs across a broader range of habitat types or management treatments would better support these more targeted ecological questions.

6-3 Future directions

6-3-1 BirdNET post-processing and threshold setting

This thesis focused on species-specific confidence thresholding as a transparent and interpretable approach to post-processing of BirdNET outputs (Chapter 2). However, threshold-based classification is not the only available strategy. Alternative approaches include using confidence scores directly without transforming them into binary detections (Wood and Kahl 2024), avoiding explicit threshold setting altogether (Navine et al. 2024), or applying more complex and adaptive post-processing (Thompson et al. 2025). Given the relatively early stage of BirdNET development, currently within its first five years, methods for interpreting and using its outputs are still actively evolving. In this context, the species-specific thresholding strategy explored in this thesis remains one of the most intuitive and accessible approaches among currently available methods. Nevertheless, continued exploration of alternative post-processing strategies, as well as further validation across different taxa, soundscapes, and research objectives, is strongly encouraged.

6-3-2 Understanding the causes of species-specific BirdNET performance

It is well established that BirdNET performance varies substantially among species (Pérez-Granados 2023). Identifying the underlying causes of this variation could provide important insights for improving model performance. For example, if reduced performance is driven by training data imbalance, increasing the availability of recordings for underrepresented species

may be an effective solution. If performance is linked to song complexity or variability, alternative feature representations or model architectures may be required. In this thesis, performance, defined by the confidence threshold required to achieve a given level of precision, was not found to be related to the vocal characteristics examined (Chapter 2). This might have resulted from not capturing the relevant metrics that drives the performance variation, or that the species included did not encompass sufficient vocal diversity. Future studies should therefore 1) consider incorporating a broader range of acoustic features, 2) focus on larger and more diverse set of species, and 3) utilize multivariate analysis methods, such as principal component analysis, for more detailed examination of performance based on holistic measures of sounds. In addition, performance differences may reflect limitations in the training dataset, particularly for species or vocalization types that are underrepresented. As BirdNET is trained in part on recordings from Xeno-Canto, continued community contributions to this database represent an important pathway for improving model performance. Uploading recordings from diverse regions and species is strongly encouraged and aligns with recommendations from the BirdNET development team (BirdNET Analyzer Online Documentation: <https://birdnet-team.github.io/BirdNET-Analyzer/>).

6-3-3 Expanding BirdNET beyond species-level classification

Beyond species-level identification, there is substantial potential to extend BirdNET toward more detailed acoustic monitoring. Recent developments allow for customized model training based on BirdNET embeddings, opening opportunities to identify biological units beyond species (Kahl et al. 2021). In parallel, this thesis demonstrated that passive acoustic monitoring could provide recordings of sufficient quality for individual owl identification (Chapter 3). Using customized training, future work could potentially combine these two strengths to develop

models capable of identifying individual animals. Such individual-level classification could enable long-term studies within the same study area, supporting research on territory fidelity, individual interactions, social structure, and population dynamics, including the detection of new individuals entering a population. Another promising direction is the identification of song types or call types within a species (McGinn et al. 2023). Although not explored in this thesis, this approach builds on the same principle that acoustic classification can extend beyond species identity. Together, these developments suggest that BirdNET-based workflows have the potential to support increasingly fine-scale biological inference.

6-3-4 Integrating passive acoustic monitoring with ecological modelling

This thesis demonstrated that passive acoustic monitoring, when combined with BirdNET and appropriate statistical methods, can provide ecological information that is difficult to obtain through traditional survey approaches. Generalized additive models were used to characterize diel and seasonal vocal activity patterns (Chapter 4), occupancy models to map spatial distributions and habitat associations (Chapter 4), and rarefaction curves to assess species richness at both site and landscape scales (Chapter 5). Future work should continue to integrate conventional ecological modelling frameworks with acoustic data. Continuous acoustic monitoring offers substantially higher temporal resolution than human-based surveys, enabling more detailed characterization of activity patterns that can inform the feasibility of survey protocols (Singer et al. 2025) and quantify the effects of climate change or urbanization on bird diel and seasonal phenology (Møller et al. 2015). The application of additional modelling approaches represents a promising direction for fully realizing the potential of passive acoustic

monitoring. Continued methodological development in this area will be essential for translating large-scale acoustic datasets into robust ecological inference.

6-3-5 Integrating passive acoustic monitoring with camera traps and other monitoring methods

Although not explored in this thesis, an important and promising future direction is the integration of passive acoustic monitoring with other wildlife monitoring approaches, particularly camera trapping. Previous studies have demonstrated the strong potential of combining these methods, but much of the existing work has focused on comparing their relative performance (e.g., Enari et al. 2019), rather than integrating them to address more complex ecological questions (Buxton et al. 2018). The combination of passive acoustic monitoring and camera trapping offers complementary strengths. Acoustic monitoring is highly effective for detecting vocalizing taxa, such as birds (Sugai et al. 2019), while camera traps are particularly well suited for monitoring non-vocalizing wildlife, especially mammals (Rowcliffe and Carbone 2008). Recent advances in passive monitoring technologies, including autonomous recording units (ARUs) and camera traps (CTs), now enable continuous, low-cost data collection across broad spatial and temporal scales. At the same time, developments in machine learning, such as BirdNET for acoustic data (Kahl et al. 2021) and MegaDetector for camera trap imagery (Beery et al. 2019), have greatly increased the efficiency and scalability of data processing for both data streams. Together, the integration of ARUs and CTs provides an unprecedented opportunity to investigate cross-taxonomic activity patterns and interactions between bird and mammal communities. Such integrated monitoring frameworks could support more comprehensive assessments of community dynamics, habitat use, and species interactions, thereby opening new

avenues for ecological inference. Future work in this area could focus on: (1) integrating acoustic and camera trap data to quantify cross-taxonomic interactions and improve understanding of key ecological processes; (2) developing and validating statistical and modelling approaches for multi-sensor wildlife data; and (3) translating integrated monitoring workflows into generalizable guidance for biodiversity assessment and conservation applications.

References

- Alcocer, Irene, Herlander Lima, Larissa Sayuri Moreira Sugai, and Diego Llusia. 2022. Acoustic Indices as Proxies for Biodiversity: A Meta-Analysis. *Biological Reviews* 97 (6): 2209–36. <https://doi.org/10.1111/brv.12890>.
- Altman, Bob. 1997. Olive-Sided Flycatcher in Western North America: Status Review. *US Fish and Wildlife Service, Oregon State Office*.
- Araya-Salas, Marcelo, and Grace Smith-Vidaurre. 2017. warbleR: An R Package to Streamline Analysis of Animal Acoustic Signals. *Methods in Ecology and Evolution* 8 (2): 184–91. <https://doi.org/10.1111/2041-210X.12624>.
- Araya-Salas, Marcelo, Grace Smith-Vidaurre, Gloriana Chaverri, et al. 2023. Ohun: An R Package for Diagnosing and Optimizing Automatic Sound Event Detection. *Methods in Ecology and Evolution* 14 (9): 2259–71. <https://doi.org/10.1111/2041-210X.14170>.
- Bakx, Tristan R. M., Zsófia Koma, Arie C. Seijmonsbergen, and W. Daniel Kissling. 2019. Use and Categorization of Light Detection and Ranging Vegetation Metrics in Avian Diversity and Species Distribution Research. *Diversity and Distributions* 25 (7): 1045–59. <https://doi.org/10.1111/ddi.12915>.
- Baptista, Luis F., and Sandra L. L. Gaunt. 1994. Advances in Studies of Avian Sound Communication. *The Condor* 96 (3): 817–30. <https://doi.org/10.2307/1369489>.
- Bartoń, Kamil. 2025. MuMIn: Multi-Model Inference. v1.48.11. <https://cran.r-project.org/web/packages/MuMIn/index.html>.
- Baum, Julia K., Margaret A. Slein, Josef C. Garen, et al. 2026. Widespread Ecological Responses and Cascading Effects of the 2021 Western North American Heatwave. *Nature Ecology & Evolution*. <https://doi.org/10.1038/s41559-026-02987-6>
- Beckett, Matthew D., and Gary Ritchison. 2010. Effects of Breeding Stage and Behavioral Context on Singing Behavior of Male Indigo Buntings. *The Wilson Journal of Ornithology* 122 (4): 655–65. <https://doi.org/10.1676/09-204.1>.
- Beery, Sara, Dan Morris, and Siyu Yang. 2019. Efficient Pipeline for Camera Trap Image Review. arXiv:1907.06772. Preprint. <https://doi.org/10.48550/arXiv.1907.06772>.
- Benedict, Lauryn, and Nadjé A Najar. 2019. Are Commonly Used Metrics of Bird Song Complexity Concordant? *The Auk* 136 (1): ukyy008. <https://doi.org/10.1093/auk/ukyy008>.
- Best, Louis B. 1981. Seasonal Changes in Detection of Individual Bird Species. *Studies in Avian Biology* 6: 252–61.

- Bibby, Colin J., Neil D. Burgess, David A. Hill, and Simon Mustoe. 2000. Bird Census Techniques. <https://shop.elsevier.com/books/bird-census-techniques/bibby/978-0-12-095831-3>.
- Billerman, S. M., B. K. Keeney, G. M. Kirwan, F. Medrano, N. D. Sly, and M. G. Smith. 2025. Birds of the World. *Cornell Laboratory of Ornithology*. <https://doi.org/10.2173/bow>.
- BirdLife International. 2025. IUCN Red List of Threatened Species: Contopus Cooperi. *IUCN Red List of Threatened Species*. <https://www.iucnredlist.org/en>.
- BirdNET Team. 2025. BirdNET-Analyzer. <https://github.com/birdnet-team/BirdNET-Analyzer>.
- Bradbury, Richard B., Ross A. Hill, David C. Mason, et al. 2005. Modelling Relationships between Birds and Vegetation Structure Using Airborne LiDAR Data: A Review with Case Studies from Agricultural and Woodland Environments. *Ibis* 147 (3): 443–52. <https://doi.org/10.1111/j.1474-919x.2005.00438.x>.
- Buxton, Rachel T., Patrick E. Lendrum, Kevin R. Crooks, and George Wittemyer. 2018. Pairing Camera Traps and Acoustic Recorders to Monitor the Ecological Impact of Human Disturbance. *Global Ecology and Conservation* 16 (October): e00493. <https://doi.org/10.1016/j.gecco.2018.e00493>.
- Chambert, Thierry, J. Hardin Waddle, David A. W. Miller, Susan C. Walls, and James D. Nichols. 2018. A New Framework for Analysing Automated Acoustic Species Detection Data: Occupancy Estimation and Optimization of Recordings Post-Processing. *Methods in Ecology and Evolution* 9 (3): 560–70. <https://doi.org/10.1111/2041-210X.12910>.
- Chao, Anne, Nicholas J. Gotelli, T. C. Hsieh, et al. 2014. Rarefaction and Extrapolation with Hill Numbers: A Framework for Sampling and Estimation in Species Diversity Studies. *Ecological Monographs* 84 (1): 45–67. <https://doi.org/10.1890/13-0133.1>.
- Chronister, Lauren M., Jeffery T. Larkin, Tessa A. Rhinehart, David King, Jeffery L. Larkin, and Justin Kitzes. 2025. Evaluating the Predictors of Habitat Use and Successful Reproduction in a Model Bird Species Using a Large-Scale Automated Acoustic Array. *Ecography* 2025 (3): e06940. <https://doi.org/10.1111/ecog.06940>.
- Cole, Jerry S., Nicole L. Michel, Shane A. Emerson, and Rodney B. Siegel. 2022. Automated Bird Sound Classifications of Long-Duration Recordings Produce Occupancy Model Outputs Similar to Manually Annotated Data. *Ornithological Applications* 124 (2): duac003. <https://doi.org/10.1093/ornithapp/duac003>.
- Culbert, Patrick D., Volker C. Radeloff, Curtis H. Flather, Josef M. Kellndorfer, Chadwick D. Rittenhouse, and Anna M. Pidgeon. 2013. The Influence of Vertical and Horizontal Habitat Structure on Nationwide Patterns of Avian Biodiversity. *The Auk* 130 (4): 656–65. <https://doi.org/10.1525/auk.2013.13007>.

- Cunningham, R.B., D. B. Lindenmayer, H. A. Nix, and B. D. Lindenmayer. 1999. Quantifying Observer Heterogeneity in Bird Counts. *Australian Journal of Ecology* 24 (3): 270–77. <https://doi.org/10.1046/j.1442-9993.1999.00971.x>.
- Czarnecki, Chapin, Lauren Chronister, Cameron Fiss, Jeffery L. Larkin, and Justin Kitzes. 2024. Uncovering Breeding Habitat Use of an Uncommon Songbird in Pennsylvania Using Large Scale Acoustic Data. *Wildlife Letters* 2 (4): 169–78. <https://doi.org/10.1002/wll2.12052>.
- Davidson, P.J.A., R.J. Cannings, A.R. Couturier, D. Lepage, and C.M. Di Corrado. 2015. The Atlas of the Breeding Birds of British Columbia, 2008-2012. <https://www.birdatlas.bc.ca/>.
- Dawson, Deanna K., and Murray G. Efford. 2009. Bird Population Density Estimated from Acoustic Signals. *Journal of Applied Ecology* 46 (6): 1201–9. <https://doi.org/10.1111/j.1365-2664.2009.01731.x>.
- Delpont, Wayne, Alan C. Kemp, and J. Willem H. Ferguson. 2002. Vocal Identification of Individual African Wood Owls *Strix Woodfordii*: A Technique to Monitor Long-Term Adult Turnover and Residency. *Ibis* 144 (1): 30–39. <https://doi.org/10.1046/j.0019-1019.2001.00019.x>.
- Doser, Jeffrey W., Andrew O. Finley, Marc Kéry, and Elise F. Zipkin. 2022. spOccupancy: An R Package for Single-Species, Multi-Species, and Integrated Spatial Occupancy Models. *Methods in Ecology and Evolution* 13 (8): 1670–78. <https://doi.org/10.1111/2041-210X.13897>.
- Enari, Hiroto, Haruka S. Enari, Kei Okuda, Tetsuya Maruyama, and Kana N. Okuda. 2019. An Evaluation of the Efficiency of Passive Acoustic Monitoring in Detecting Deer and Primates in Comparison with Camera Traps. *Ecological Indicators* 98 (March): 753–62. <https://doi.org/10.1016/j.ecolind.2018.11.062>.
- Environment and Climate Change Canada. 2019. Olive-Sided Flycatcher (*Contopus Cooperi*): COSEWIC Assessment and Status Report 2018. Program descriptions;research. February 25. <https://www.canada.ca/en/environment-climate-change/services/species-risk-public-registry/cosewic-assessments-status-reports/olive-sided-flycatcher-2018.html>.
- Fairbairn, Andrew J., Josija-Simeon Burmeister, Wolfgang W. Weisser, and Sebastian T. Meyer. 2025. BirdNET Can Be as Good as Experts for Acoustic Bird Monitoring in a European City. *PLOS ONE* 20 (9): e0330836. <https://doi.org/10.1371/journal.pone.0330836>.
- Francis, Anthony P., and David J. Currie. 2003. A Globally Consistent Richness-Climate Relationship for Angiosperms. *The American Naturalist* 161 (4): 523–36. <https://doi.org/10.1086/368223>.

- Freeman, Pamela L. 2000. Identification of Individual Barred Owls Using Spectrogram Analysis and Auditory Cues. *Journal of Raptor Research* 34 (2): 85–92.
- Galeotti, P., and G. Pavan. 1991. Individual Recognition of Male Tawny Owls (*Strix Aluco*) Using Spectrograms of Their Territorial Calls. *Ethology Ecology & Evolution* 3 (2): 113–26. <https://doi.org/10.1080/08927014.1991.9525378>.
- Gibb, Rory, Ella Browning, Paul Glover-Kapfer, and Kate E. Jones. 2019. Emerging Opportunities and Challenges for Passive Acoustics in Ecological Assessment and Monitoring. *Methods in Ecology and Evolution* 10 (2): 169–85. <https://doi.org/10.1111/2041-210X.13101>.
- Gilbert, Gillian, Glen A. Tyler, and Ken W. Smith. 2002. Local Annual Survival of Booming Male Great Bittern *Botaurus Stellaris* in Britain, in the Period 1990–1999. *Ibis* 144 (1): 51–61. <https://doi.org/10.1046/j.0019-1019.2001.00012.x>.
- Goldstein, Benjamin R., Alex J. Jensen, Roland Kays, et al. 2024. Guidelines for Estimating Occupancy from Autocorrelated Camera Trap Detections. *Methods in Ecology and Evolution* 15 (7): 1177–91. <https://doi.org/10.1111/2041-210X.14359>.
- Gordo, Oscar. 2007. Why Are Bird Migration Dates Shifting? A Review of Weather and Climate Effects on Avian Migratory Phenology. *Climate Research* 35 (1/2): 37–58.
- Gottschalk, T. K., F. Huettmann, and M. Ehlers. 2005. Review Article: Thirty Years of Analysing and Modelling Avian Habitat Relationships Using Satellite Imagery Data: A Review. *International Journal of Remote Sensing* 26 (12): 2631–56. <https://doi.org/10.1080/01431160512331338041>.
- Grava, Thibault, Nicolas Mathevon, Emelyne Place, and Patrick Balluet. 2008. Individual Acoustic Monitoring of the European Eagle Owl *Bubo Bubo*. *Ibis* 150 (2): 279–87. <https://doi.org/10.1111/j.1474-919X.2007.00776.x>.
- Hartwig, Simone. 2005. Individual Acoustic Identification as a Non-Invasive Conservation Tool: An Approach to the Conservation of the African Wild Dog *Lycaon Pictus* (Temminck, 1820). *Bioacoustics* 15 (1): 35–50. <https://doi.org/10.1080/09524622.2005.9753537>.
- Hawkins, Bradford A., Richard Field, Howard V. Cornell, et al. 2003. Energy, Water, and Broad-Scale Geographic Patterns of Species Richness. *Ecology* 84 (12): 3105–17. <https://doi.org/10.1890/03-8006>.
- Hill, Andrew P., Peter Prince, Evelyn Piña Covarrubias, C. Patrick Doncaster, Jake L. Snaddon, and Alex Rogers. 2018. AudioMoth: Evaluation of a Smart Open Acoustic Device for Monitoring Biodiversity and the Environment. *Methods in Ecology and Evolution* 9 (5): 1199–211. <https://doi.org/10.1111/2041-210X.12955>.
- Hill, Andrew P., Peter Prince, Jake L. Snaddon, C. Patrick Doncaster, and Alex Rogers. 2019. AudioMoth: A Low-Cost Acoustic Device for Monitoring Biodiversity and the

- Environment. *HardwareX* 6 (October): e00073.
<https://doi.org/10.1016/j.ohx.2019.e00073>.
- Hill, Samuel D., Dianne H. Brunton, Michael G. Anderson, and Weihong Ji. 2018. Fighting Talk: Complex Song Elicits More Aggressive Responses in a Vocally Complex Songbird. *Ibis* 160 (2): 257–68. <https://doi.org/10.1111/ibi.12542>.
- Holschuh, Carmen I, and Ken A Otter. 2005. Using Vocal Individuality to Monitor Queen Charlotte Saw-whet Owls (*aegolius Acadicus Brooksi*). *Journal of Raptor Research* 39 (2): 134–41.
- Hsieh, T. C., K. H. Ma, and Anne Chao. 2016. iNEXT: An R Package for Rarefaction and Extrapolation of Species Diversity (Hill Numbers). *Methods in Ecology and Evolution* 7 (12): 1451–56. <https://doi.org/10.1111/2041-210X.12613>.
- Incze, Ágnes, Henrietta-Bernadett Jancsó, Zoltán Szilágyi, Attila Farkas, and Csaba Sulyok. 2018. Bird Sound Recognition Using a Convolutional Neural Network. *2018 IEEE 16th International Symposium on Intelligent Systems and Informatics (SISY)*, September, 000295–000300. <https://doi.org/10.1109/SISY.2018.8524677>.
- Jackson, Rodney M., Jerry D. Roe, Rinchen Wangchuk, and Don O. Hunter. 2006. Estimating Snow Leopard Population Abundance Using Photography and Capture-Recapture Techniques. *Wildlife Society Bulletin* 34 (3): 772–81. [https://doi.org/10.2193/0091-7648\(2006\)34%255B772:ESLPAU%255D2.0.CO;2](https://doi.org/10.2193/0091-7648(2006)34%255B772:ESLPAU%255D2.0.CO;2).
- Janik, Vincent M. 2009. Chapter 4 Acoustic Communication in Delphinids. In *Advances in the Study of Behavior*, vol. 40. *Academic Press*. [https://doi.org/10.1016/S0065-3454\(09\)40004-4](https://doi.org/10.1016/S0065-3454(09)40004-4).
- K. Lisa Yang Center for Conservation Bioacoustics at the Cornell Lab of Ornithology. 2023. Raven Lite: Interactive Sound Analysis Software. v2.0.4. *The Cornell Lab of Ornithology*. <https://ravensoundsoftware.com/>.
- Kahl, Stefan, Tom Denton, Holger Klinck, et al. 2023. Overview of BirdCLEF 2023: Automated Bird Species Identification in Eastern Africa 4.0 International (CC BY 4.0).
- Kahl, Stefan, Connor M. Wood, Maximilian Eibl, and Holger Klinck. 2021. BirdNET: A Deep Learning Solution for Avian Diversity Monitoring. *Ecological Informatics* 61 (March): 101236. <https://doi.org/10.1016/j.ecoinf.2021.101236>.
- Karanth, K. Ullas. 1995. Estimating Tiger Panthera Tigris Populations from Camera-Trap Data Using Capture—Recapture Models. *Biological Conservation* 71 (3): 333–38. [https://doi.org/10.1016/0006-3207\(94\)00057-W](https://doi.org/10.1016/0006-3207(94)00057-W).
- Kassambara, Alboukadel. 2017. Practical Guide to Principal Component Methods in R: PCA, M (CA), FAMD, MFA, HCPC, Factoextra. v2. *Statistical tools for high-throughput data analysis*.

- Knight, Elly, Tessa Rhinehart, Devin R. de Zwaan, et al. 2024. Individual Identification in Acoustic Recordings. *Trends in Ecology & Evolution* 39 (10): 947–60. <https://doi.org/10.1016/j.tree.2024.05.007>.
- Koh, Chih-Yuan, Jaw-Yuan Chang, Chiang-Lin Tai, Da-Yo Huang, Han-Hsing Hsieh, and Yi-Wen Liu. 2019. Bird Sound Classification Using Convolutional Neural Networks. *CLEF 2019 Working Notes*.
- Kojima, Ryosuke, Osamu Sugiyama, Kotaro Hoshiba, Reiji Suzuki, and Kazuhiro Nakadai. 2018. HARK-Bird-Box: A Portable Real-Time Bird Song Scene Analysis System. *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, October, 2497–502. <https://doi.org/10.1109/IROS.2018.8594070>.
- Küçüktopcu, Okan, Engin Masazade, Cem Ünsalan, and Pramod K. Varshney. 2019. A Real-Time Bird Sound Recognition System Using a Low-Cost Microcontroller. *Applied Acoustics* 148 (May): 194–201. <https://doi.org/10.1016/j.apacoust.2018.12.028>.
- Langtimm, Catherine A., Cathy A. Beck, Holly H. Edwards, et al. 2004. Survival Estimates for Florida Manatees from the Photo-Identification of Individuals. *Marine Mammal Science* 20 (3): 438–63. <https://doi.org/10.1111/j.1748-7692.2004.tb01171.x>.
- Larkin, Jeffery T., Darin J. McNeil, Lauren Chronister, et al. 2024. A Large-Scale Assessment of Eastern Whip-Poor-Will (*Antrostomus Vociferus*) Occupancy across a Gradient of Forest Management Intensity Using Autonomous Recording Units. *Journal of Environmental Management* 366 (August): 121786. <https://doi.org/10.1016/j.jenvman.2024.121786>.
- Lemenkova, Polina. 2019. K-Means Clustering in R Libraries {cluster} and {factoextra} for Grouping Oceanographic Data. *International Journal of Informatics and Applied Mathematics* 2 (1): 1–26.
- Maes, Dirk, Dirk Bauwens, Luc De Bruyn, et al. 2005. Species Richness Coincidence: Conservation Strategies Based on Predictive Modelling. *Biodiversity & Conservation* 14 (6): 1345–64. <https://doi.org/10.1007/s10531-004-9662-x>.
- Massimino, Dario, Sarah J. Harris, and Simon Gillings. 2021. Phenological Mismatch between Breeding Birds and Their Surveyors and Implications for Estimating Population Trends. *Journal of Ornithology* 162 (1): 143–54. <https://doi.org/10.1007/s10336-020-01821-5>.
- Mattmüller, Ramona M., Karolin Thomisch, Ilse Van Opzeeland, Kristin L. Laidre, and Malene Simon. 2022. Passive Acoustic Monitoring Reveals Year-Round Marine Mammal Community Composition off Tasiilaq, Southeast Greenland. *The Journal of the Acoustical Society of America* 151 (2): 1380–92. <https://doi.org/10.1121/10.0009429>.
- Mazur, K. M., and P. C. James. 2021. Barred Owl (*Strix Varia*). In *Birds of the World*, edited by A. F. Poole and F. B. Gill. *Cornell Lab of Ornithology, Ithaca, NY, USA*.

- McGinn, Kate, Stefan Kahl, M. Zachariah Peery, Holger Klinck, and Connor M. Wood. 2023. Feature Embeddings from the BirdNET Algorithm Provide Insights into Avian Ecology. *Ecological Informatics* 74 (May): 101995. <https://doi.org/10.1016/j.ecoinf.2023.101995>.
- McGregor, Peter, and Tom Peake. 1998. The Role of Individual Identification in Conservation Biology. In *Behavioral Ecology and Conservation Biology*, edited by Tim Caro. *Oxford University Press*.
- Møller, Anders Pape, Mario Díaz, Tomáš Grim, et al. 2015. Effects of Urbanization on Bird Phenology: A Continental Study of Paired Urban and Rural Populations. *Climate Research* 66 (December): 185–99. <https://doi.org/10.3354/cr01344>.
- Monteiro, N. M., R. M. Silva, M. Cunha, A. Antunes, A. G. Jones, and M. N. Vieira. 2014. Validating the Use of Colouration Patterns for Individual Recognition in the Worm Pipefish Using a Novel Set of Microsatellite Markers. *Molecular Ecology Resources* 14 (1): 150–56. <https://doi.org/10.1111/1755-0998.12151>.
- Nagy, Christopher M., and Robert F. Rockwell. 2012. Identification of Individual Eastern Screech-Owls *Megascops asio* via Vocalization Analysis. *Bioacoustics* 21 (2): 127–40. <https://doi.org/10.1080/09524622.2011.651829>.
- Navine, Amanda K., Tom Denton, Matthew J. Weldy, and Patrick J. Hart. 2024. All Thresholds Barred: Direct Estimation of Call Density in Bioacoustic Data. *Frontiers in Bird Science* 3 (April). <https://doi.org/10.3389/fbirs.2024.1380636>.
- Odom, Karan J., Jonathan C. Slaght, and R.J. Gutiérrez. 2013. Distinctiveness in the Territorial Calls of Great Horned Owls within and among Years. *Journal of Raptor Research* 47 (1): 21–30. <https://doi.org/10.3356/JRR-12-11.1>.
- Opaev, Alexey. 2016. Relationships between Repertoire Size and Organization of Song Bouts in the Grey-Crowned Warbler (*Seicercus tephrocephalus*). *Journal of Ornithology* 157 (4): 949–60. <https://doi.org/10.1007/s10336-016-1342-6>.
- Otter, Ken. 1996. Individual Variation in the Advertising Call of Male Northern Saw-Whet Owls. *Journal of Field Ornithology* 67 (3): 398–405.
- Pérez-Granados, Cristian. 2023. BirdNET: Applications, Performance, Pitfalls and Future Opportunities. *Ibis* 165 (3): 1068–75. <https://doi.org/10.1111/ibi.13193>.
- Pérez-Granados, Cristian. 2025. Birdnet Confidence Scores Decrease with Bird Distance from the Recorder: Revisiting Pérez-Granados (2023). *Ardeola* 72 (2): 149–59. <https://doi.org/10.13157/arla.72.2.2025.f01>.
- Pérez-Granados, Cristian, Karl-L. Schuchmann, and Marinez I. Marques. 2021. Vocal Activity of the Ferruginous Pygmy-Owl (*Glaucidium brasilianum*) Is Strongly Correlated with Moon Phase and Nocturnal Temperature. *Ethology Ecology & Evolution* 33 (1): 62–72. <https://doi.org/10.1080/03949370.2020.1820582>.

- Pérez-Granados, Cristian, and Juan Traba. 2021. Estimating Bird Density Using Passive Acoustic Monitoring: A Review of Methods and Suggestions for Further Research. *Ibis* 163 (3): 765–83. <https://doi.org/10.1111/ibi.12944>.
- Picciulin, Marta, Loïc Kéver, Eric Parmentier, and Marta Bolgan. 2019. Listening to the Unseen: Passive Acoustic Monitoring Reveals the Presence of a Cryptic Fish Species. *Aquatic Conservation: Marine and Freshwater Ecosystems* 29 (2): 202–10. <https://doi.org/10.1002/aqc.2973>.
- Pollock, Laura J., Justin Kitzes, Sara Beery, et al. 2025. Harnessing Artificial Intelligence to Fill Global Shortfalls in Biodiversity Knowledge. *Nature Reviews Biodiversity* 1 (3): 166–82. <https://doi.org/10.1038/s44358-025-00022-3>.
- Rahbek, Carsten, Nicholas J Gotelli, Robert K Colwell, Gary L Entsminger, Thiago Fernando L.V.B Rangel, and Gary R Graves. 2006. Predicting Continental-Scale Patterns of Bird Species Richness with Spatially Explicit Models. *Proceedings of the Royal Society B: Biological Sciences* 274 (1607): 165–74. <https://doi.org/10.1098/rspb.2006.3700>.
- Robertson, Bruce A., and Richard L. Hutto. 2007. Is Selectively Harvested Forest an Ecological Trap for Olive-Sided Flycatchers? *The Condor: Ornithological Applications* 109 (1): 109–21. <https://doi.org/10.1093/condor/109.1.109>.
- Rognan, Cameron B., Joseph M. Szewczak, and Michael L. Morrison. 2009. Vocal Individuality of Great Gray Owls in the Sierra Nevada. *The Journal of Wildlife Management* 73 (5): 755–60. <https://doi.org/10.2193/2008-124>.
- Ross, Samuel R. P.-J., Darren P. O’Connell, Jessica L. Deichmann, et al. 2023. Passive Acoustic Monitoring Provides a Fresh Perspective on Fundamental Ecological Questions. *Functional Ecology* 37 (4): 959–75. <https://doi.org/10.1111/1365-2435.14275>.
- Rowcliffe, Marcus, and Chris Carbone. 2008. Surveys Using Camera Traps: Are We Looking to a Brighter Future? *Animal Conservation* 11 (June): 185–86. <https://doi.org/10.1111/j.1469-1795.2008.00180.x>.
- Sadhukhan, Sougata, Holly Root-Gutteridge, and Bilal Habib. 2021. Identifying Unknown Indian Wolves by Their Distinctive Howls: Its Potential as a Non-Invasive Survey Method. *Scientific Reports* 11 (1): 1. <https://doi.org/10.1038/s41598-021-86718-w>.
- Santos, Ednei B. dos, Paulo E. Llambías, and Drew Rendall. 2018. Male Song Diversity and Its Relation to Breeding Success in Southern House Wrens Troglodytes Aedon Chilensis. *Journal of Avian Biology* 49 (6). <https://doi.org/10.1111/jav.01606>.
- Shonfield, Julia, and Erin Bayne. 2017. Avian Conservation and Ecology: Autonomous Recording Units in Avian Ecological Research: Current Use and Future Applications. *Avian Conservation and Ecology* 12 (1). <https://doi.org/10.5751/ACE-00974-120114>.

- Singer, David, Johannes Kamp, Hermann Hondong, Andreas Schuldt, and Jonas Hagge. 2025. Diel and Seasonal Vocal Activity Patterns Revealed by Passive Acoustic Monitoring Suggest Expert Recommendations for Breeding Bird Surveys Need Adjustment. *Journal of Ornithology*, ahead of print, July 9. <https://doi.org/10.1007/s10336-025-02307-y>.
- Stowell, Dan. 2022. Computational Bioacoustics with Deep Learning: A Review and Roadmap. *PeerJ* 10 (March): e13152. <https://doi.org/10.7717/peerj.13152>.
- Stowell, Dan, Michael D. Wood, Hanna Pamuła, Yannis Stylianou, and Hervé Glotin. 2019. Automatic Acoustic Detection of Birds through Deep Learning: The First Bird Audio Detection Challenge. *Methods in Ecology and Evolution* 10 (3): 368–80. <https://doi.org/10.1111/2041-210X.13103>.
- Sueur, Jérôme, Almo Farina, Amandine Gasc, Nadia Pieretti, and Sandrine Pavoine. 2014. Acoustic Indices for Biodiversity Assessment and Landscape Investigation. *Acta Acustica United with Acustica* 100 (August). <https://doi.org/10.3813/AAA.918757>.
- Sugai, Larissa Sayuri Moreira, Thiago Sanna Freire Silva, José Wagner Ribeiro Jr, and Diego Llusia. 2019. Terrestrial Passive Acoustic Monitoring: Review and Perspectives. *BioScience* 69 (1): 15–25. <https://doi.org/10.1093/biosci/biy147>.
- Swanepoel, D. G. J. 1996. Identification of the Nile Crocodile *Crocodylus Niloticus* by the Use of Natural Tail Marks. *Koedoe* 39 (1): 1. <https://doi.org/10.4102/koedoe.v39i1.287>.
- Takagi, Masaaki. 2020. Vocalizations of the Ryukyu Scops Owl *Otus Elegans*: Individually Recognizable and Stable. *Bioacoustics* 29 (1): 28–44. <https://doi.org/10.1080/09524622.2018.1539925>.
- Tattoni, Clara, Franco Rizzolli, and Paolo Pedrini. 2012. Can LiDAR Data Improve Bird Habitat Suitability Models? *Ecological Modelling*, 7th European Conference on Ecological Modelling (ECEM), vol. 245 (October): 103–10. <https://doi.org/10.1016/j.ecolmodel.2012.03.020>.
- Thompson, Michael C., Mark J. Ducey, John S. Gunn, and Rebecca J. Rowe. 2025. A Post-Processing Framework for Assessing BirdNET Identification Accuracy and Community Composition. *Ibis* 167 (2): 530–42. <https://doi.org/10.1111/ibi.13357>.
- Tingley, Morgan W., Christopher P. Nadeau, Manette E. Sandor. 2020. Multi-species occupancy models as robust estimators of community richness. *Methods in Ecology and Evolution* 11:633–642. <https://doi.org/10.1111/2041-210X.13378>.
- Trapanotto, Martino, Loris Nanni, Sheryl Brahnham, and Xiang Guo. 2022. Convolutional Neural Networks for the Identification of African Lions from Individual Vocalizations. *Journal of Imaging* 8 (4): 4. <https://doi.org/10.3390/jimaging8040096>.

- Tripp, T.M., and K.A. Otter. 2006. Vocal Individuality as a Potential Long-Term Monitoring Tool for Western Screech-Owls, *Megascops Kennicottii*. *Canadian Journal of Zoology* 84 (5): 744–53. <https://doi.org/10.1139/z06-055>.
- Tseng, Sunny, Dexter P. Hodder, and Ken A. Otter. 2025. Setting BirdNET Confidence Thresholds: Species-Specific vs. Universal Approaches. *Journal of Ornithology* 166 (4): 1123–35. <https://doi.org/10.1007/s10336-025-02260-w>.
- Tseng, Yi-Chin, Bianca N. I. Eskelson, Kathy Martin, and Valerie LeMay. 2021. Automatic Bird Sound Detection: Logistic Regression Based Acoustic Occupancy Model. *Bioacoustics* 30 (3): 324–40. <https://doi.org/10.1080/09524622.2020.1730241>.
- Venables, W. N., and B. D. Ripley. 2002. Modern Applied Statistics with S. v4. *Springer*. <https://www.stats.ox.ac.uk/pub/MASS4/>.
- Verner, Jared. 1980. Bird Communities of Mixed-Conifer Forests of the Sierra Nevada. *USDA Forest Service*, Salt Lake City, Utah.
- Vierling, Kerri T, Lee A Vierling, William A Gould, Sebastian Martinuzzi, and Rick M Clawges. 2008. Lidar: Shedding New Light on Habitat Characterization and Modeling. *Frontiers in Ecology and the Environment* 6 (2): 90–98. <https://doi.org/10.1890/070001>.
- Ware, Lena, C. Lisa Mahon, Logan McLeod, and Jean-François Jetté. 2023. Artificial Intelligence (BirdNET) Supplements Manual Methods to Maximize Bird Species Richness from Acoustic Data Sets Generated from Regional Monitoring. *Canadian Journal of Zoology* 101 (12): 1031–51. <https://doi.org/10.1139/cjz-2023-0044>.
- Weihls, Claus, Uwe Ligges, Karsten Luebke, and Nils Raabe. 2005. klaR Analyzing German Business Cycles. In *Data Analysis and Decision Support*, edited by D. Baier, R. Decker, and L. Schmidt-Thieme. Springer-Verlag.
- Weiss, Michael, Henrike Hultsch, Iris Adam, Constance Scharff, and Silke Kipper. 2014. The Use of Network Analysis to Study Complex Animal Communication Systems: A Study on Nightingale Song. *Proceedings. Biological Sciences* 281 (1785): 20140460. <https://doi.org/10.1098/rspb.2014.0460>.
- Wheelhouse, Lauren M, Dexter P Hodder, and Ken A Otter. 2022. The Retention of Non-Commercial Hardwoods in Mixed Stands Maintains Higher Avian Biodiversity than Clear-Cutting. *Forestry: An International Journal of Forest Research* 95 (4): 572–81. <https://doi.org/10.1093/forestry/cpac012>.
- Wikipedia*. 2024. Biogeoclimatic zones of British Columbia. https://en.wikipedia.org/w/index.php?title=Biogeoclimatic_zones_of_British_Columbia&oldid=1211770511.

- Wood, Connor M., and Stefan Kahl. 2024. Guidelines for Appropriate Use of BirdNET Scores and Other Detector Outputs. *Journal of Ornithology* 165 (3): 777–82. <https://doi.org/10.1007/s10336-024-02144-5>.
- Wood, Simon N. 2001. mgcv: GAMs and Generalized Ridge Regression for R. *R News*.
- Wright, John M. 1997. Preliminary Study of Olive-Sided Flycatchers, July 1994-April 1997. Final Research Report SE-3-3, 4, and 5. *Alaska Department of Fish and Game Division of Wildlife Conservation*.
- Xanthopoulos, Petros, Panos M. Pardalos, and Theodore B. Trafalis. 2013. Linear Discriminant Analysis. In *Robust Data Mining*, edited by Petros Xanthopoulos, Panos M. Pardalos, and Theodore B. Trafalis. SpringerBriefs in Optimization. *Springer*. https://doi.org/10.1007/978-1-4419-9878-1_4.
- Yee, Siew Ann, Chong Leong Puan, Phooi Kuan Chang, and Badrul Azhar. 2016. Vocal Individuality of Sunda Scops-Owl (*Otus Lempiji*) in Peninsular Malaysia. *Journal of Raptor Research* 50 (4): 379–90. <https://doi.org/10.3356/JRR-15-76.1>.
- Yip, Daniel A., Lionel Leston, Erin M. Bayne, Péter Sólymos, and Alison Grover. 2017. Experimentally Derived Detection Distances from Audio Recordings and Human Observers Enable Integrated Analysis of Point Count Data. *Avian Conservation and Ecology* 12 (1): art11. <https://doi.org/10.5751/ACE-00997-120111>.
- Zhou, Bing, Can-Wei Xia, Ze-Ru Chen, and Wen-Hong Deng. 2020. Individual Identification of Male Ural Owls Based On Territorial Calls. *Journal of Raptor Research* 54 (1): 57–65. <https://doi.org/10.3356/0892-1016-54.1.57>.
- Zuur, Alain F., Elena N. Ieno, Neil Walker, Anatoly A. Saveliev, and Graham M. Smith. 2009. Mixed Effects Models and Extensions in Ecology with R. Statistics for Biology and Health. *Springer*. <https://doi.org/10.1007/978-0-387-87458-6>.

Appendix

Appendix for Chapter 2

Online repository. R code and example datasets for this chapter are available at:

<https://link.springer.com/article/10.1007/s10336-025-02260-w>

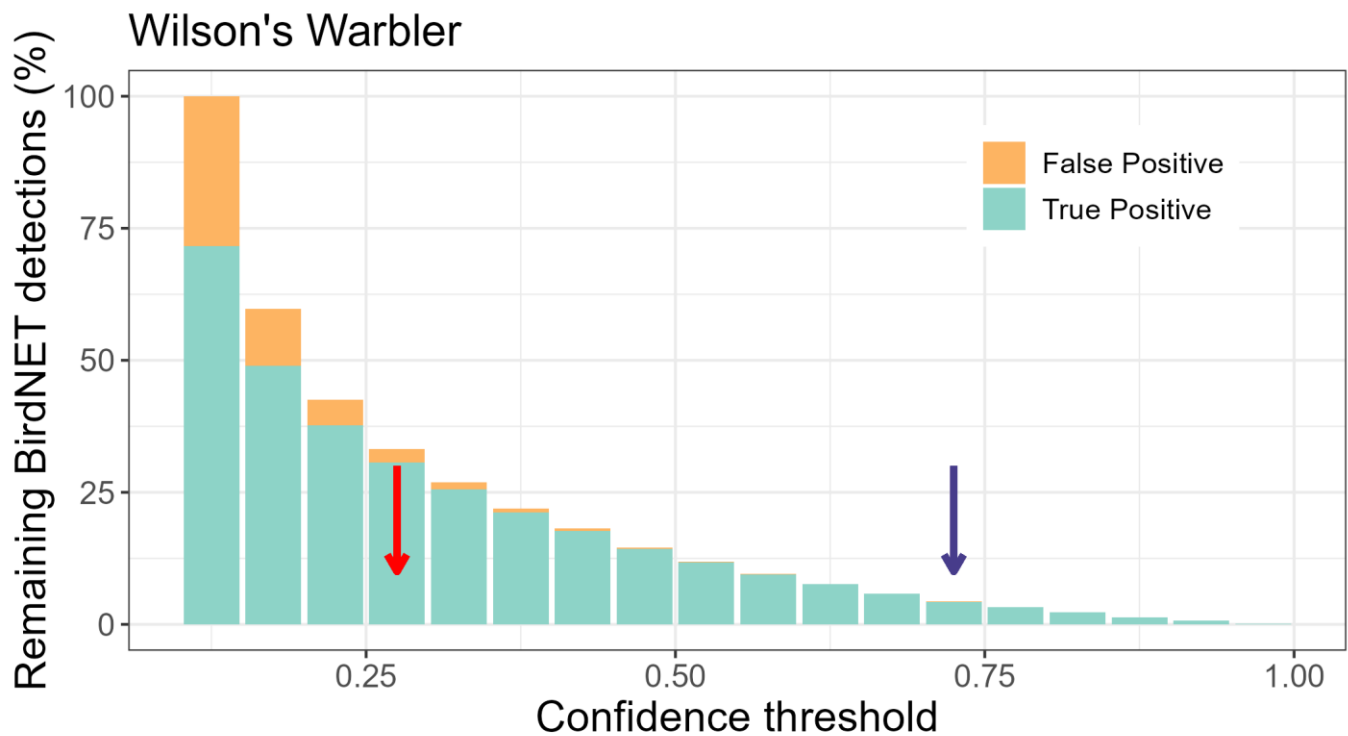


Fig. A2-1 The number of detections identified by BirdNET after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment, and categorized these detections as true positives (green) or false positives (orange), for the target species in this study. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9.

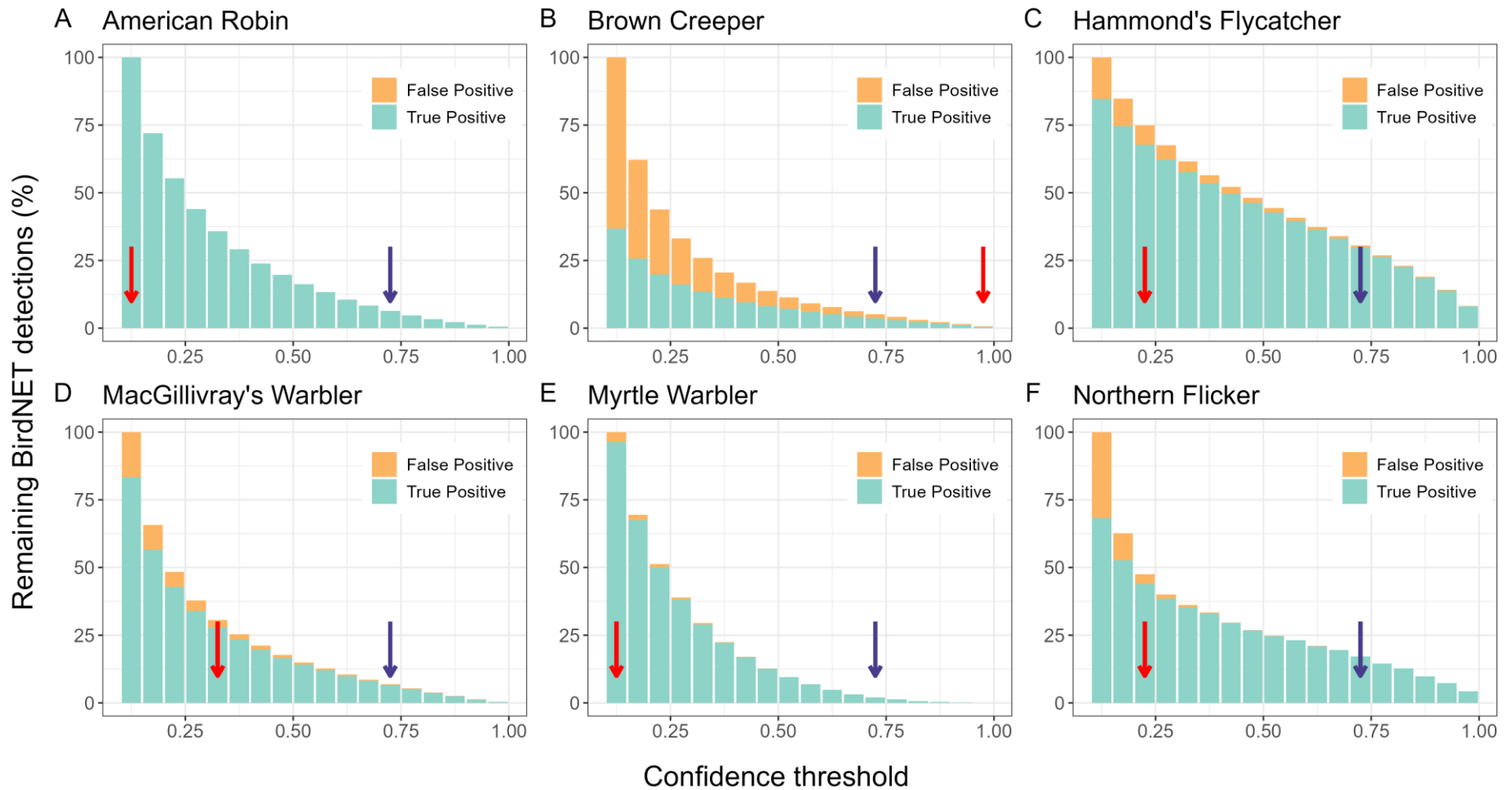


Fig. A2-2 The number of detections identified by BirdNET after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment, and categorized these detections as true positives (green) or false positives (orange), for the target species in this study. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9.

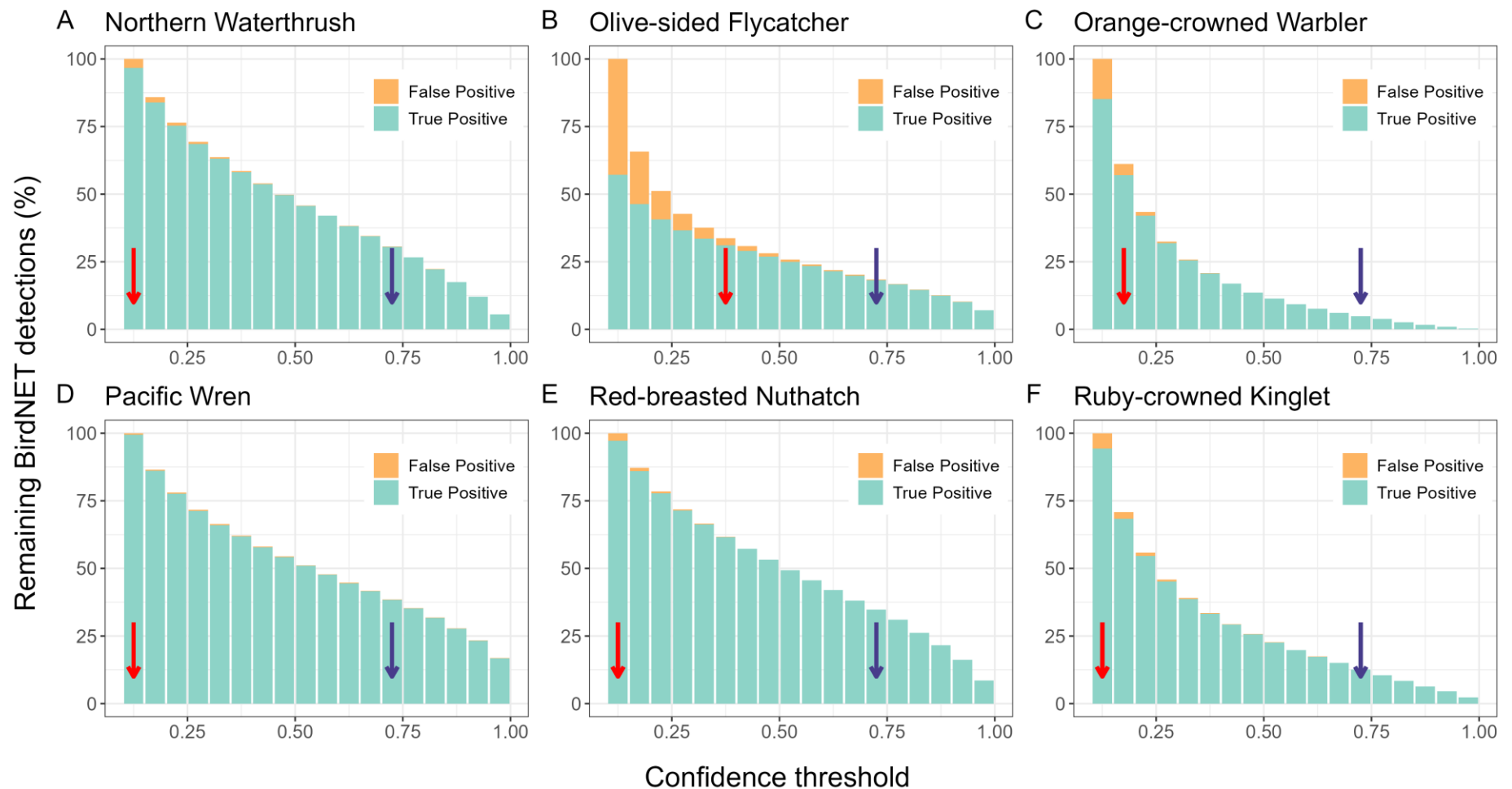


Fig. A2-3 The number of detections identified by BirdNET after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment, and categorized these detections as true positives (green) or false positives (orange), for the target species in this study. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9.

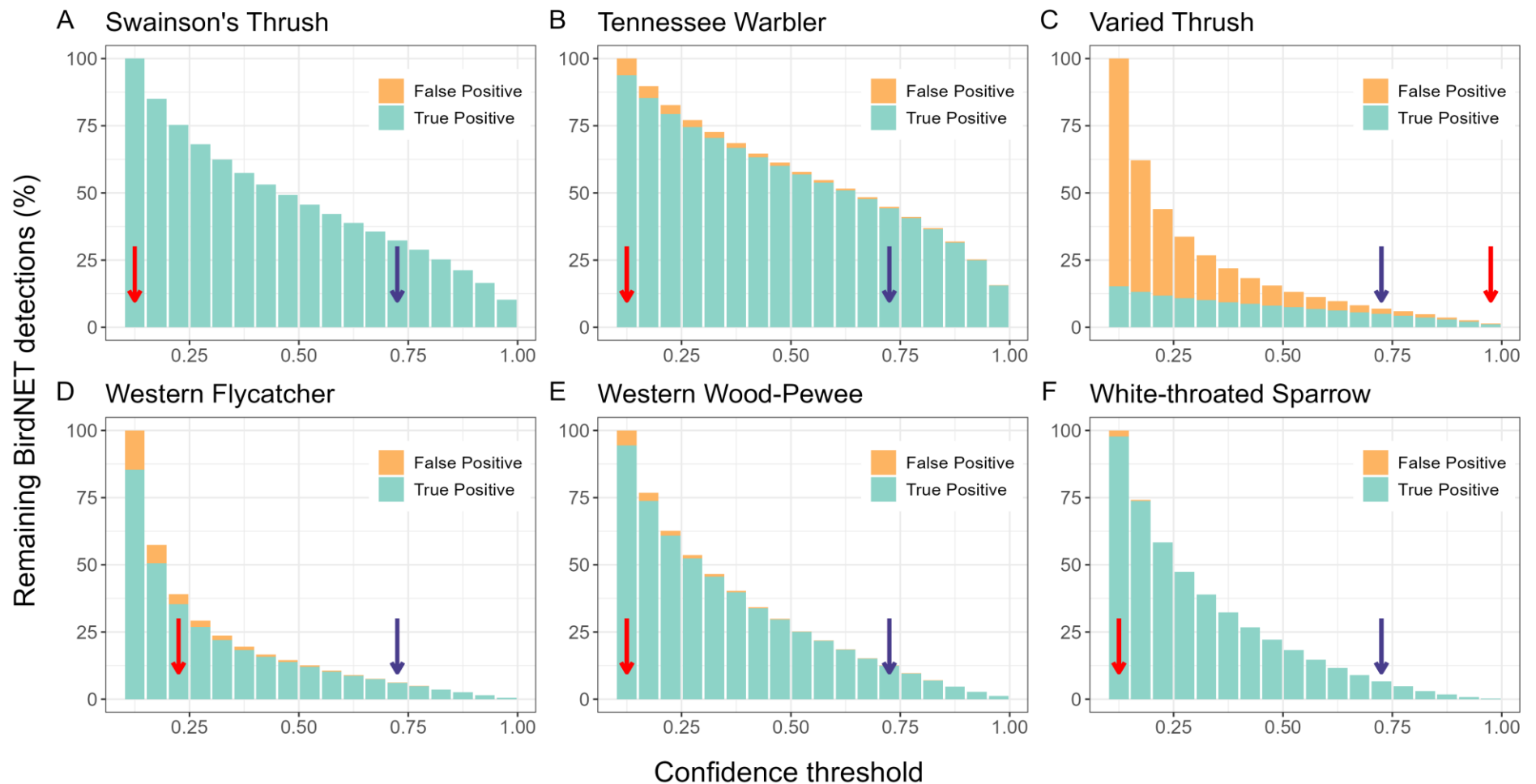


Fig. A2-4 The number of detections identified by BirdNET after applying various confidence thresholds, ranging from 0.1 to 0.95, with 0.05 increment, and categorized these detections as true positives (green) or false positives (orange), for the target species in this study. Blue arrows represented the universal threshold of 0.7 and the red arrows represented the species-specific thresholds for each species, defined as the minimum confidence threshold achieving a precision of at least 0.9.

Appendix for Chapter 3

We conducted a principal component analysis on the 30 sound features for 454 calls. We determined the number of principal components using broken stick method proposed by Jackson (1993) and selected the first 3 principal components (**Table A3-1**). These three components collectively explained 55% of the variance. Notably, the first component showed a higher correlation with frequency features, while the second and third components exhibited correlations with temporal features (**Table A3-1**; **Fig. A3-1**). The observed overlaps in the clusters align with expectations, as principal component analysis excels in reorienting original variables rather than distinctly separating clusters (**Fig. A3-2**).

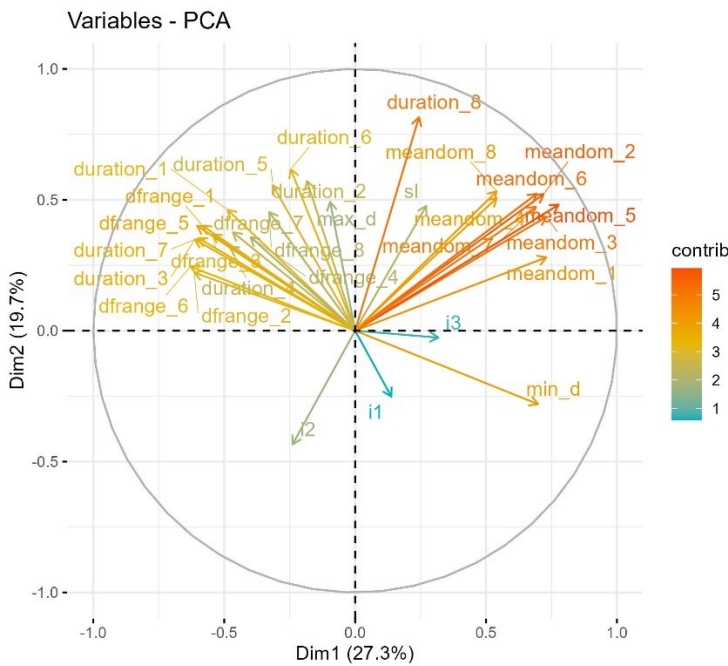


Fig. A3-1 PCA visualization of all the original variables on the PC1 and PC2.

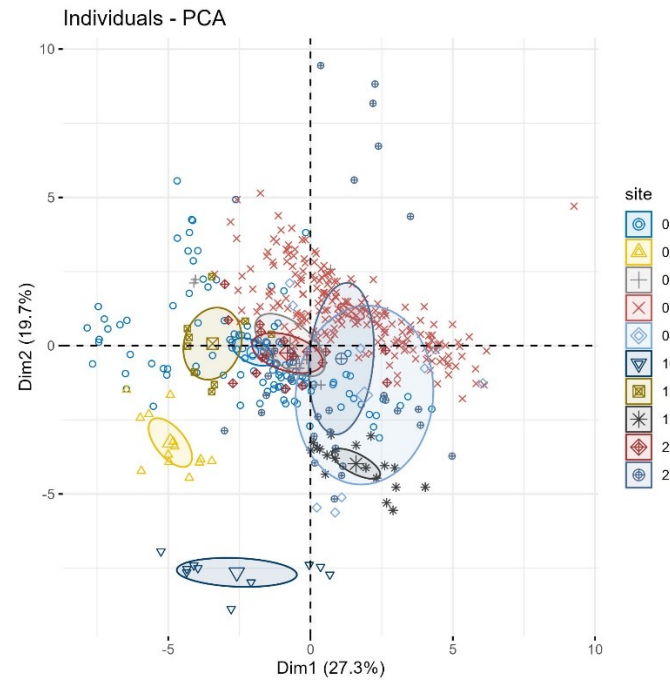


Fig. A3-2 PCA visualization of all the observations reprojected on the PC1 and PC2.

Table A3-1 Statistics of the PCA analysis on the 30 sound features.

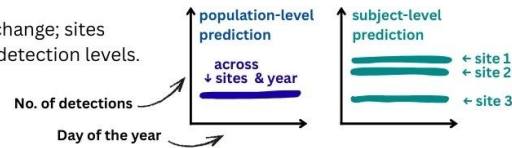
Components	Variance explained	Accumulated variance	Broken stick value	Correlated variables (absolute cor > 0.6)
Comp1	27%	27%	0.13	min_d, md_1, md_2, md_3, md_5, md_6, md_7
Comp2	20%	47%	0.09	d_6, d_8
Comp3	8%	55%	0.08	cl, i1, i3

Appendix for Chapter 4

m0 constant detection rate, with random site intercept

$1 + s(\text{site}, \text{bs} = "re")$

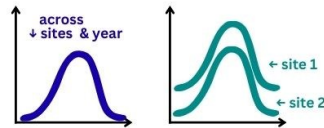
Assumes no seasonal change; sites differ only in baseline detection levels.



m1 one seasonal curve, with random site intercept

$s(\text{yday}, \text{bs} = "cc", k = 20) + s(\text{site}, \text{bs} = "re")$

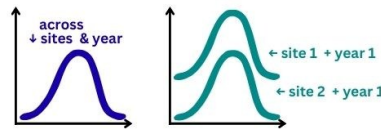
All sites and years share the same seasonal activity pattern; each site has a different baseline detection level.



m2 one seasonal curve, with random site + year intercept

$s(\text{yday}, \text{bs} = "cc", k = 20) + s(\text{year}, \text{bs} = "re") + s(\text{site}, \text{bs} = "re")$

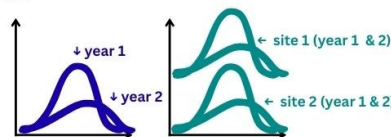
All years follow the same seasonal pattern, but years and sites differ in their overall detection levels.



m3 seasonal curve by year, with random site intercept

$s(\text{yday}, \text{by} = \text{year}, \text{bs} = "cc", k = 20) + s(\text{site}, \text{bs} = "re")$

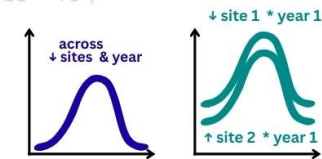
Each year has its own seasonal activity pattern (shape and peak timing), and sites differ in baseline detection rates.



m4 one seasonal curve, with random site*year intercept

$s(\text{yday}, \text{bs} = "cc", k = 20) + s(\text{yday_site}, \text{bs} = "re")$

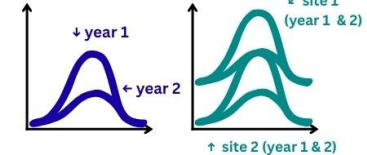
All sites and years follow the same seasonal pattern, but each site-year combination has its own baseline detection level (to capture annual conditions unique to each site).



m5 one seasonal curve and smooth seasonal deviation by year, with random site intercept

$s(\text{yday}, \text{bs} = "cc", k = 20) + s(\text{yday_year}, \text{bs} = "fs", k = 10) + s(\text{site}, \text{bs} = "re")$

There is one overall seasonal pattern, but the seasonal curve can change gradually across years, allowing flexible inter-annual variation in shape. Sites differ in baseline detection.



m6 seasonal curve by year, with random site*year intercept

$s(\text{yday}, \text{by} = \text{year}, \text{bs} = "cc", k = 20) + s(\text{yday_site}, \text{bs} = "re")$

Each year has its own seasonal activity pattern, and each site-year combination has its own baseline detection rate (e.g., accounting for weather, vegetation).

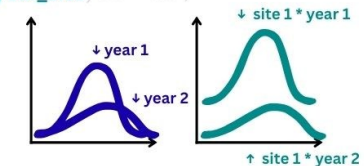


Figure A4-1 Specifications of the candidate GAM models, from the null model to the most complex form. All models included Julian day as a cyclic smooth to capture seasonal detection patterns and random effects to account for variation among sites and years. For each model, we provide the model definition, corresponding R code, ecological interpretation, and illustrative predictions: population-level predictions are shown as dark blue curves, and site-year (subject-level) predictions as light blue curves.

Appendix for Chapter 5

Table A5-1. The full list of 122 bird species detected from 66 sites at the John Prince Research Forest during the three years monitoring of songbird breeding season. For each species, the table includes the total number of detections identified by BirdNET and the number of sites where the species was detected. The values were calculated before applying any filters such as species-specific threshold or site selection.

Family	Scientific name	Common name	No. detections	No. sites
Accipitridae (Hawks, Eagles, and Kites)	<i>Astur atricapillus</i>	American Goshawk	4713	54
	<i>Buteo jamaicensis</i>	Red-tailed Hawk	296	50
	<i>Buteo platypterus</i>	Broad-winged Hawk	3302	57
	<i>Haliaeetus leucocephalus</i>	Bald Eagle	398	30
	<i>Accipiter striatus</i>	Sharp-shinned Hawk	116	38
Pandionidae (Osprey)	<i>Pandion haliaetus</i>	Osprey	238	39
Anatida (Ducks, Geese, and Waterfowl)	<i>Anas platyrhynchos</i>	Mallard	478	40
	<i>Branta canadensis</i>	Canada Goose	788	39
	<i>Mergus merganser</i>	Common Merganser	866	32
	<i>Bucephala clangula</i>	Common Goldeneye	82	21
	<i>Anas crecca</i>	Green-winged Teal	124	25
	<i>Lophodytes cucullatus</i>	Hooded Merganser	75	13
	<i>Bucephala islandica</i>	Barrow's Goldeneye	19	7
	<i>Cygnus buccinator</i>	Trumpeter Swan	97	18
	<i>Spatula discors</i>	Blue-winged Teal	24	7
	<i>Mareca americana</i>	American Wigeon	23	4
	<i>Aythya collaris</i>	Ring-necked Duck	144	14
Trochilidae (Hummingbirds)	<i>Selasphorus rufus</i>	Rufous Hummingbird	177	24

Family	Scientific name	Common name	No. detections	No. sites
Caprimulgidae (Nightjars and Allies)	<i>Chordeiles minor</i>	Common Nighthawk	469	45
Laridae (Gulls, Terns, and Skimmers)	<i>Chroicocephalus philadelphia</i>	Bonaparte's Gull	31	6
Scolopacidae (Sandpipers and Allies)	<i>Actitis macularius</i>	Spotted Sandpiper	297	33
	<i>Gallinago delicata</i>	Wilson's Snipe	11401	37
	<i>Tringa solitaria</i>	Solitary Sandpiper	94	23
	<i>Tringa melanoleuca</i>	Greater Yellowlegs	994	48
Alcedinidae (Kingfishers)	<i>Megaceryle alcyon</i>	Belted Kingfisher	462	51
Falconidae (Falcons and Caracaras)	<i>Falco sparverius</i>	American Kestrel	143	42
	<i>Falco columbarius</i>	Merlin	528	32
Phasianidae (Pheasants, Grouse, and Allies)	<i>Canachites canadensis</i>	Spruce Grouse	15906	58
	<i>Bonasa umbellus</i>	Ruffed Grouse	1000	49
	<i>Lagopus lagopus</i>	Willow Ptarmigan	62	23
Gaviidae (Loons)	<i>Gavia immer</i>	Common Loon	5530	57
Gruidae (Cranes)	<i>Antigone canadensis</i>	Sandhill Crane	535	35
Rallidae (Rails, Gallinules, and Coots)	<i>Porzana carolina</i>	Sora	3448	17
Bombycillidae (Waxwings)	<i>Bombycilla cedrorum</i>	Cedar Waxwing	5771	59
	<i>Bombycilla garrulus</i>	Bohemian Waxwing	151	40
Cardinalidae (Cardinals and Allies)	<i>Piranga ludoviciana</i>	Western Tanager	39616	61
	<i>Pheucticus ludovicianus</i>	Rose-breasted Grosbeak	3278	59
Certhiidae (Treecreepers)	<i>Certhia americana</i>	Brown Creeper	15142	61

Family	Scientific name	Common name	No. detections	No. sites
Cinclidae (Dippers)	<i>Cinclus mexicanus</i>	American Dipper	1364	27
Corvidae (Crows, Jays, and Magpies)	<i>Perisoreus canadensis</i>	Canada Jay	634	55
	<i>Corvus corax</i>	Common Raven	4672	54
	<i>Corvus brachyrhynchos</i>	American Crow	1477	30
Fringillidae (Finches, Euphonias, and Allies)	<i>Pinicola enucleator</i>	Pine Grosbeak	1139	61
	<i>Spinus pinus</i>	Pine Siskin	6400	60
	<i>Loxia leucoptera</i>	White-winged Crossbill	6125	60
	<i>Coccothraustes vespertinus</i>	Evening Grosbeak	552	58
	<i>Haemorhous purpureus</i>	Purple Finch	10420	62
	<i>Leucosticte tephrocotis</i>	Gray-crowned Rosy-Finch	1210	51
	<i>Haemorhous mexicanus</i>	House Finch	4431	59
	<i>Loxia curvirostra</i>	Red Crossbill	165	45
Hirundinidae (Swallows)	<i>Tachycineta thalassina</i>	Violet-green Swallow	485	39
	<i>Tachycineta bicolor</i>	Tree Swallow	1094	27
Icteridae (Troupials and Allies)	<i>Agelaius phoeniceus</i>	Red-winged Blackbird	204	23
	<i>Molothrus ater</i>	Brown-headed Cowbird	87	28
	<i>Euphagus carolinus</i>	Rusty Blackbird	1395	36
Motacillidae (Wagtails and Pipits)	<i>Anthus rubescens</i>	American Pipit	4368	58
Paridae (Tits, Chickadees, and Titmice)	<i>Poecile hudsonicus</i>	Boreal Chickadee	4430	57
	<i>Poecile gambeli</i>	Mountain Chickadee	12994	61
	<i>Poecile atricapillus</i>	Black-capped Chickadee	1615	54
Parulidae (New World Warblers)	<i>Setophaga magnolia</i>	Magnolia Warbler	74084	62
	<i>Setophaga coronata</i>	Yellow-rumped Warbler	111092	62
	<i>Setophaga townsendi</i>	Townsend's Warbler	7361	61

Family	Scientific name	Common name	No. detections	No. sites
	<i>Setophaga striata</i>	Blackpoll Warbler	4619	61
	<i>Geothlypis tolmiei</i>	MacGillivray's Warbler	30304	61
	<i>Leiothlypis peregrina</i>	Tennessee Warbler	36766	58
	<i>Setophaga ruticilla</i>	American Redstart	74872	59
	<i>Seiurus aurocapilla</i>	Ovenbird	5286	59
	<i>Cardellina pusilla</i>	Wilson's Warbler	4738	60
	<i>Leiothlypis celata</i>	Orange-crowned Warbler	11951	60
	<i>Parkesia noveboracensis</i>	Northern Waterthrush	146613	60
	<i>Setophaga petechia</i>	Yellow Warbler	3407	42
	<i>Geothlypis trichas</i>	Common Yellowthroat	4153	46
Passerellidae (New World Sparrows)	<i>Spizella passerina</i>	Chipping Sparrow	5162	54
	<i>Zonotrichia albicollis</i>	White-throated Sparrow	42421	60
	<i>Zonotrichia leucophrys</i>	White-crowned Sparrow	69534	62
	<i>Junco hyemalis</i>	Dark-eyed Junco	48344	62
	<i>Passerella iliaca</i>	Fox Sparrow	2861	60
	<i>Melospiza lincolnii</i>	Lincoln's Sparrow	19790	59
	<i>Melospiza melodia</i>	Song Sparrow	3390	23
	<i>Spizella pallida</i>	Clay-colored Sparrow	228	39
	<i>Melospiza georgiana</i>	Swamp Sparrow	1846	56
	<i>Zonotrichia atricapilla</i>	Golden-crowned Sparrow	140	35
Regulidae (Kinglets)	<i>Pooecetes gramineus</i>	Vesper Sparrow	456	32
	<i>Regulus satrapa</i>	Golden-crowned Kinglet	84683	61
Sittidae (Nuthatches)	<i>Corthylio calendula</i>	Ruby-crowned Kinglet	13918	59
	<i>Sitta canadensis</i>	Red-breasted Nuthatch	23207	61
Troglodytidae (Wrens)	<i>Troglodytes pacificus</i>	Pacific Wren	60989	59

Family	Scientific name	Common name	No. detections	No. sites
Turdidae (Thrushes and Allies)	<i>Catharus ustulatus</i>	Swainson's Thrush	327431	62
	<i>Ixoreus naevius</i>	Varied Thrush	32529	61
	<i>Catharus guttatus</i>	Hermit Thrush	52026	61
	<i>Turdus migratorius</i>	American Robin	37737	62
	<i>Myadestes townsendi</i>	Townsend's Solitaire	787	55
	<i>Sialia currucoides</i>	Mountain Bluebird	1238	54
Tyrannidae (Tyrant Flycatchers)	<i>Empidonax difficilis</i>	Western Flycatcher	12474	62
	<i>Empidonax hammondi</i>	Hammond's Flycatcher	141949	62
	<i>Empidonax minimus</i>	Least Flycatcher	44654	58
	<i>Empidonax oberholseri</i>	Dusky Flycatcher	9356	58
	<i>Contopus cooperi</i>	Olive-sided Flycatcher	17061	61
	<i>Empidonax flaviventris</i>	Yellow-bellied Flycatcher	5815	61
	<i>Empidonax alnorum</i>	Alder Flycatcher	21412	49
	<i>Tyrannus tyrannus</i>	Eastern Kingbird	48	26
Vireonidae (Vireos, Shrike-Babblers, and Erpornis)	<i>Contopus sordidulus</i>	Western Wood-Pewee	10992	50
	<i>Vireo gilvus</i>	Warbling Vireo	87002	62
	<i>Vireo cassinii</i>	Cassin's Vireo	6188	55
Ardeidae (Herons, Egrets, and Bitterns)	<i>Vireo olivaceus</i>	Red-eyed Vireo	600	35
	<i>Botaurus lentiginosus</i>	American Bittern	359	34
Picidae (Woodpeckers)	<i>Sphyrapicus varius</i>	Yellow-bellied Sapsucker	5261	57
	<i>Sphyrapicus nuchalis</i>	Red-naped Sapsucker	5182	57
	<i>Picoides arcticus</i>	Black-backed Woodpecker	2928	60
	<i>Picoides dorsalis</i>	American Three-toed Woodpecker	6417	62
	<i>Dryobates villosus</i>	Hairy Woodpecker	1589	54

Family	Scientific name	Common name	No. detections	No. sites
	<i>Dryocopus pileatus</i>	Pileated Woodpecker	1910	56
	<i>Dryobates pubescens</i>	Downy Woodpecker	331	49
	<i>Colaptes auratus</i>	Northern Flicker	950	43
Podicipedidae (Grebes)	<i>Podiceps grisegena</i>	Red-necked Grebe	26	8
Strigidae (Owls)	<i>Strix varia</i>	Barred Owl	498	43
	<i>Bubo virginianus</i>	Great Horned Owl	4902	57
	<i>Asio otus</i>	Long-eared Owl	294	40
	<i>Strix nebulosa</i>	Great Gray Owl	419	36
	<i>Aegolius acadicus</i>	Northern Saw-whet Owl	2223	54
	<i>Glaucidium gnoma</i>	Northern Pygmy-Owl	876	48
	<i>Aegolius funereus</i>	Boreal Owl	5826	44

Table A5-2. Estimated asymptotic species richness for each site. ARU days refers to the number of days an autonomous recording unit (ARU) operated at a given site over the three years of surveying period. Species ARU days represents the cumulative number of days a species was detected at that site. LCL and UCL denote the lower and upper 95% confidence limits of the estimated asymptotic richness. Sites are ordered by the number of ARU days and grouped into three categories: 0–90 days, 90–120 days, and >120 days.

Site	ARU days	Species ARU days	Observed richness	Asymptotic richness	No. of species added	LCL	UCL
N_15	32	510	34	34.24	+ 0.24	34.01	38.62
N_25	50	463	31	35.41	+ 4.41	31.48	71.14
N_11	50	831	34	34.16	+ 0.16	34.01	37.49
14_29	52	606	31	34.92	+ 3.92	31.55	59.22
14_05	59	628	33	38.90	+ 5.9	34.05	66.10
N_19	65	641	31	48.72	+ 17.72	33.49	157.04
14_02	66	115	19	19.49	+ 0.49	19.03	27.34
N_23	69	571	31	33.22	+ 2.22	31.26	49.80
14_17	70	753	34	39.91	+ 5.91	35.05	67.18
14_42	71	775	29	29.66	+ 0.66	29.06	36.72
14_23	76	863	37	37.99	+ 0.99	37.09	47.94
14_16	81	1100	34	34.00	+ 0	34.00	35.61
N_05	82	862	31	31.16	+ 0.16	31.01	34.50
14_03	82	990	37	37.40	+ 0.4	37.03	42.26
N_08	83	453	32	36.45	+ 4.45	32.49	72.45
N_10	83	1034	35	41.17	+ 6.17	35.96	74.68
14_35	85	570	31	31.10	+ 0.1	31.00	33.52
14_41	85	959	32	34.63	+ 2.63	32.37	50.67
N_22	87	926	35	36.48	+ 1.48	35.17	47.76
14_08	89	746	34	35.98	+ 1.98	34.27	48.23
14_30	91	453	31	43.12	+ 12.12	33.18	98.45
14_34	91	1134	34	34.49	+ 0.49	34.04	40.19
14_21	92	870	36	37.48	+ 1.48	36.17	48.77
N_16	94	1012	33	37.45	+ 4.45	33.49	73.51
14_15	98	641	31	35.12	+ 4.12	31.67	56.54
14_19	101	1145	38	38.89	+ 0.89	38.10	46.30

Site	ARU days	Species ARU days	Observed richness	Asymptotic richness	No. of species added	LCL	UCL
14_04	102	492	28	48.79	+ 20.79	33.02	114.06
14_18	102	807	30	33.96	+ 3.96	30.55	58.46
14_10	103	865	29	33.46	+ 4.46	29.49	69.54
14_11	104	1077	34	42.91	+ 8.91	35.51	86.79
14_31	105	1247	35	35.66	+ 0.66	35.06	42.75
N_14	105	1515	35	35.99	+ 0.99	35.07	49.30
N_21	108	1068	33	35.23	+ 2.23	33.26	51.88
14_20	109	1015	35	38.96	+ 3.96	35.55	63.48
14_24	109	1166	37	41.13	+ 4.13	37.67	62.56
N_18	111	533	29	31.64	+ 2.64	29.37	47.72
N_09	111	1421	36	41.95	+ 5.95	36.92	74.61
14_28	114	985	34	37.10	+ 3.1	34.50	53.15
14_06	115	748	27	31.46	+ 4.46	27.49	67.58
N_04	115	1654	35	36.98	+ 1.98	35.18	56.95
N_12	118	1645	35	39.46	+ 4.46	35.49	75.59
14_01	121	915	31	38.93	+ 7.93	31.98	95.30
N_06	121	1665	36	45.92	+ 9.92	37.85	89.05
14_33	124	1115	34	34.99	+ 0.99	34.09	44.99
N_24	137	1642	37	37.25	+ 0.25	37.01	41.71
N_13	139	834	35	36.49	+ 1.49	35.17	47.81
14_07	139	1404	36	36.12	+ 0.12	36.01	38.91
N_17	139	1597	34	34.50	+ 0.5	34.03	42.39
14_37	143	1300	36	37.49	+ 1.49	36.17	48.81
14_40	143	1455	35	35.08	+ 0.08	35.00	37.26
14_39	145	888	32	38.21	+ 6.21	32.97	71.87
14_38	147	1393	35	36.49	+ 1.49	35.17	47.81
N_07	155	1180	33	34.49	+ 1.49	33.17	45.82
14_25	155	1841	37	37.99	+ 0.99	37.07	51.35
N_01	155	1893	35	35.50	+ 0.5	35.03	43.40
14_09	164	1990	37	37.99	+ 0.99	37.09	48.01
14_14	167	1668	33	37.47	+ 4.47	33.49	73.68
14_13	169	1877	35	42.95	+ 7.95	35.98	99.45
14_26	182	1934	36	39.98	+ 3.98	36.55	64.58